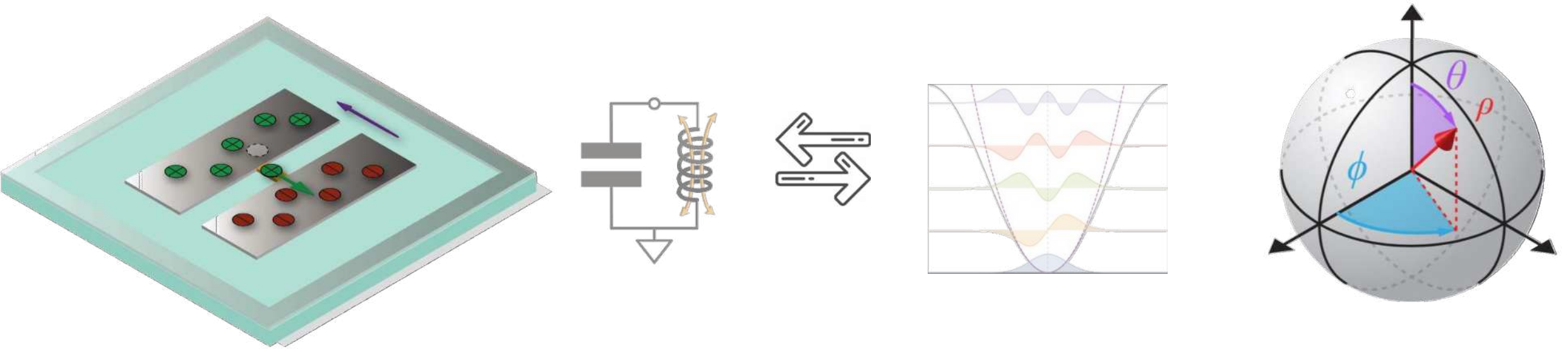


Quantum Hardware Design

Energy, Circuits, and Metal



Zlatko K. Minev

IBM Quantum

IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev

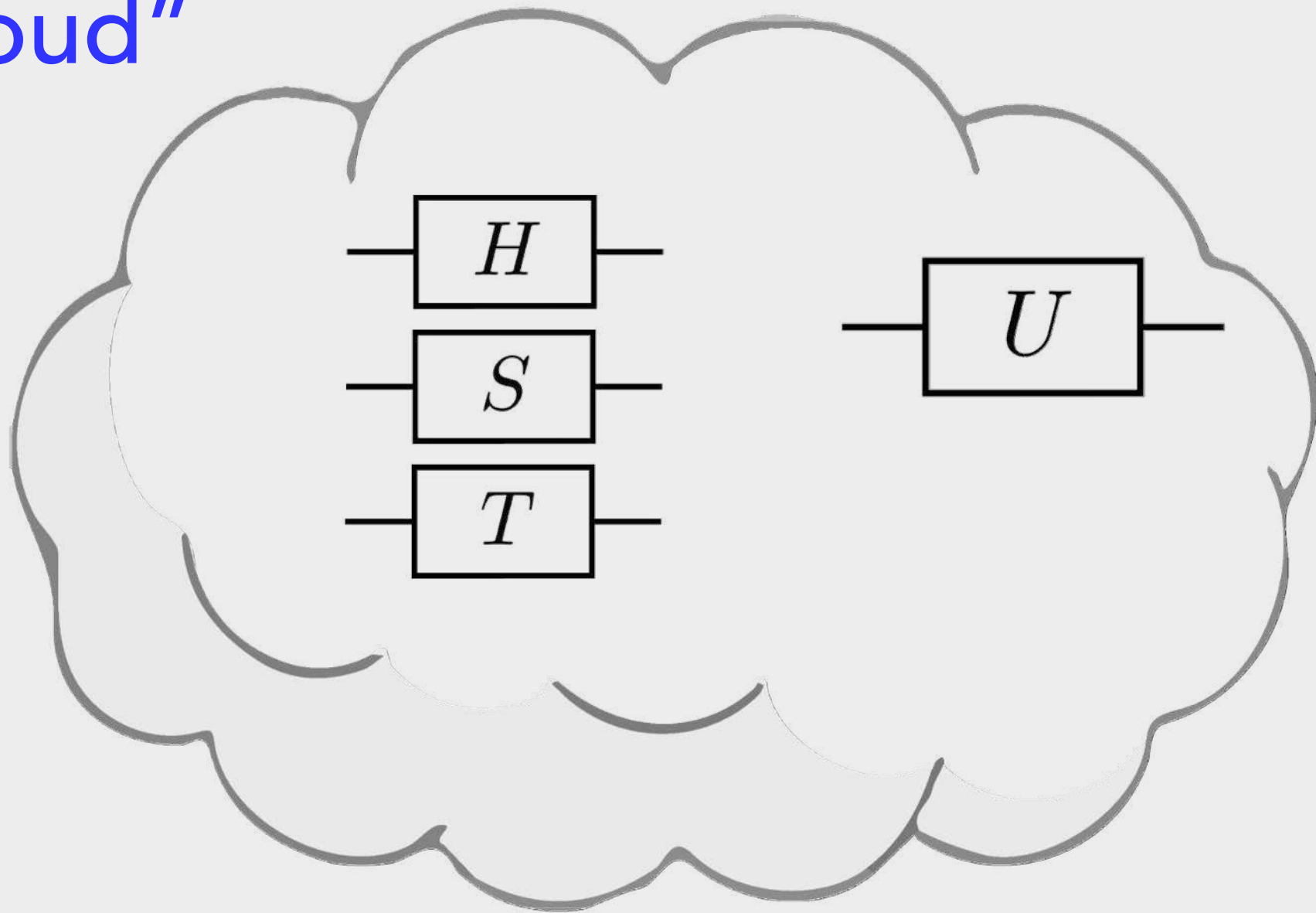


zlatko-minev.com



qiskit.org/metal

"Quantum cloud"





"Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory."

Asher Peres



Quantum in the cloud or lab

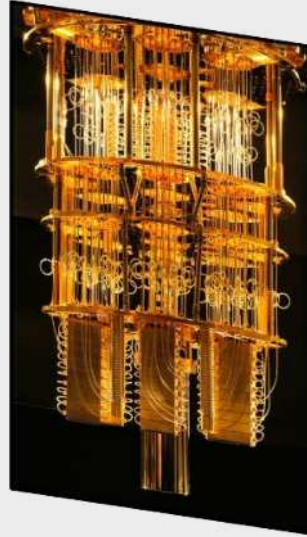
Superconducting qubits



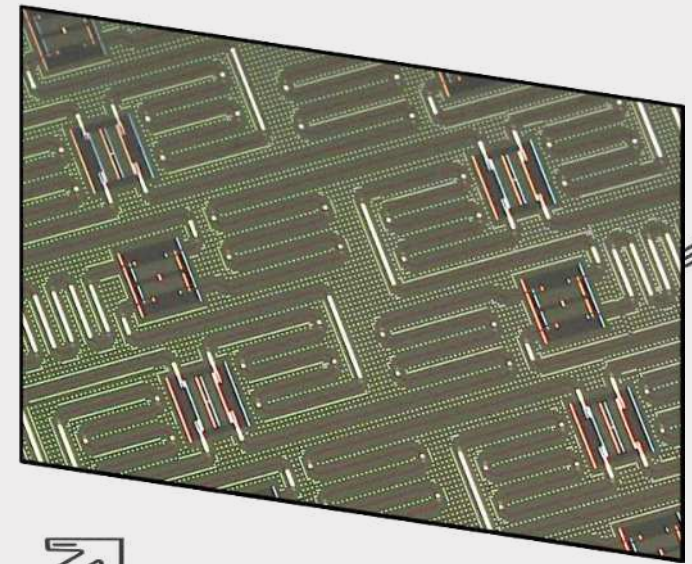
Quantum community



Lab / cloud facility



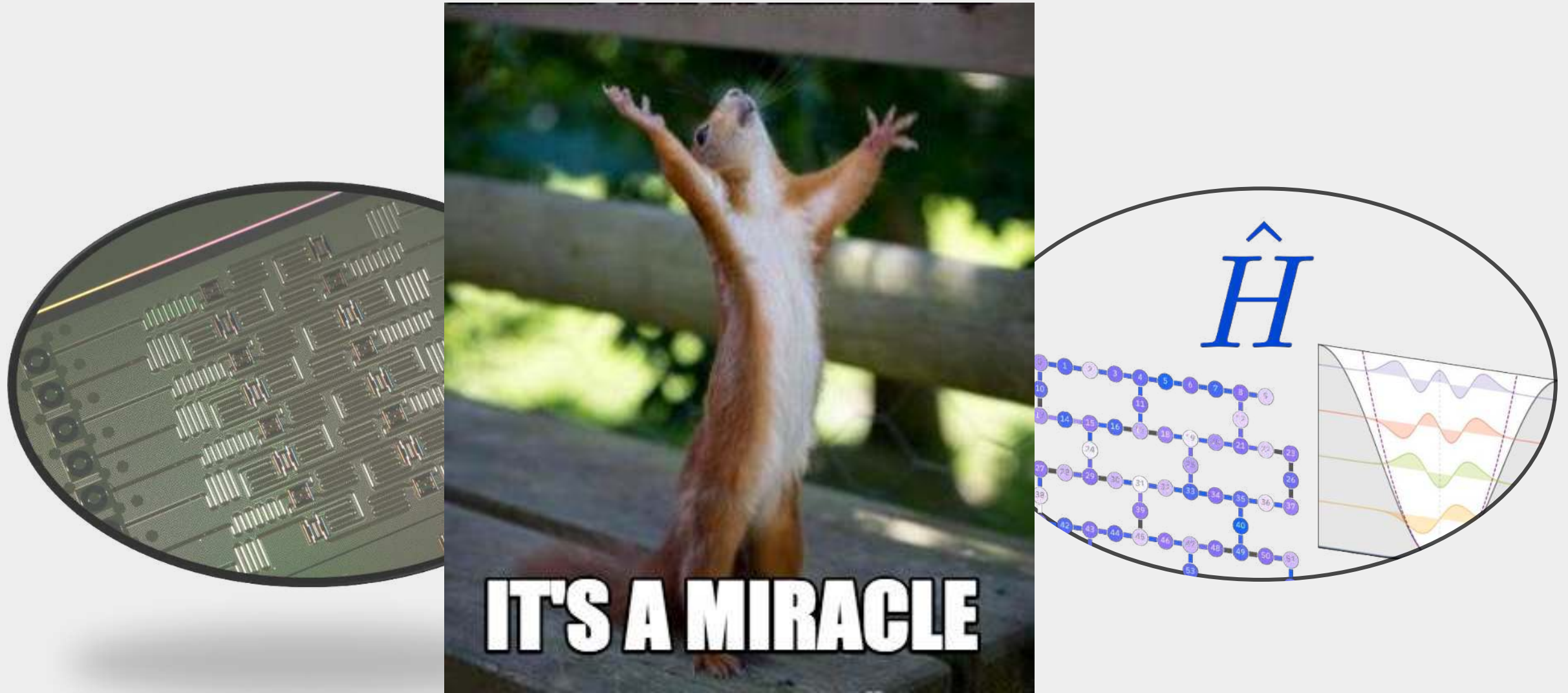
Cryogenic environment



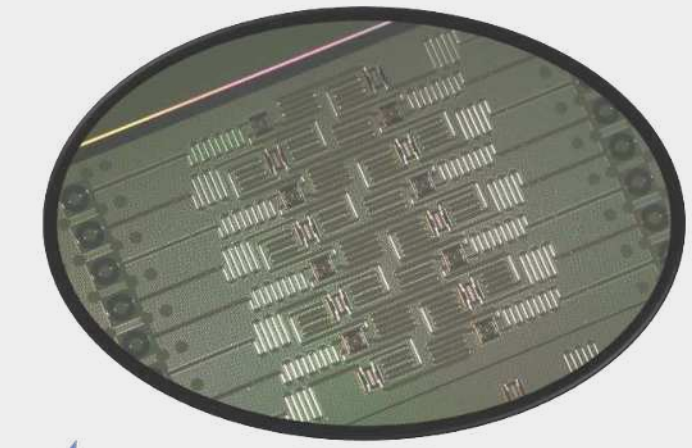
Quantum-device chip



Physical Devices $\leftrightarrow$ Quantum Hamiltonian



Physical Devices ↔ Quantum Hamiltonian



Overview of design

I just want the answer: Qiskit Metal it for me

You can design your own quantum chip! Open-source
(IBM)

Energy-participation ratio (EPR)

Minev et al. arXiv:1902.10355 & arXiv:2010.00620
(Yale)

cQED with quasi-lumped models (LOM)

Minev et al. arXiv:2103.10344
(IBM)

$\hat{H}$

Qiskit | quantum device design

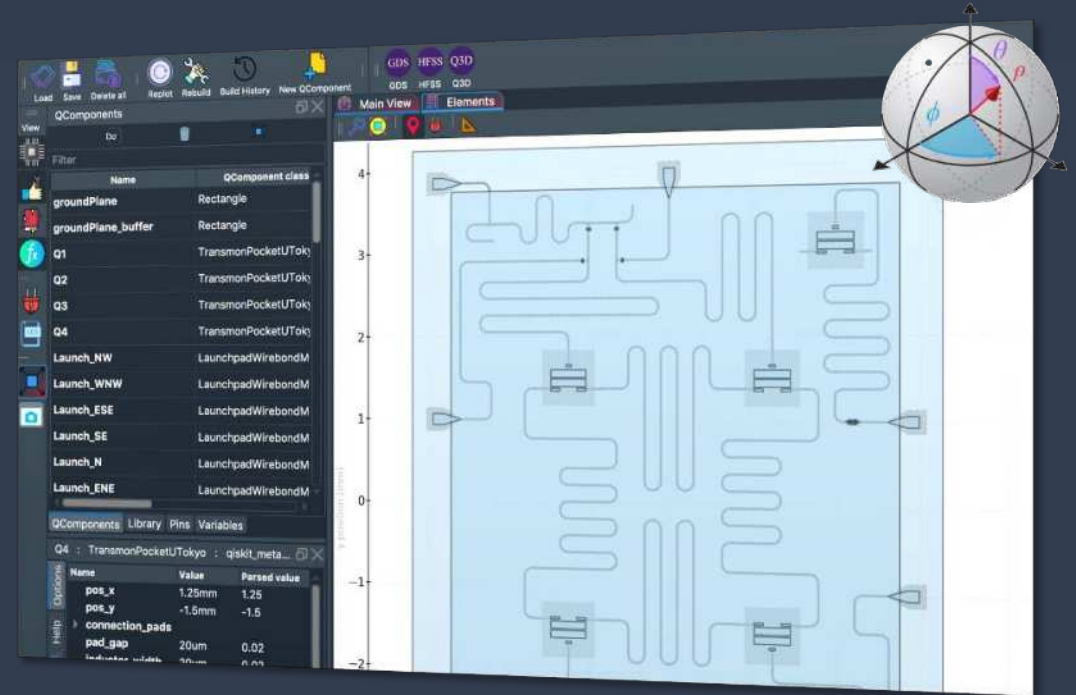


Zlatko Minev
PI & Lead



Thomas McConkey
Co-PI

Open source
Apache 2.0!
qiskit.org/metal



Jeremy Drysdale



Priti Shah



Dennis Wang



Yehan Liu



Grace Harper

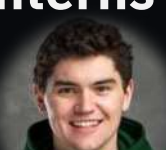


Marco Facchini



John Blair

Interns



Olivia Lanes



Will Shanks

Power-Ups



Close collab



Quantum in the cloud or lab

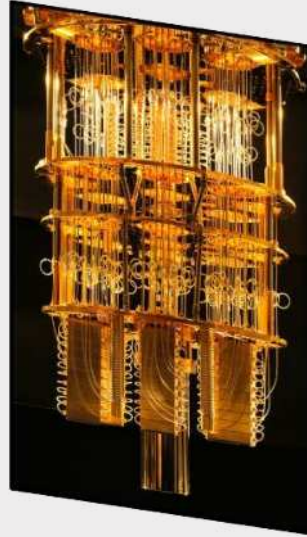
Superconducting qubits



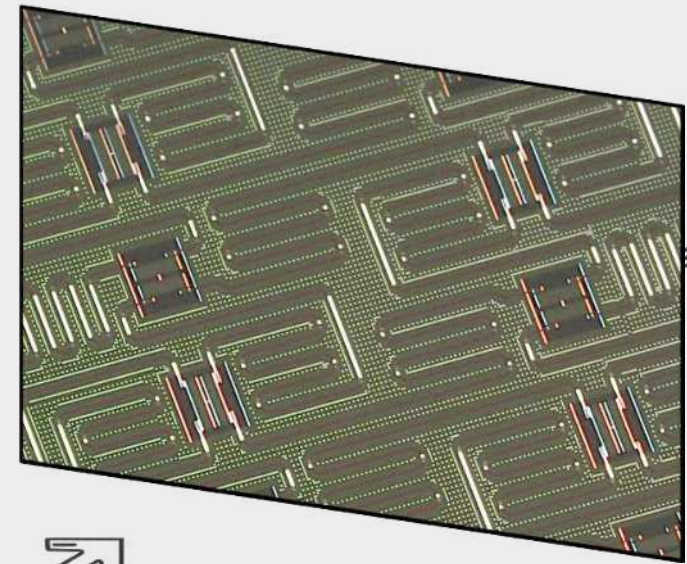
Quantum community



Lab / cloud facility

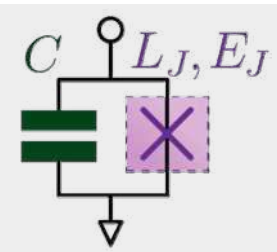


Cryogenic environment

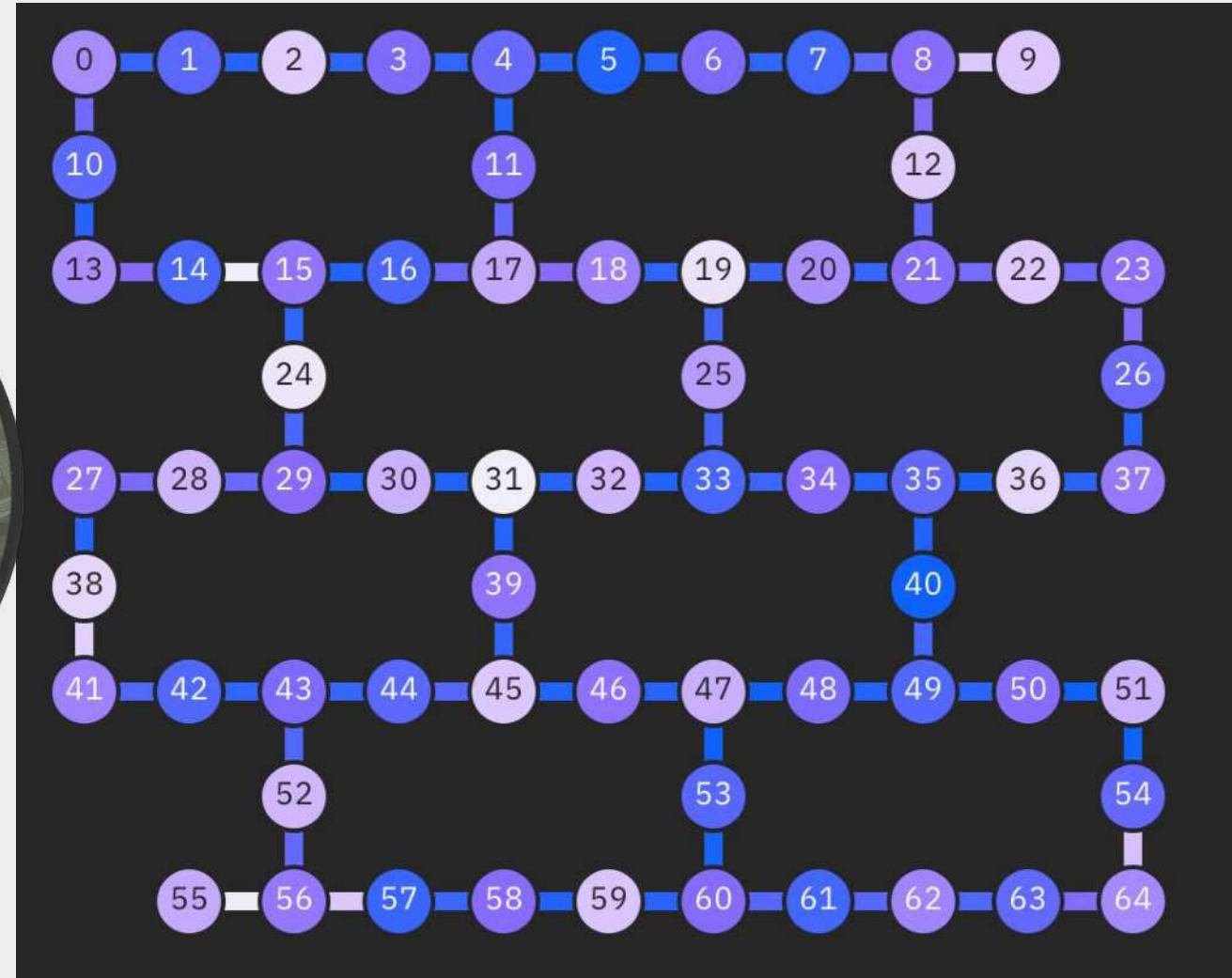
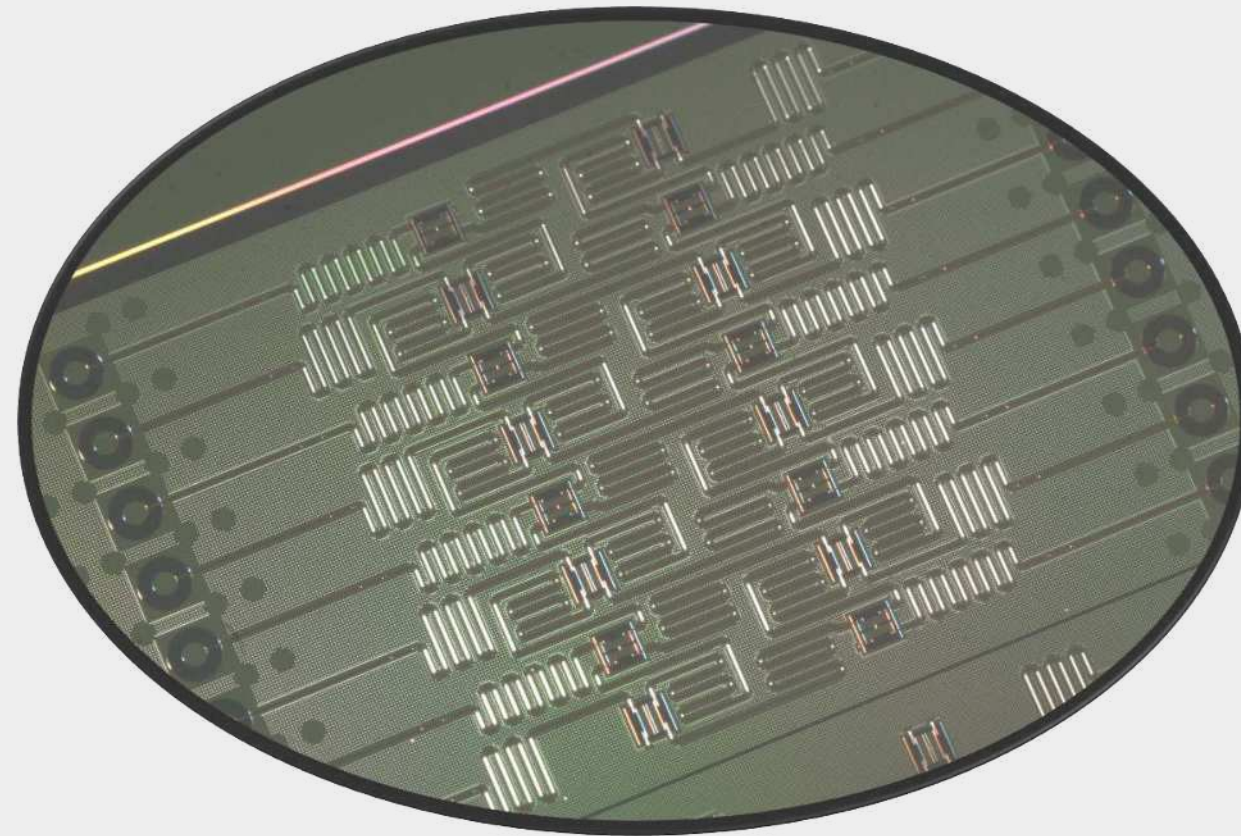


Quantum-device chip



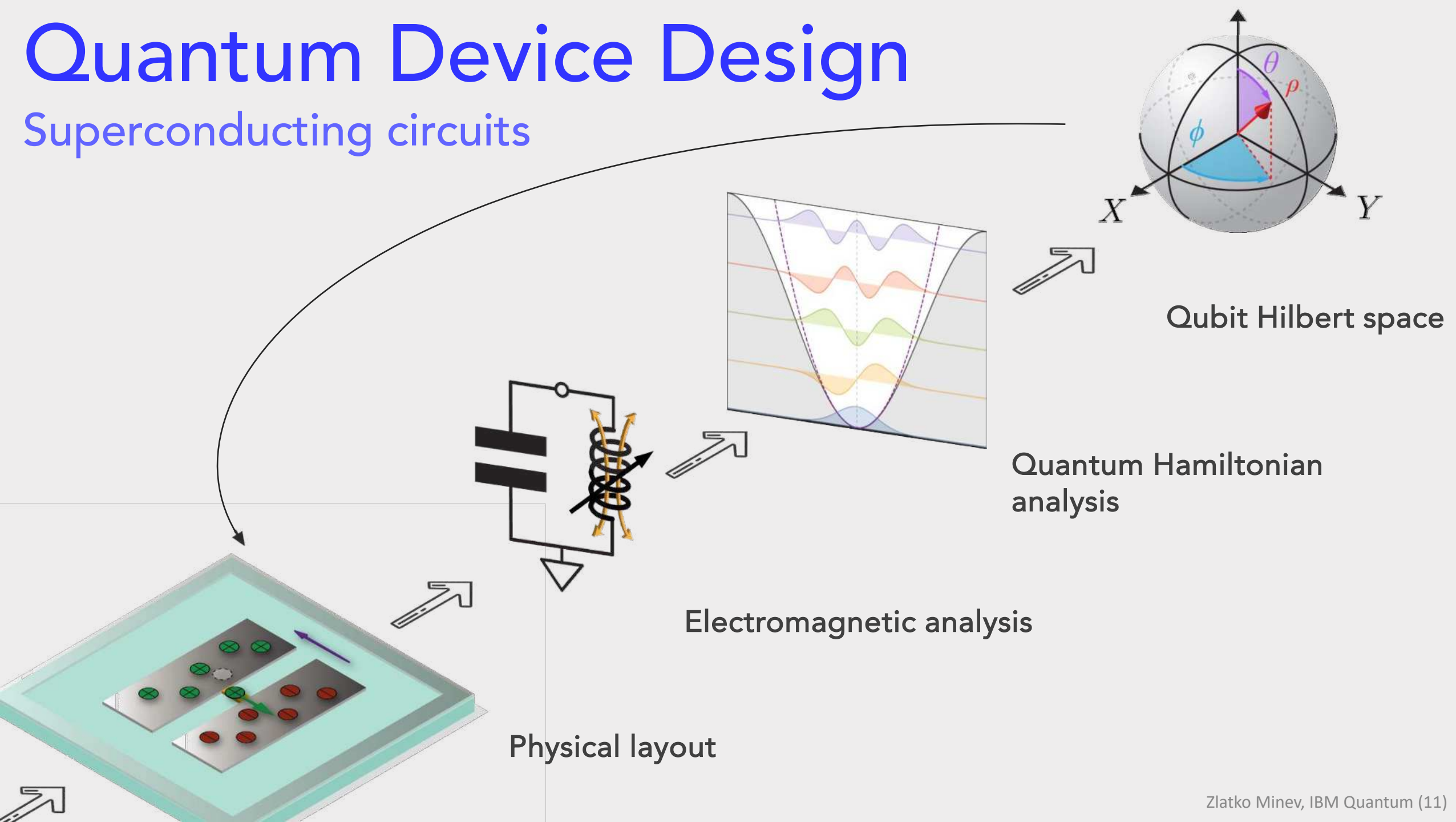


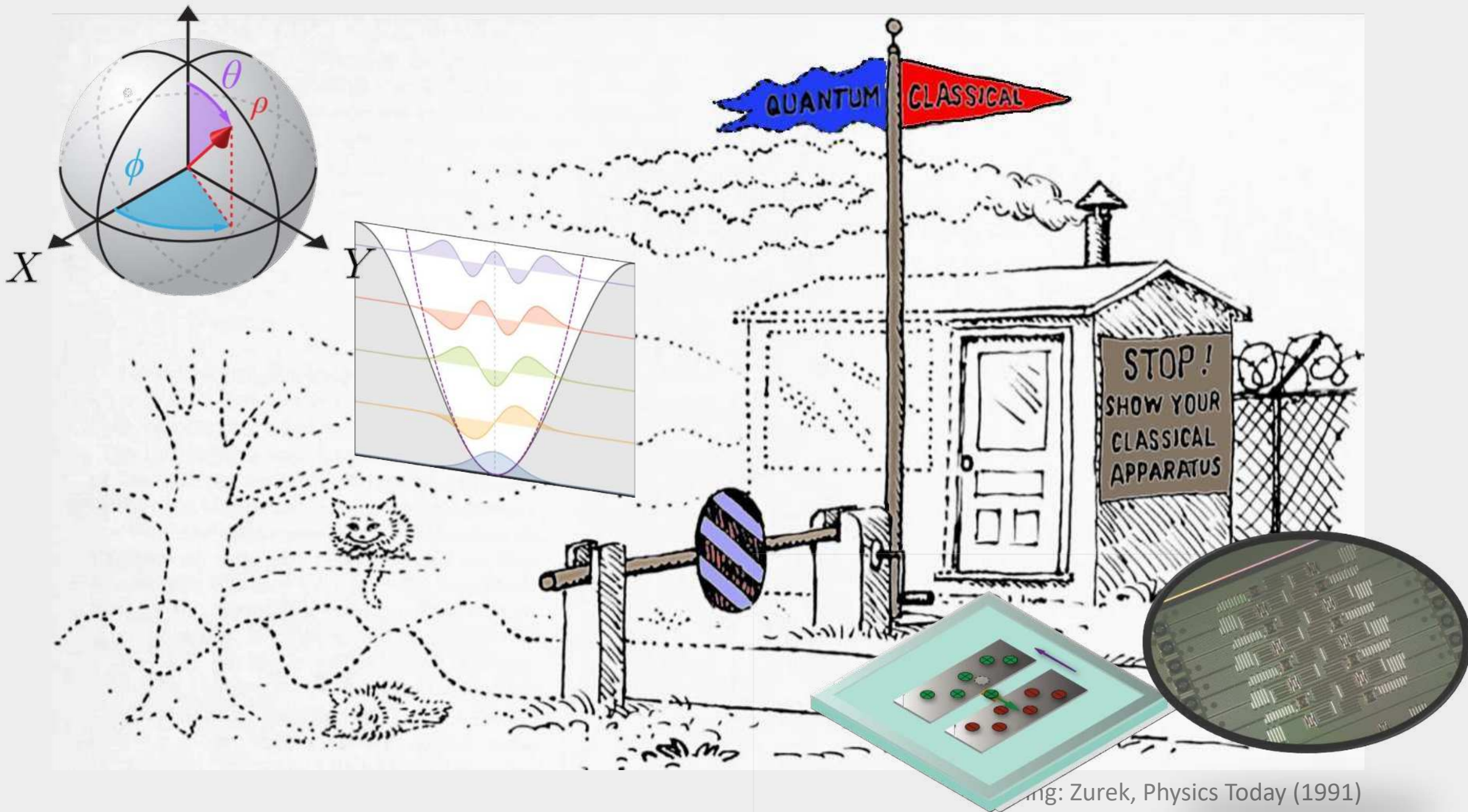
Quantum computer made from 'artificial atoms'



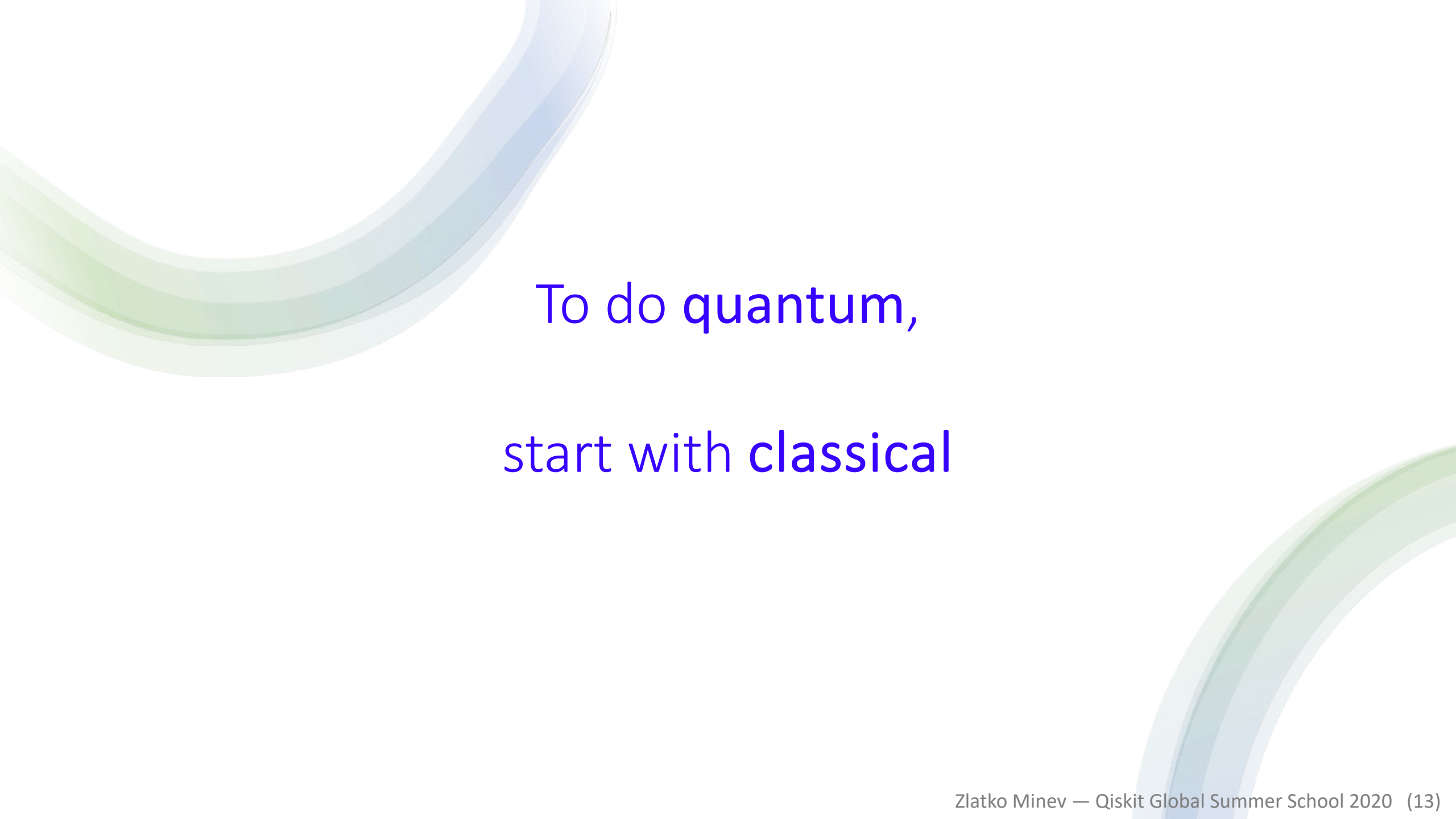
Quantum Device Design

Superconducting circuits





ing: Zurek, Physics Today (1991)



To do quantum,
start with classical

Summer School Lectures

Minev, Z., Lec. 16-22, *Introduction to Quantum Computing and Quantum Hardware*,
url: qiskit.org/learn/intro-qc-qh (2020)



Magnetic flux and inductance

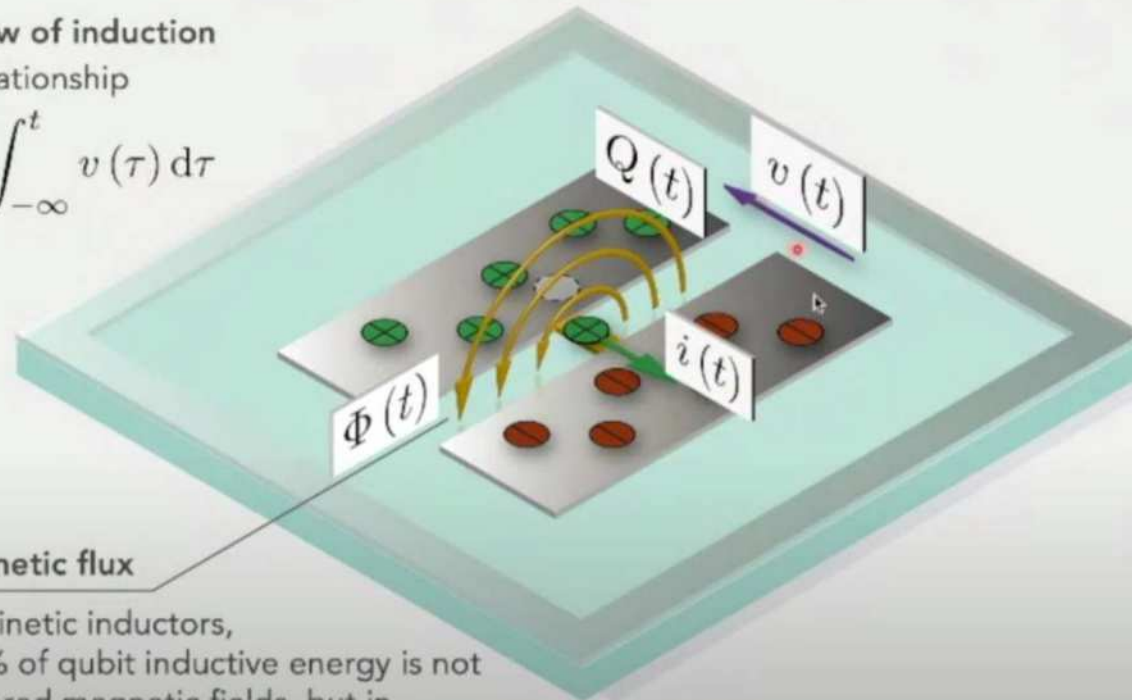
Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic induct



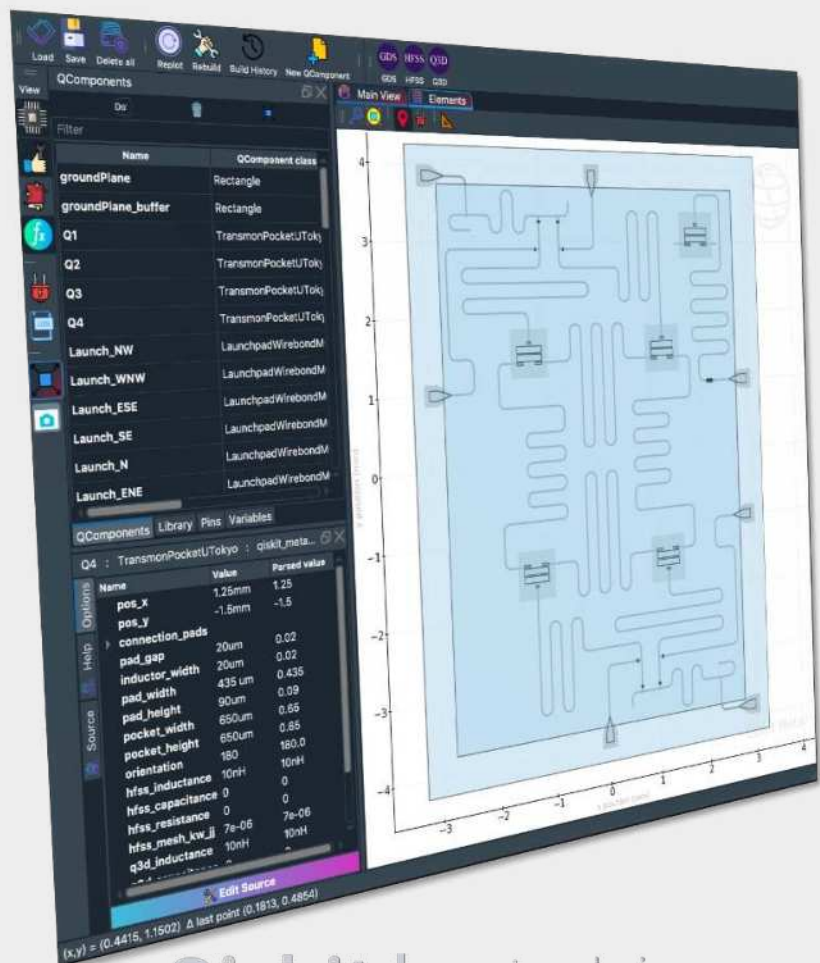
It can be related to the voltage across

Image: Zlatko Minev

For more information, review the **Introduction to Quantum Computing and Quantum Hardware** lectures below

• Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions Part 1	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions Part 2	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions Part 3	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Part 1	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Part 2	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Part 3	Lecture Video	Lecture Notes	Lab

There's a description for every job



Qiskit | quantum device
design



$$\hat{H}_{\text{tot}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$

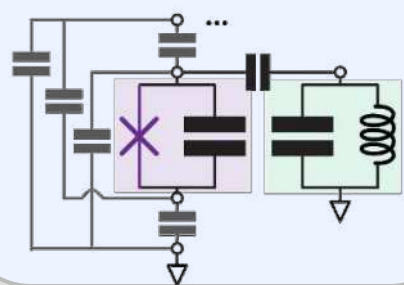


Landscape of quantization methods

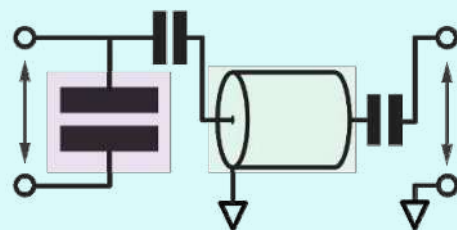
quasi-static

full-wave

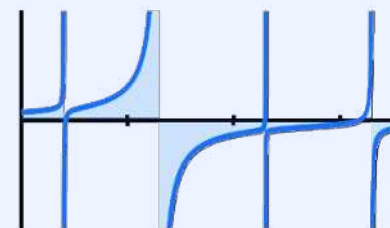
lumped



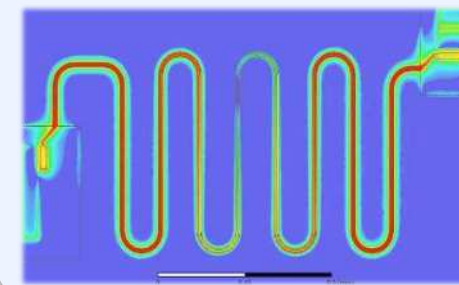
quasi-lumped



impedance



energy



fast

more information, complexity, accuracy

Qiskit | quantum device
Metal design

Yurke & Denker (1984),
Devoret (1997), Burkard
et al. (2004), Koch *et al.*...

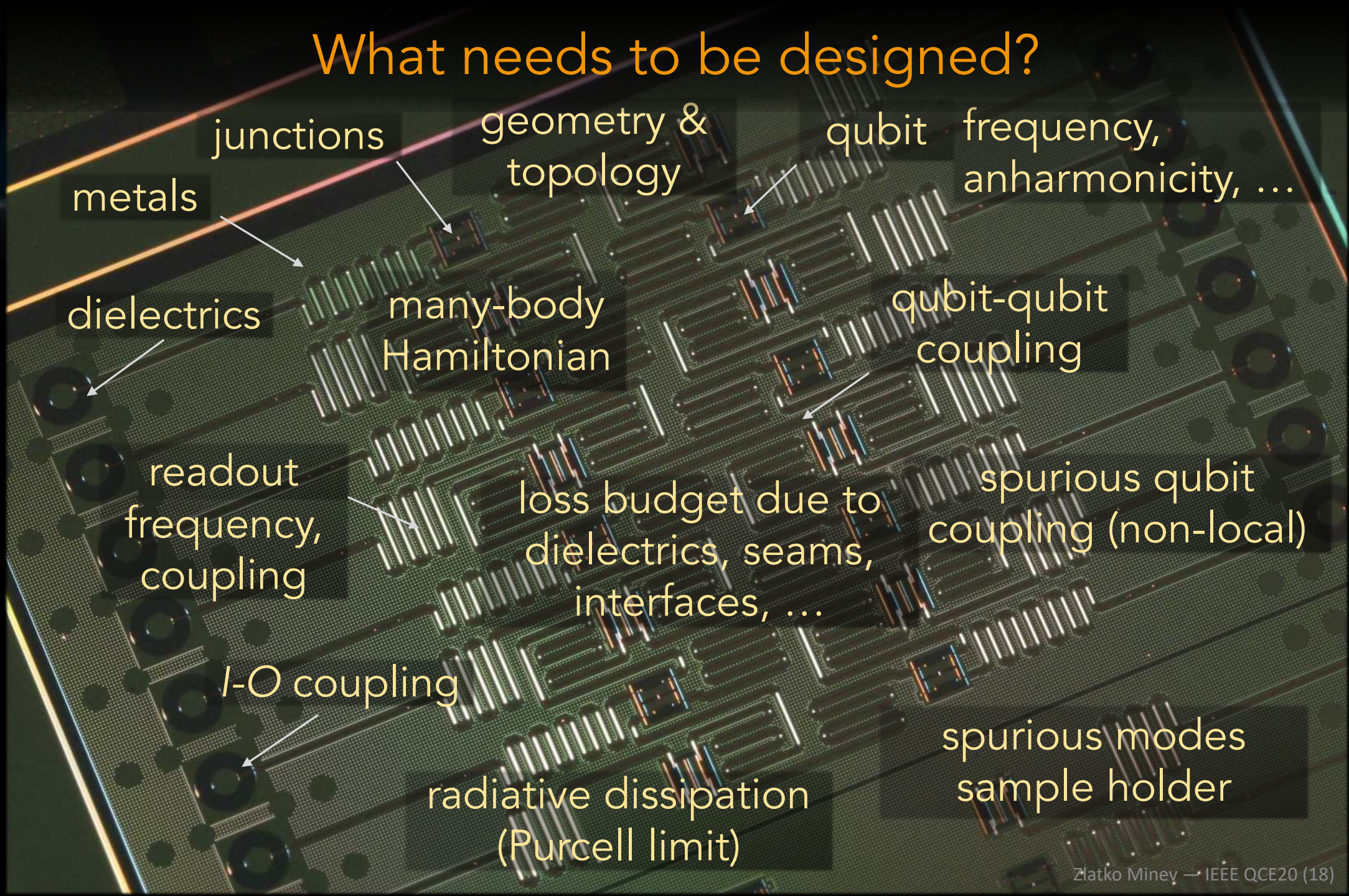
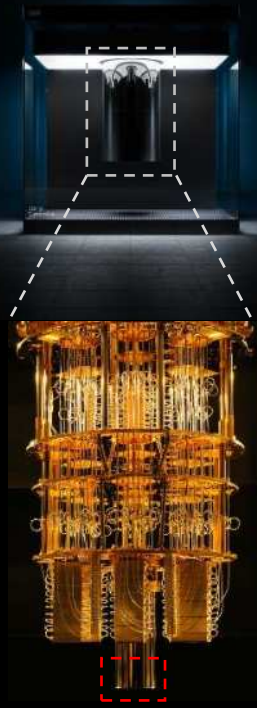
Malekakhlagh *et al.* (2017, 2019),
Gely *et al.* (2019), Parra-Rodriguez
et al. (2019), Minev *et al.* (2021), ...

Nigg *et al.* (2012), Bourassa
et al. (2012), Solgun *et al.*
(2014, 2015, 2017) ...

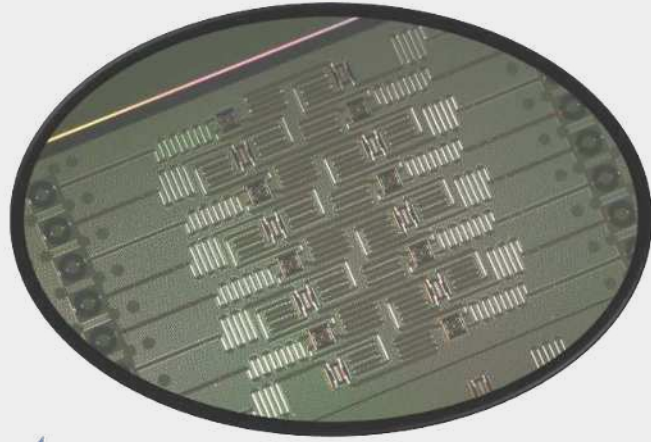
Minev (2018)
Minev *et al.* (2020)

Zlatko Minev, IBM Quantum (17)

What needs to be designed?



Physical Devices ↔ Quantum Hamiltonian



You can design your own quantum chip!

Open source
(IBM)

$\hat{H}$

You Don't Have To Be A Rocket (Or Quantum) Scientist To Design A Quantum Computer Chip Using IBM's New Tool Called Qiskit Metal

Paul Smith-Goodson Contributor



Moor Insights and Strategy Contributor Group ©

Cloud

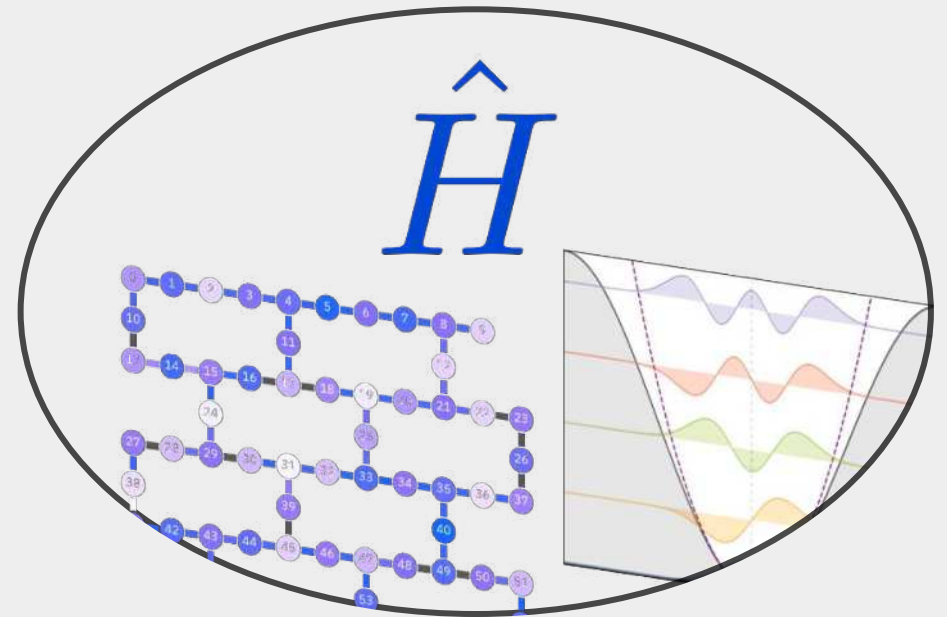
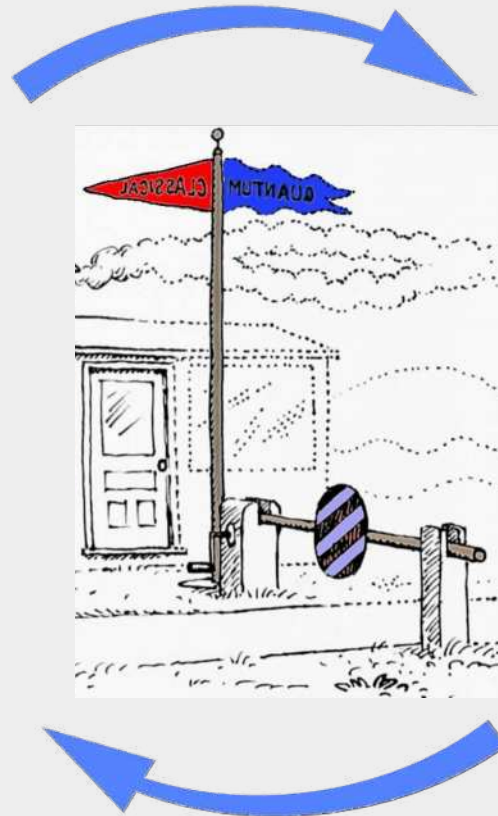
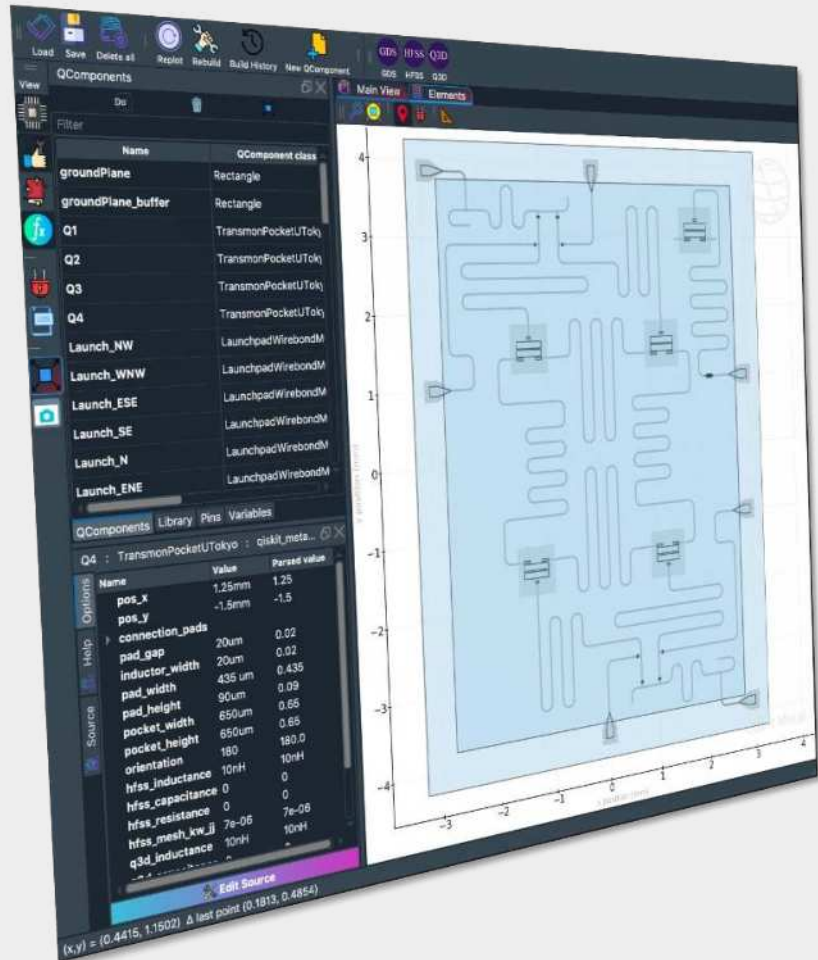
Analyst-in-residence, Quantum Computing



IBM Qiskit IBM

Intuitively, almost everyone can appreciate how difficult and knowledge-intensive it is to design, develop, analyze, and simulate a quantum computer chip. Without years

Make easy?



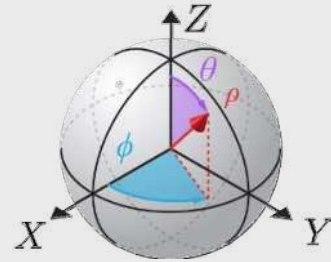
Quantum chip design flow

Complexity, Information, Accuracy
Risk, Cost, Time, Resources

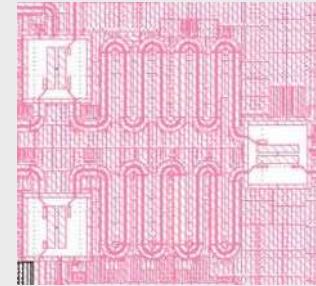
Concept



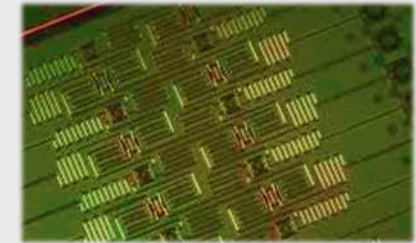
Hamiltonian



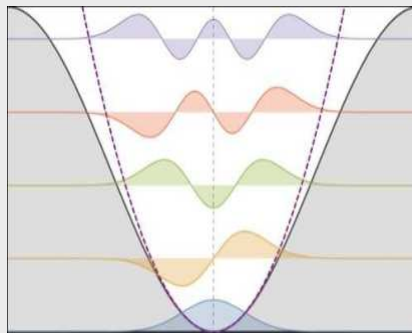
Layout



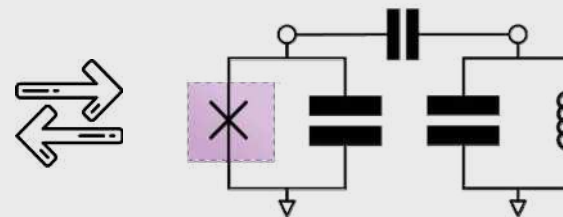
Fabrication



Quantum Analysis



Electromagnetic
Analysis



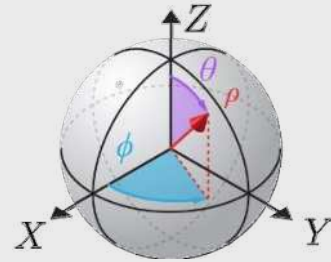
Quantum chip design flow

Complexity, Information, Accuracy
Risk, Cost, Time, Resources

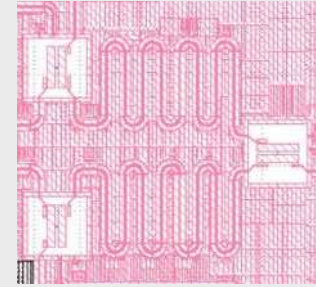
Concept



Hamiltonian



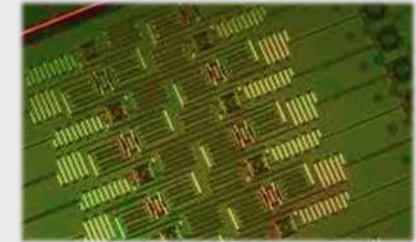
Layout



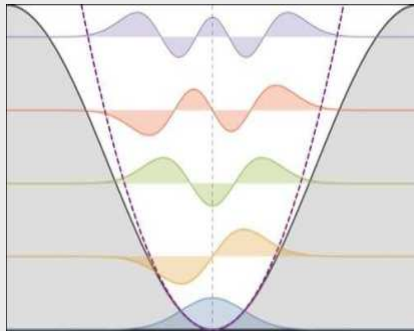
DRC
LVS



Fabrication

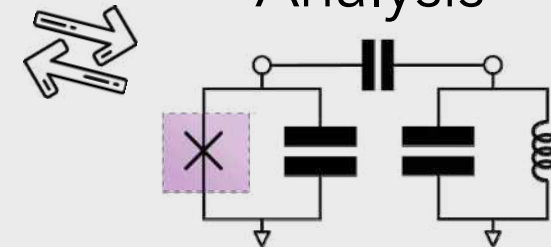


Quantum Analysis



Project
Metal

Electromagnetic
Analysis

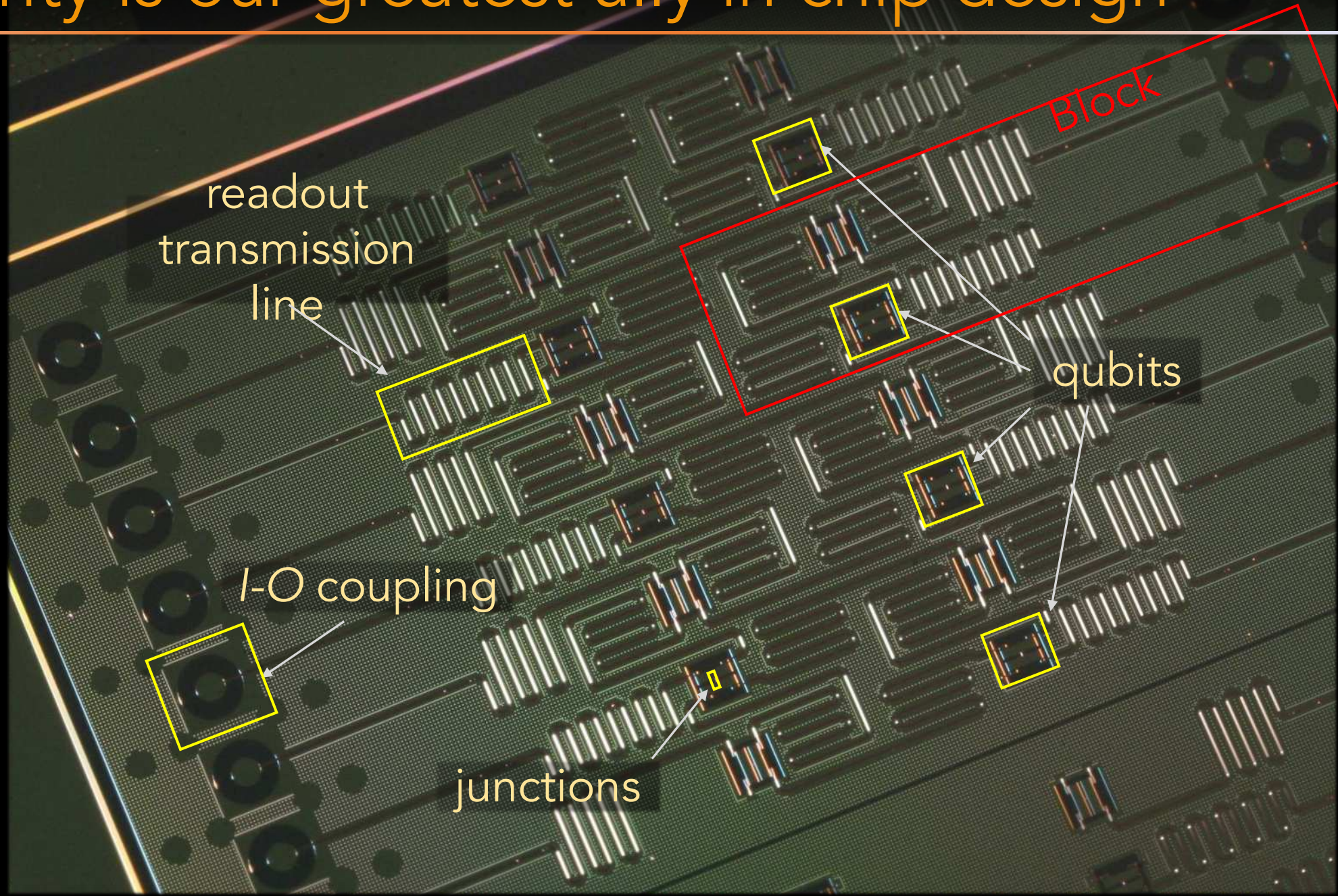


Let's layout, analyze, and
optimize a four-qubit chip...

Open source
qiskit.org/metal

Regularity is our greatest ally in chip design

Reuse
Fine-tune
Automate
Extend



Qiskit

Learning Resources Slack Support IBM Quantum Dashboard GitHub

0.0.2

Search Docs

Home
 Installing Qiskit Metal
 Frequently Asked Questions
 Roadmap
 Qiskit Metal Overview
 Question/Feedback Channel

Contributor Guide

Contributing to Qiskit Metal
 About This Site
 Reporting Bugs and Requesting
 Enhancements
 Contributing Code
 Contributing to Documentation

Docs » Qiskit Metal » Quantum Device Design & Analysis (Q-EDA) 0.0.2

Qiskit Metal | Quantum Device Design & Analysis (Q-EDA) 0.0.2

ATTENTION

This is an alpha version of Qiskit Metal, the code is still under development. Please let us know about anything you might want us to add or elaborate upon in the Slack channel #metal in the qiskit workspace.

NOTE

You can open this documentation using

```
import qiskit_metal
qiskit_metal.open_docs()
```

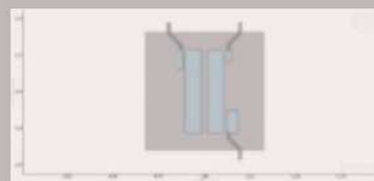
Device Library

Qiskit | quantum device design

Qubits



Single Transmon - Grounded
(xmon)



Single Transmon - Floating



Concentric Transmon



Simple CPW Meander

Resonators

Circuit Example Library

Qubits

Resonators

Composite Bi-Partite Systems

Qubit Couplers

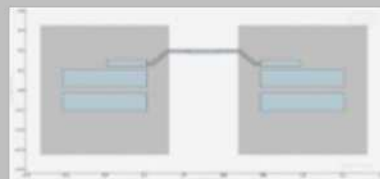
Input-Output Coupling

Small Quantum Chips

Qubit Couplers



Tunable Coupler (MIT)



Direct Coupler (transmon-
transmon)



Bus Resonator Coupler
(transmon-transmon)



CPW Launch Pad



Readout line

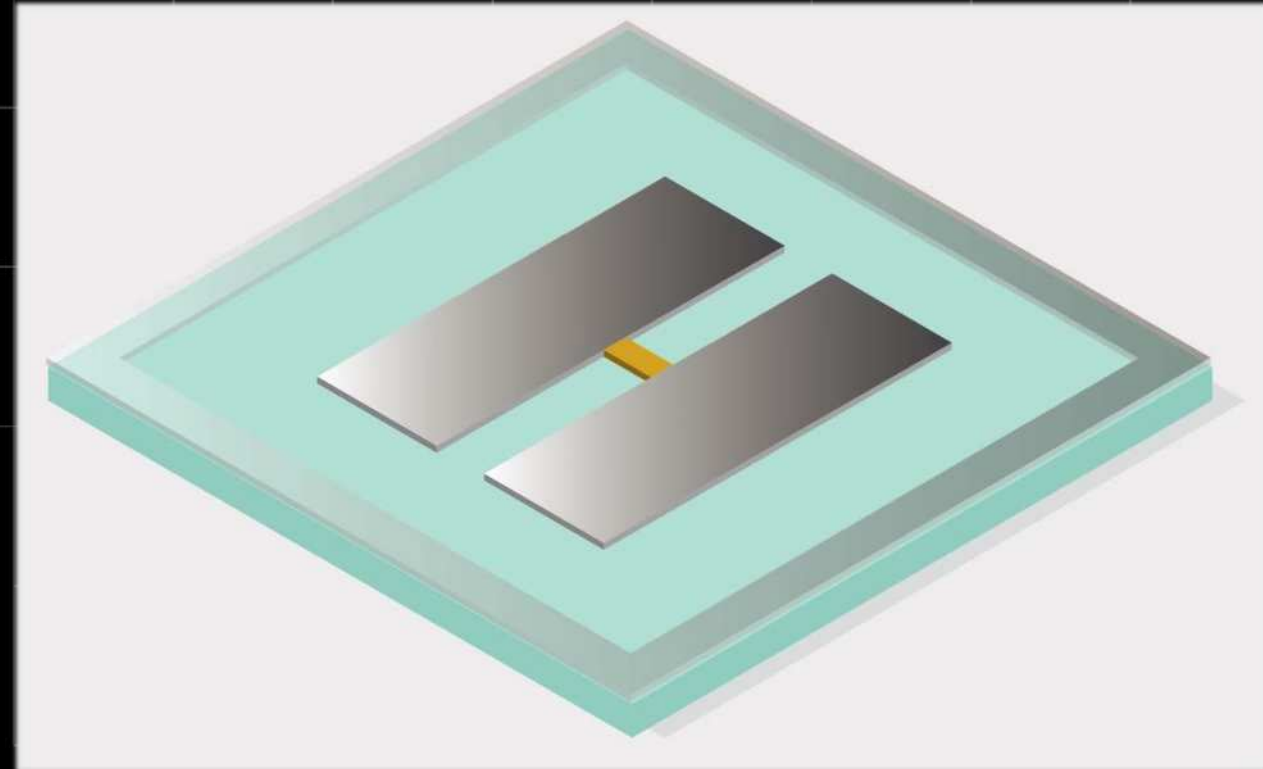
Input-Output Coupling

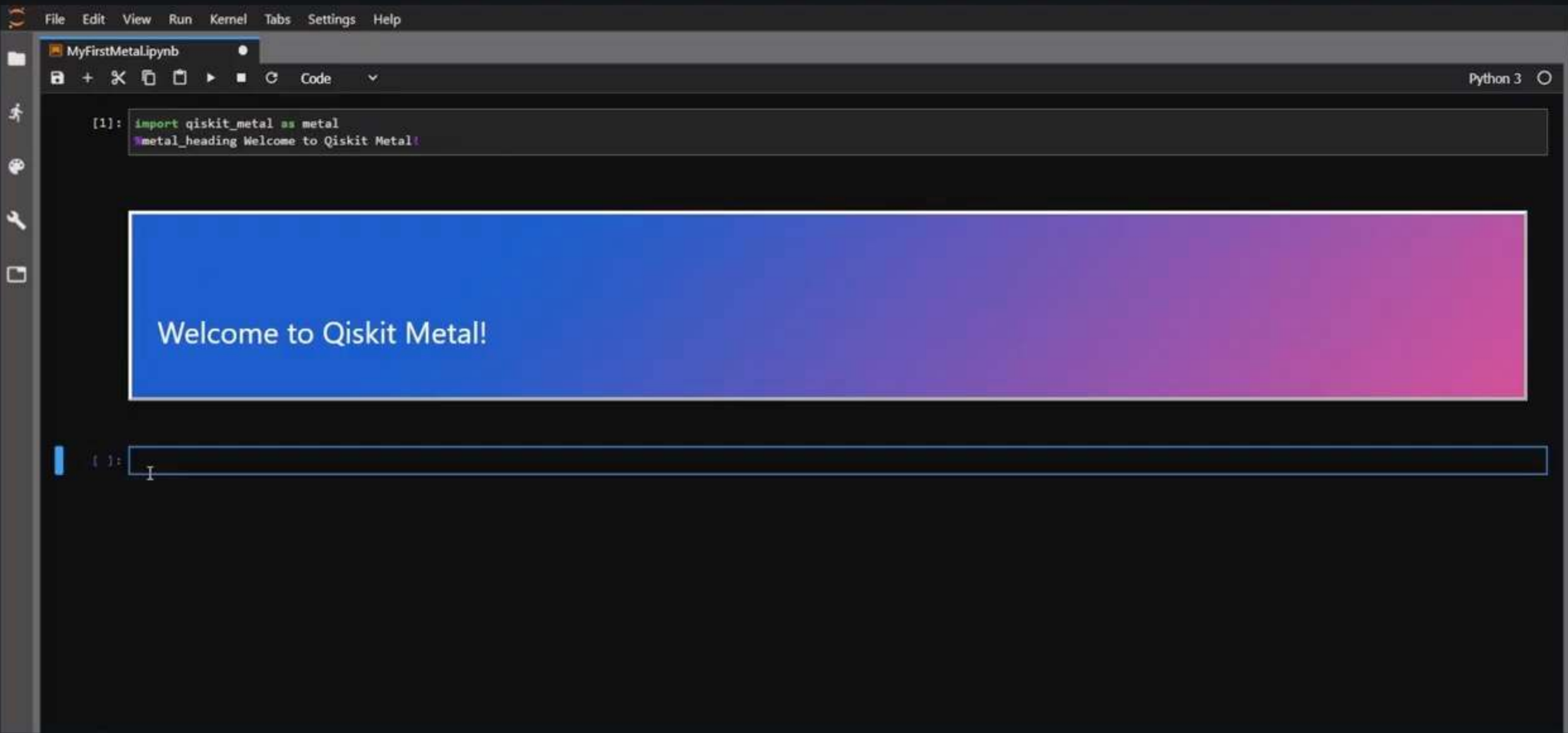
...

Create a transmon qubit

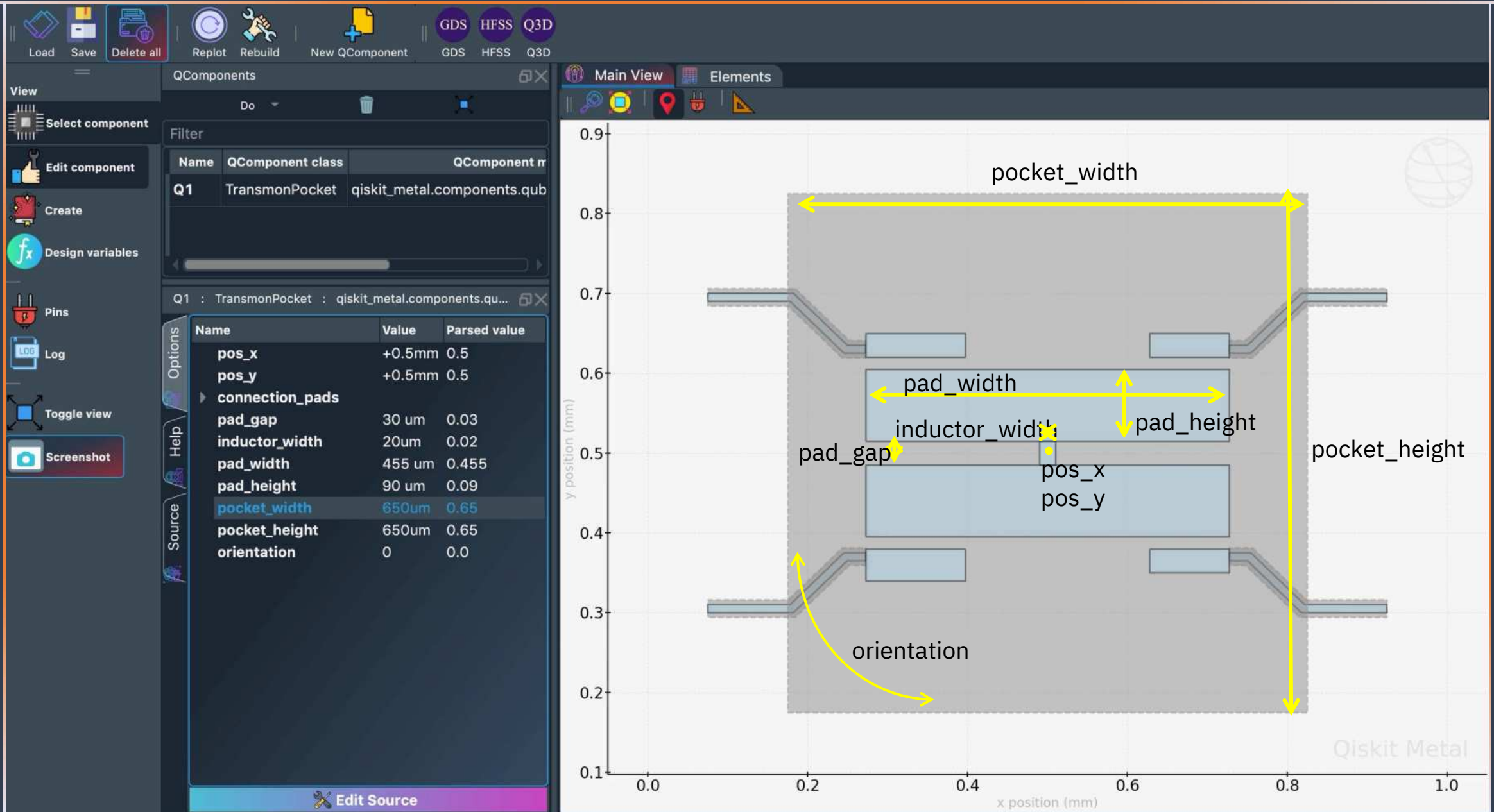
```
from qiskit_metal.qlibrary import qubits  
q1 = qubits.TransmonPocket('Q1', options=dict(...))
```

```
from qiskit_metal import MetalGUI  
MetalGUI()
```

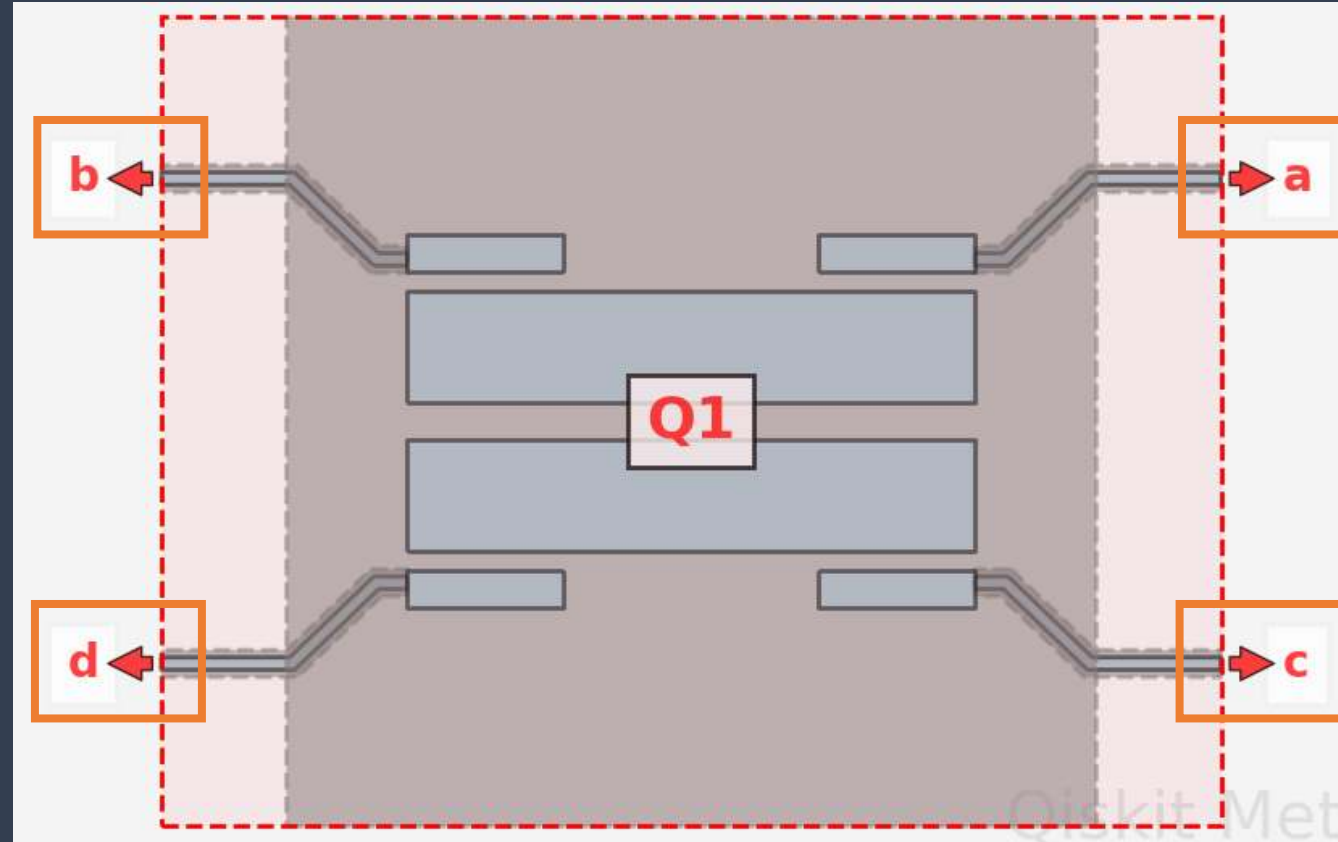


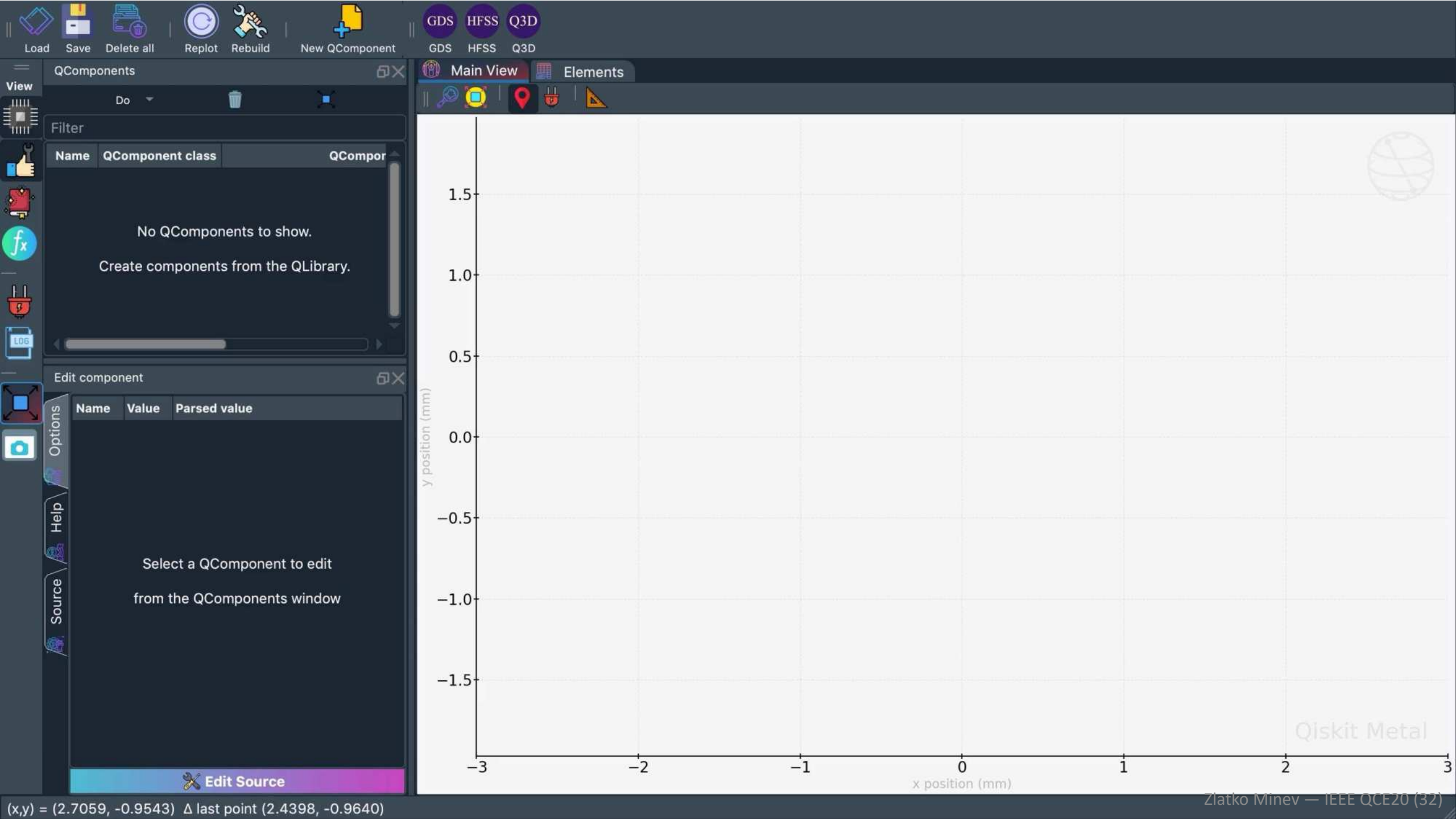


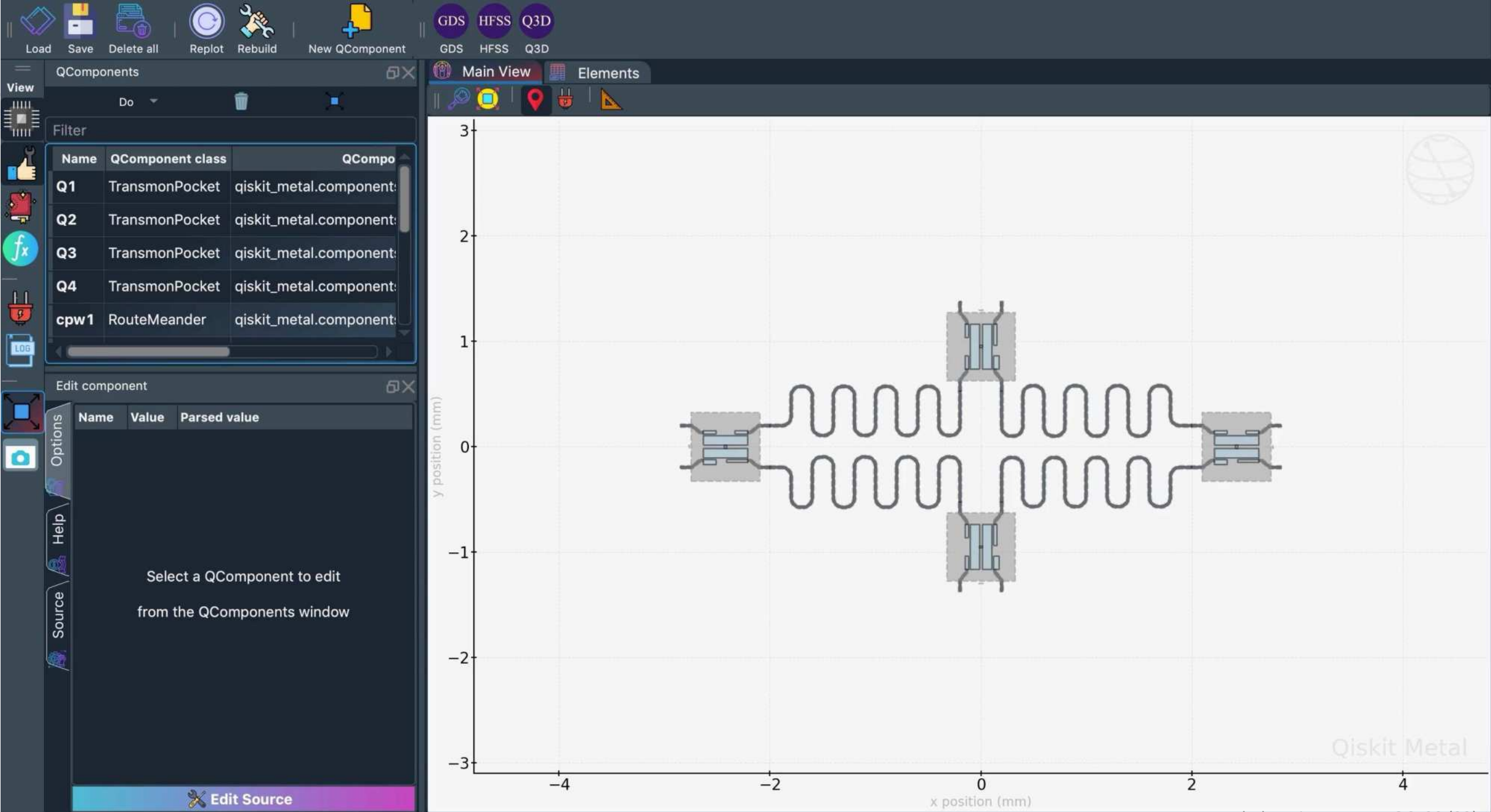
Fine-tune and automate parameters



Dynamically –connected quantum devices







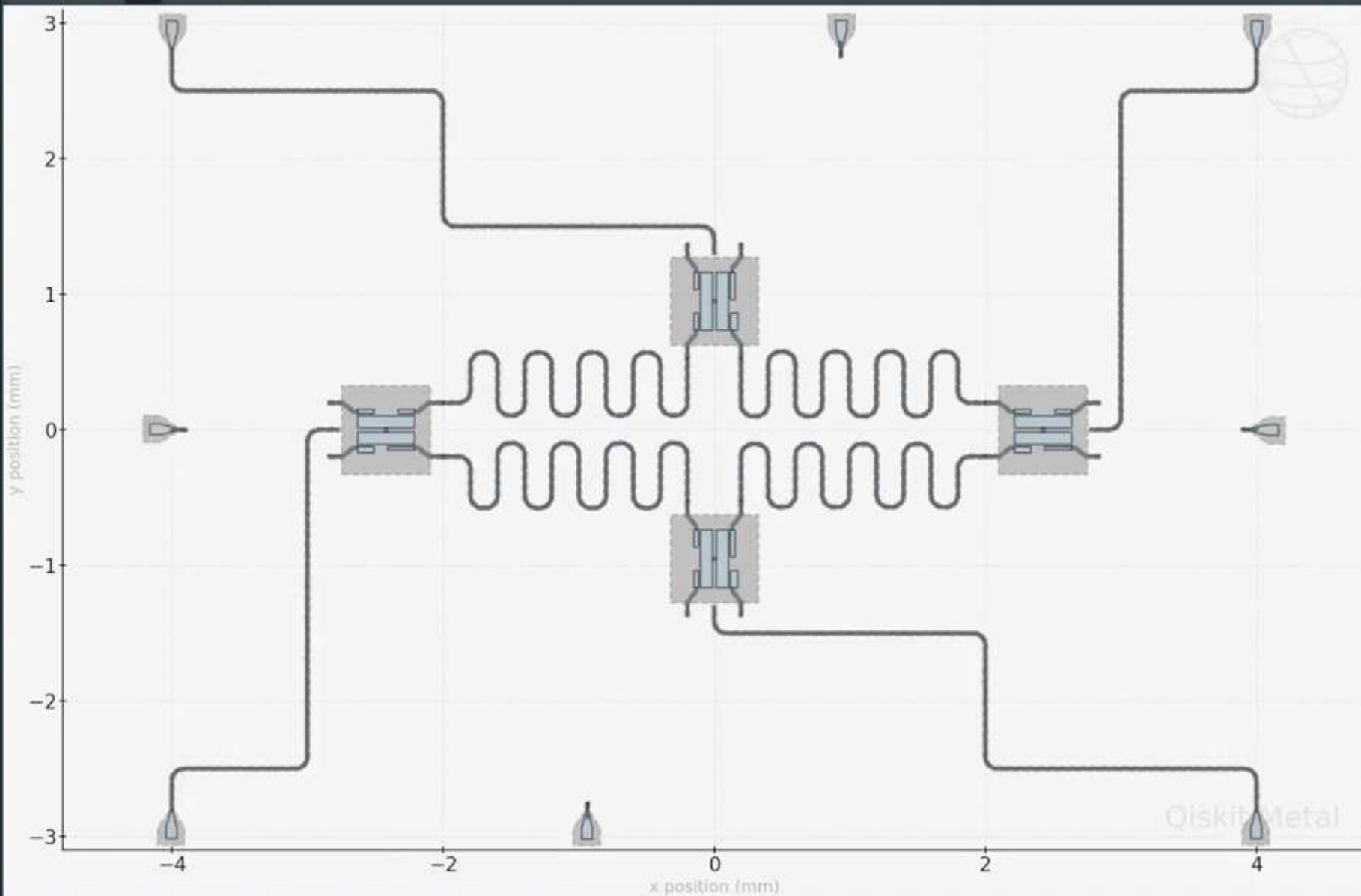
Filter

Name	QComponent class	QCompo
Q1	TransmonPocket	qiskit_metal.component:
Q2	TransmonPocket	qiskit_metal.component:
Q3	TransmonPocket	qiskit_metal.component:
Q4	TransmonPocket	qiskit_metal.component:
cpw1	RouteMeander	qiskit_metal.component:

Edit component

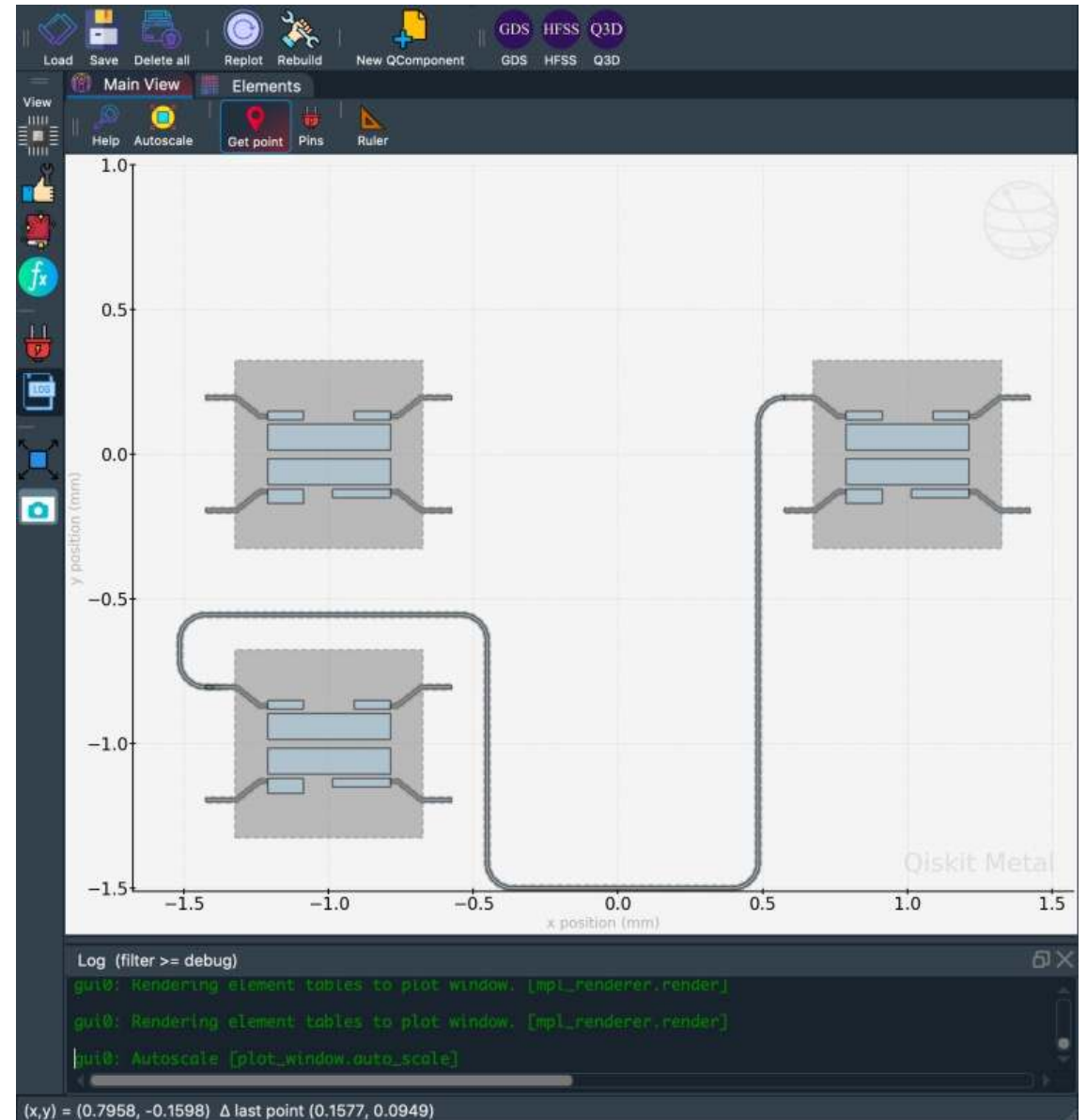
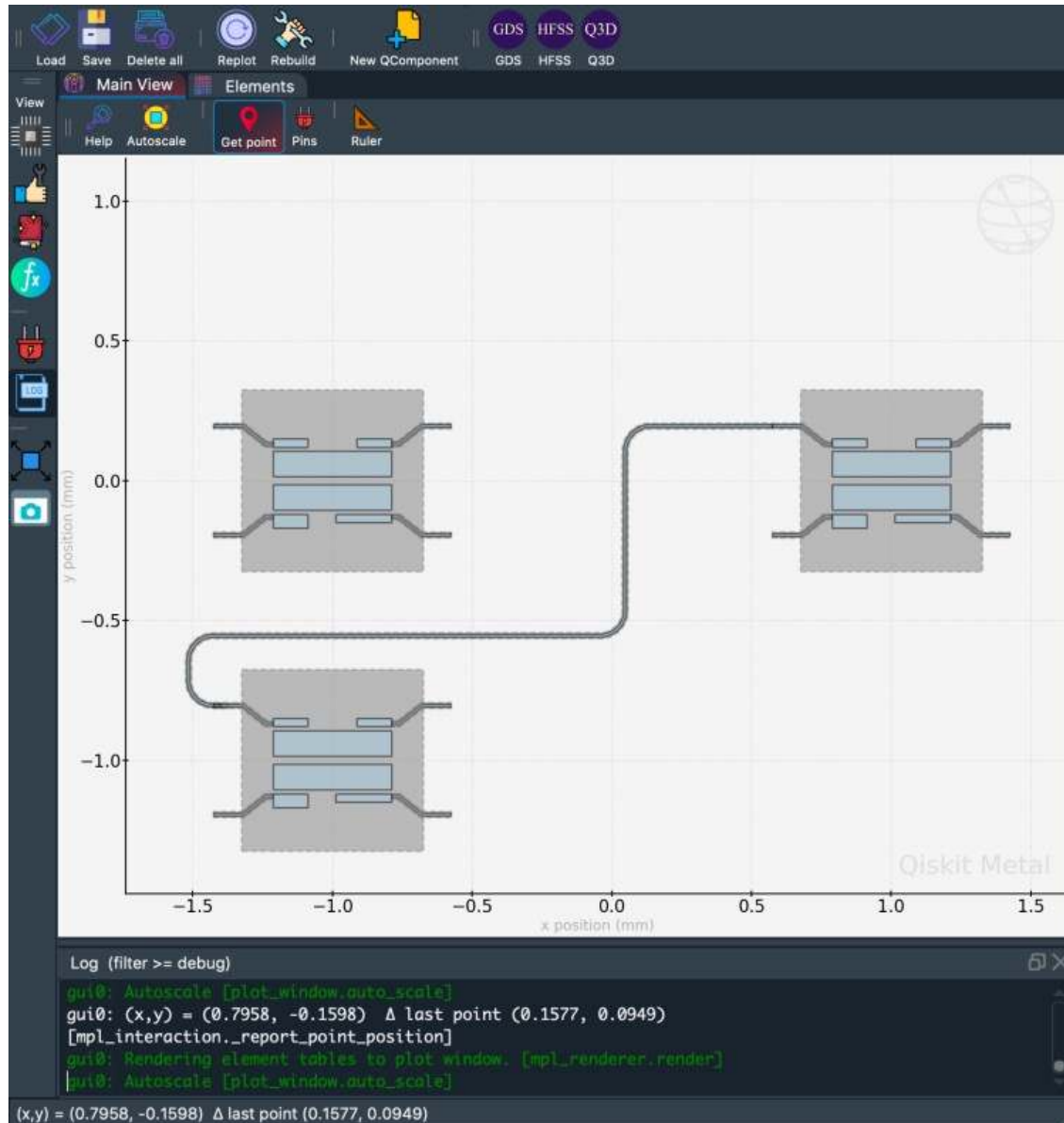
Name	Value	Parsed value
Select a QComponent to edit from the QComponents window		

Edit Source



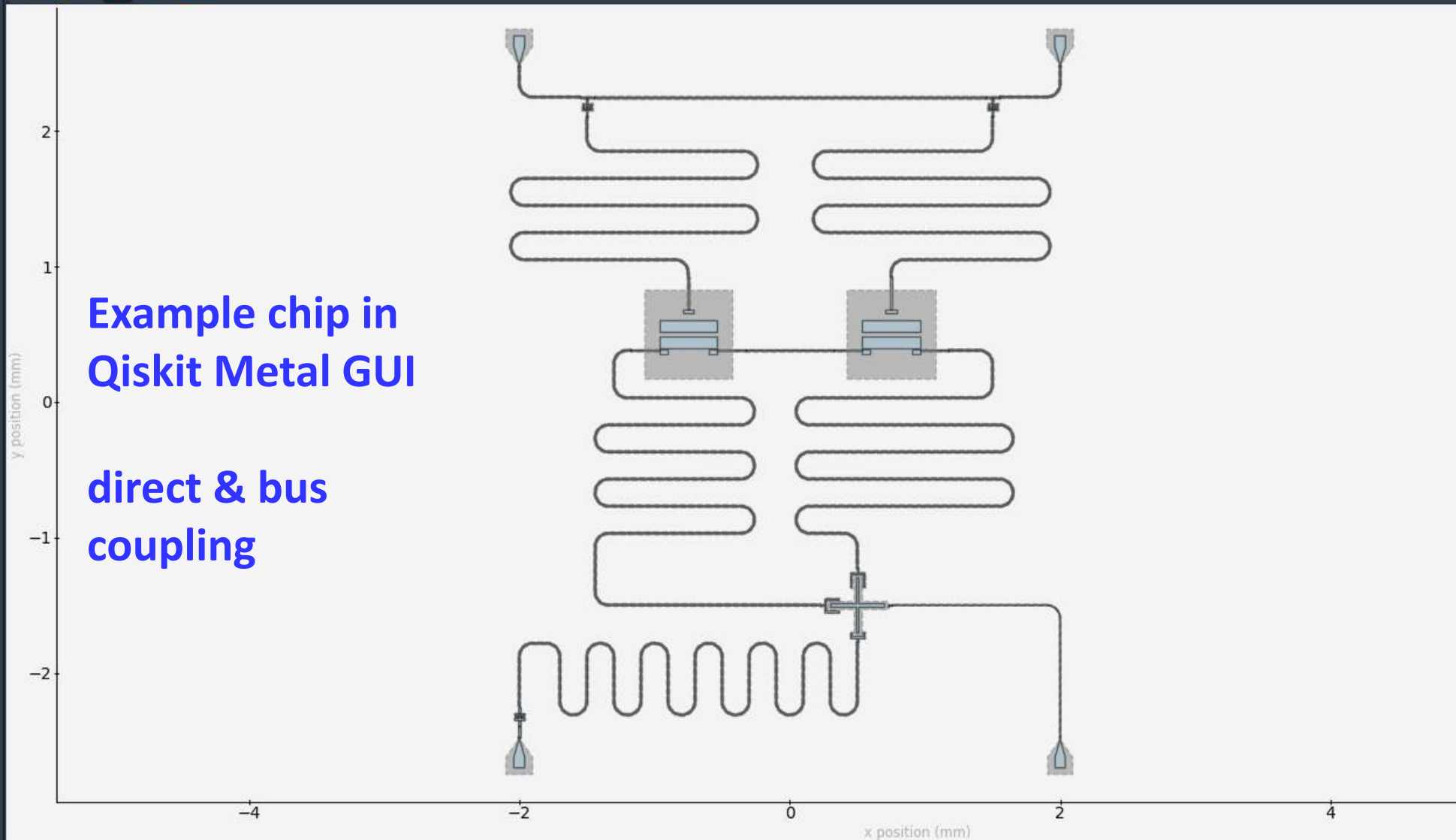
Demo Notebook for Hybrid CPW Algorithm with Anchors (Dennis Wang Talk in Session J30)

CPW Hybrid Auto and AStar.ipynb



Name	QComponent class	
Q_0	TransmonPocket6	qiskit_metal.ql
Q_1	TransmonPocket6	qiskit_metal.ql
Q_2	TransmonCrossFL	qiskit_metal.ql
Bus_01	RouteStraight	qiskit_metal.ql
Bus_02	RouteMeander	qiskit_metal.ql
Bus_12	RouteMeander	qiskit_metal.ql
Launch_Readout_Q_0	LaunchpadWirebond	qiskit_metal.ql
Launch_Readout_Q_1	LaunchpadWirebond	qiskit_metal.ql
Launch_Readout_Q_2	LaunchpadWirebond	qiskit_metal.ql
Launch_FL_Q_2	LaunchpadWirebond	qiskit_metal.ql

Name	Value	Parsed value
layer	1	1.0
trace_width	cpw_width	0.01
trace_gap	cpw_gap	0.006
lead_length	25um	0.025
pos_x	-2mm	-2
pos_y	2.5mm	2.5
orientation	-90	-90.0



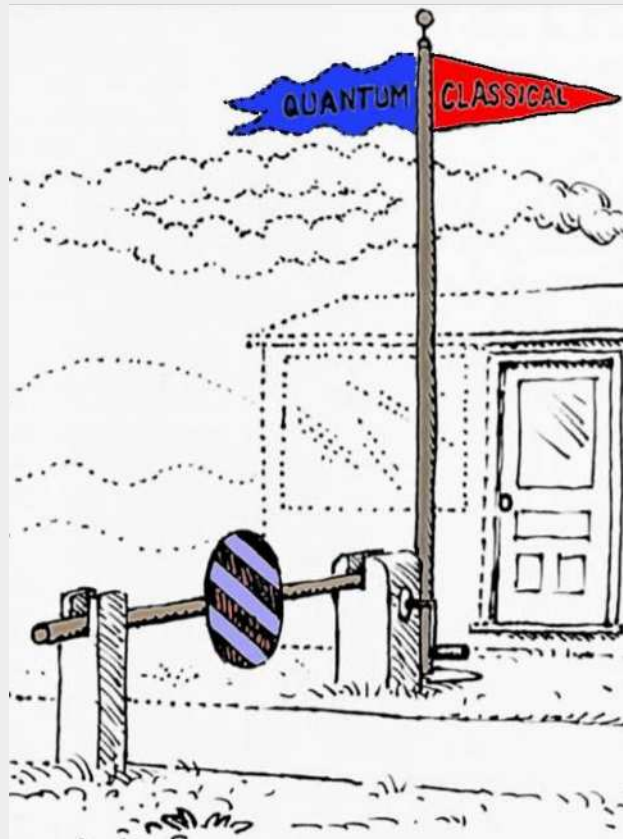
Can I make my own qubit?

```
smile = draw.shapely.geometry.Point(0, 0).buffer(0.8)
cut_sq = draw.shapely.geometry.box(-1, -0.3, 1, 1)
smile = draw.subtract(smile, cut_sq)
face = draw.subtract(face, smile)
face = draw.subtract(face, eye_r)
face = draw.subtract(face, eye_l)
face
```

[7]:



Analysis

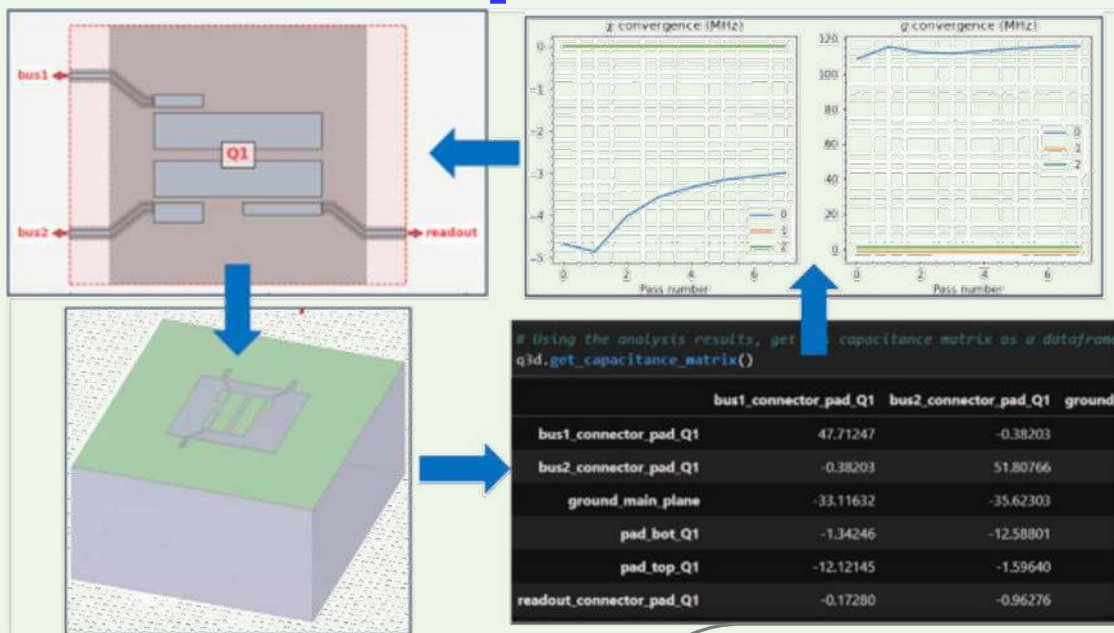


Quantum analysis library

arXiv: 2010.00620

arXiv: 1902.10355

Capacitive

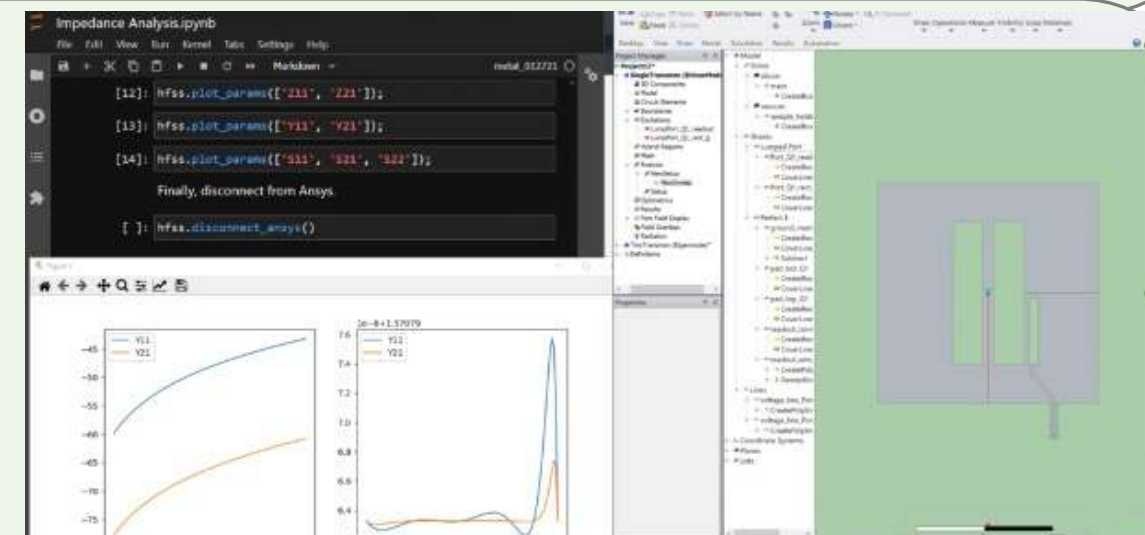
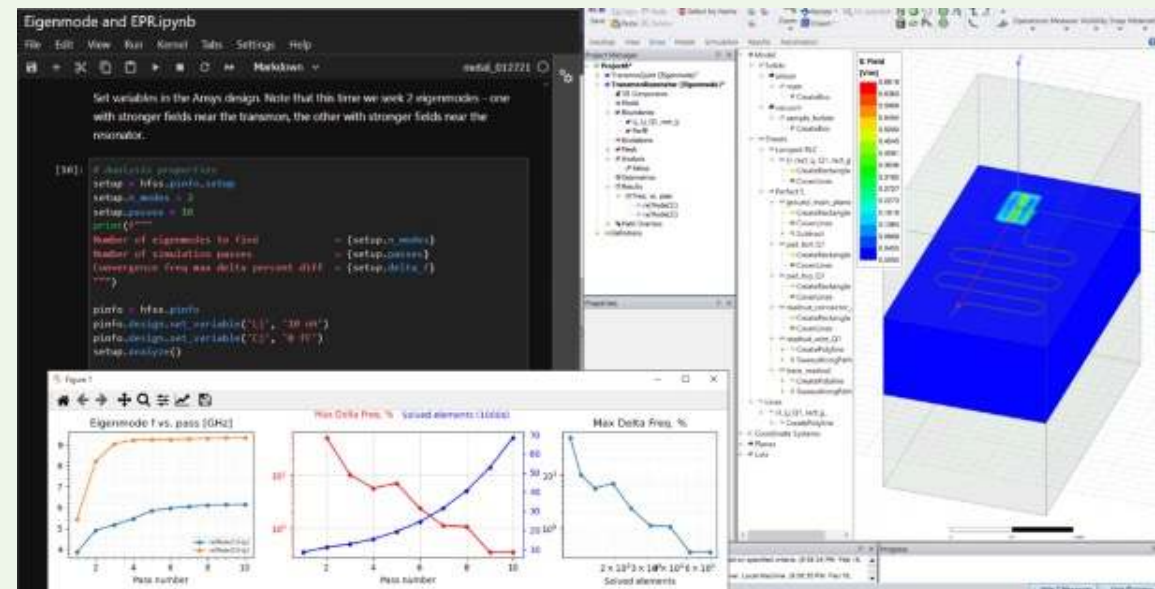


arXiv:2103.10344 ...

S, Z, Y Impedance Scattering

arXiv:1204.0587 ...

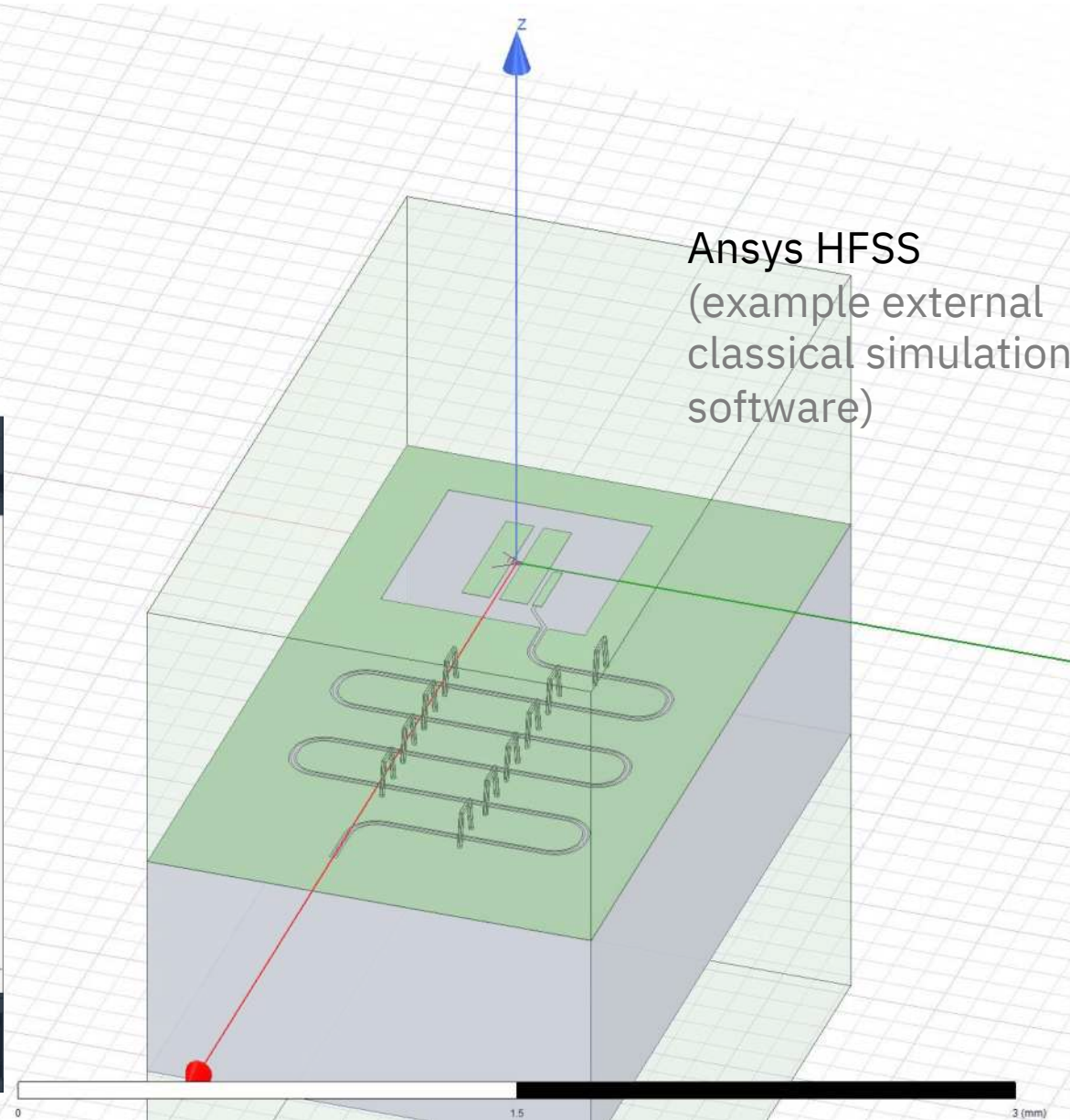
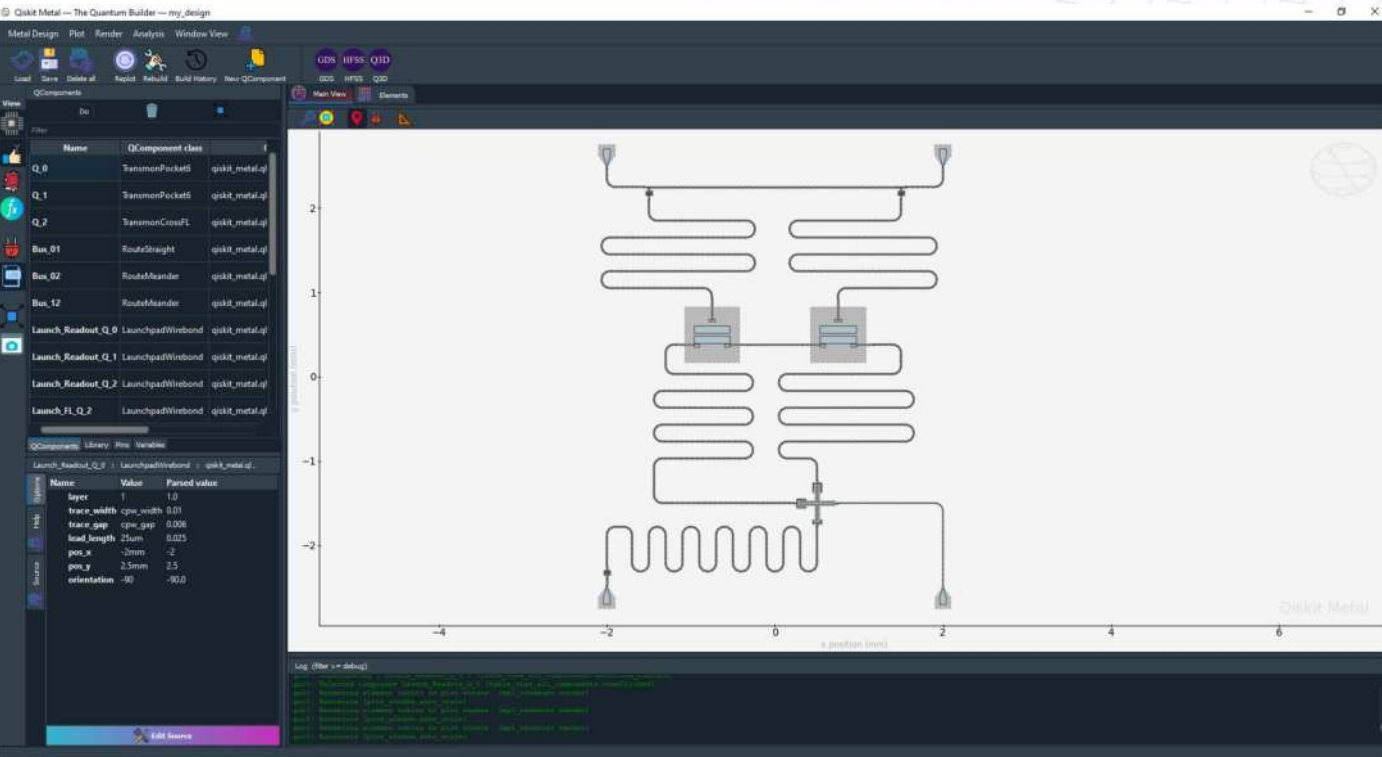
Eigenmode



Automated with
Qiskit | quantum device
design

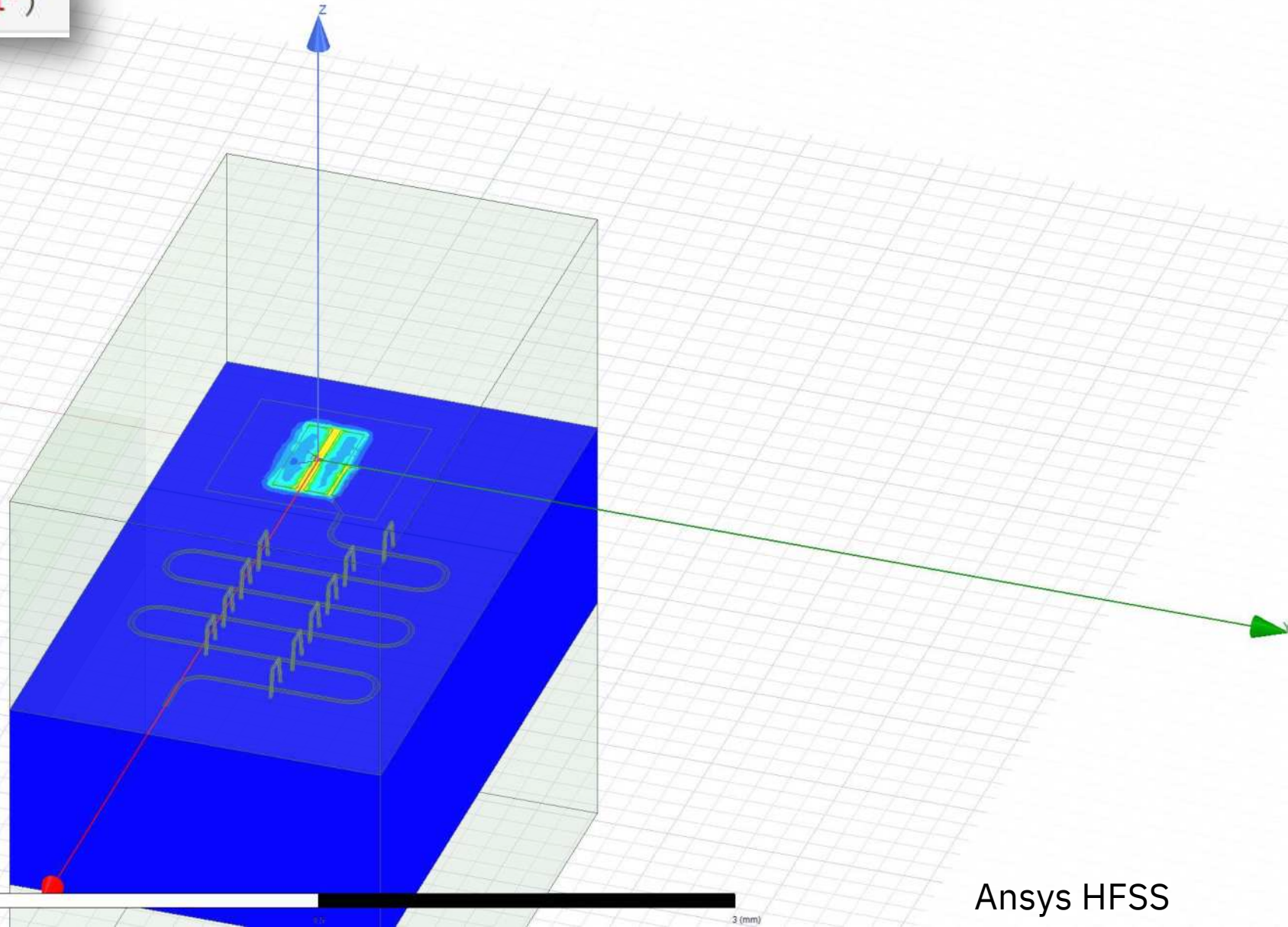
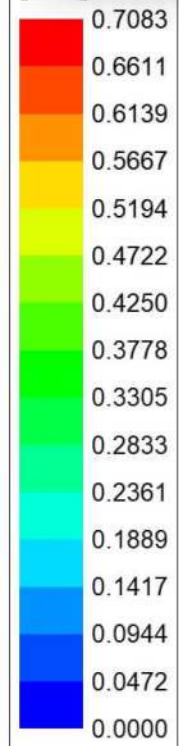


Ansys HFSS
(example external
classical simulation
software)




```
metal.analysis.lumped_model.analyze('Q1')
```

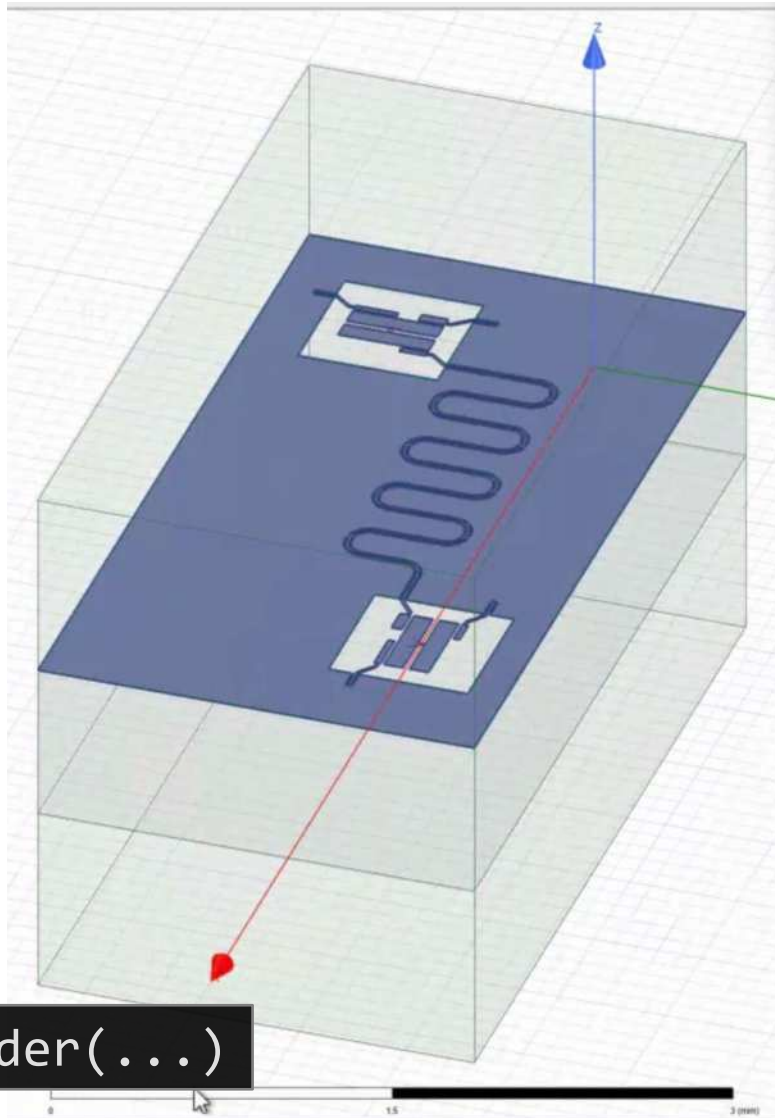
[V/m]



Ansyz HFSS

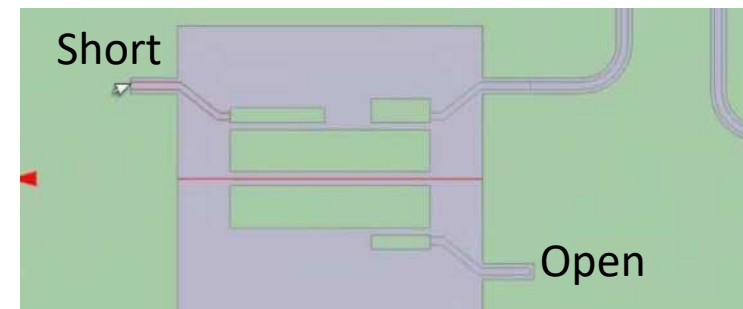
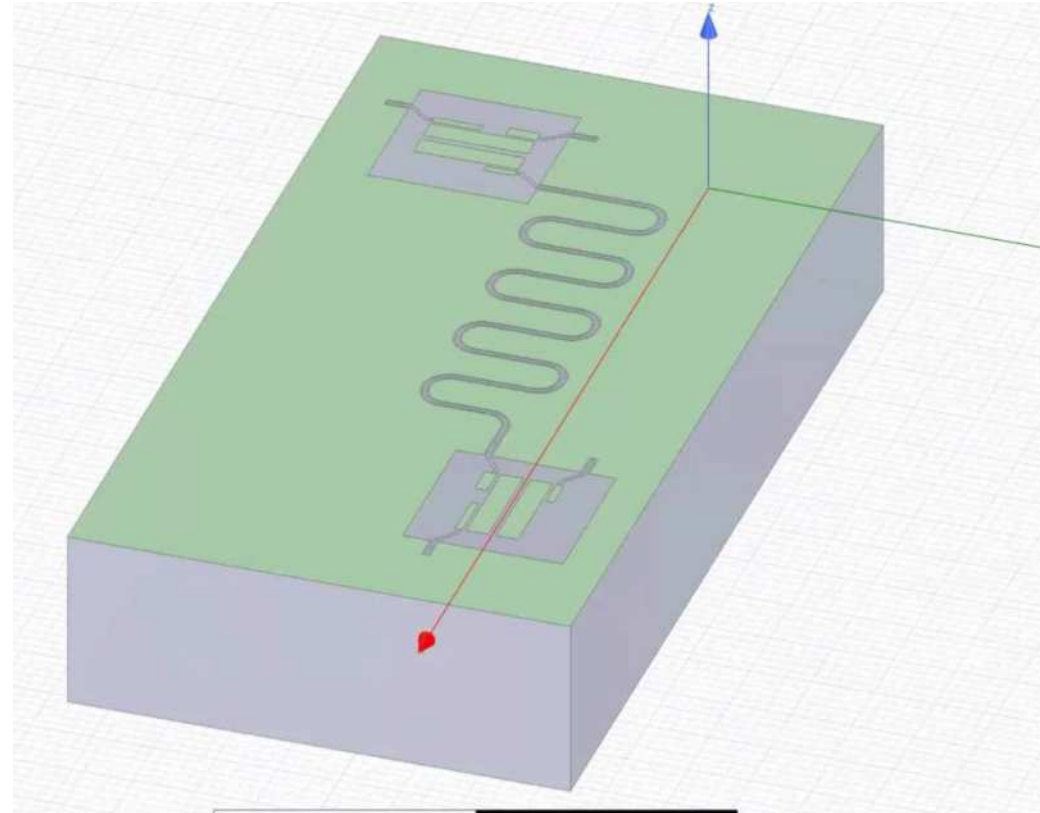
Qiskit Metal render & electromagnetic analysis

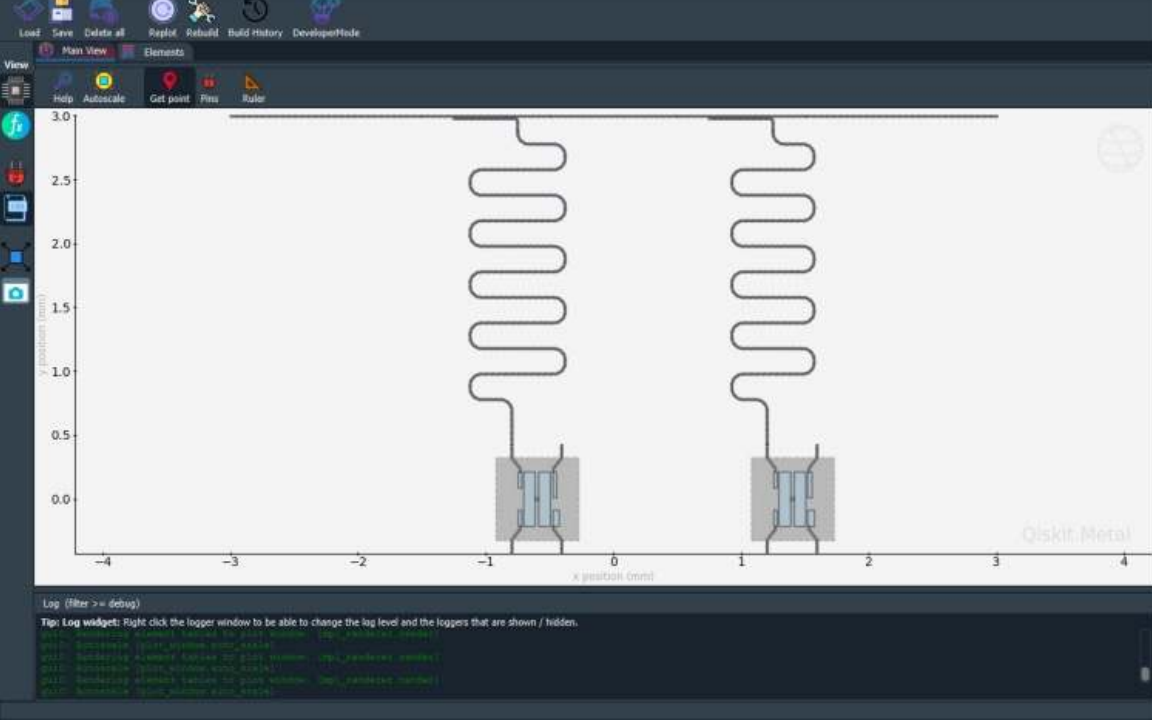
For example: Ansys HFSS



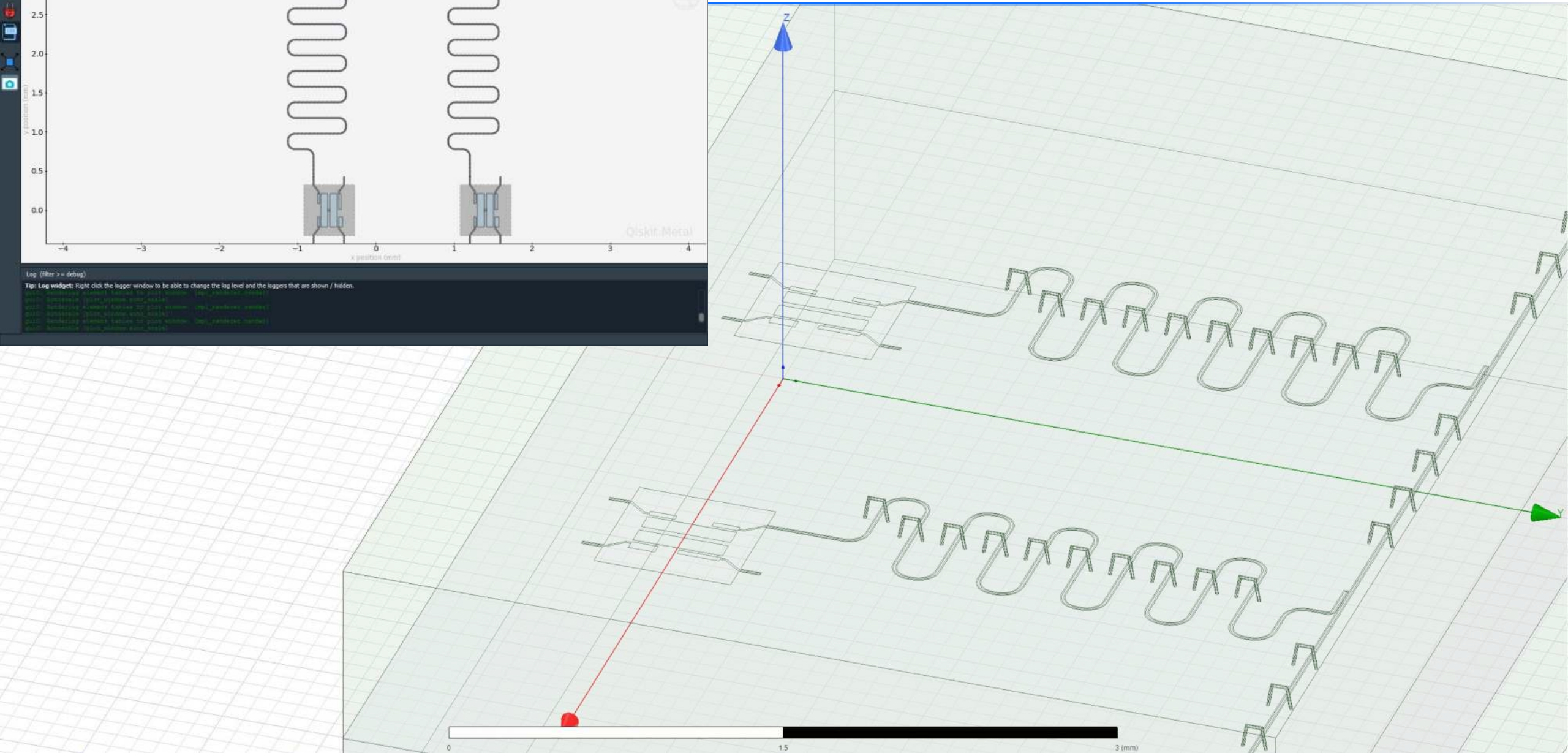
```
design.render(...)
```

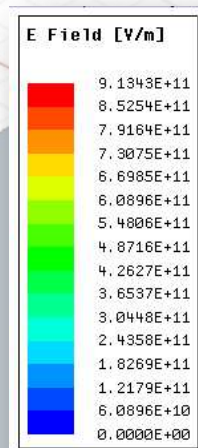
Q3D





More complex designs





Best practices

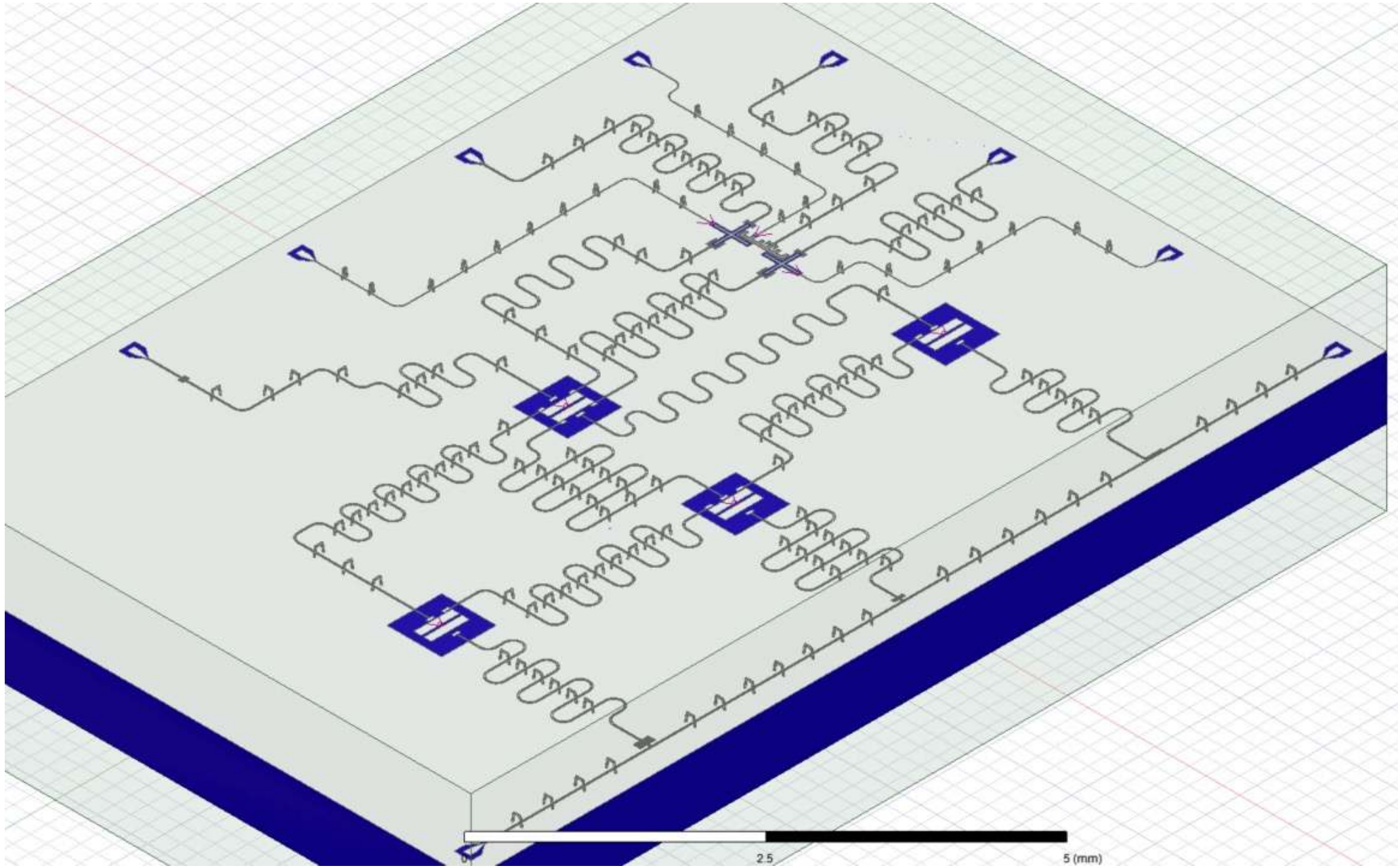
Native optimized curves

Smaller mesh RAM

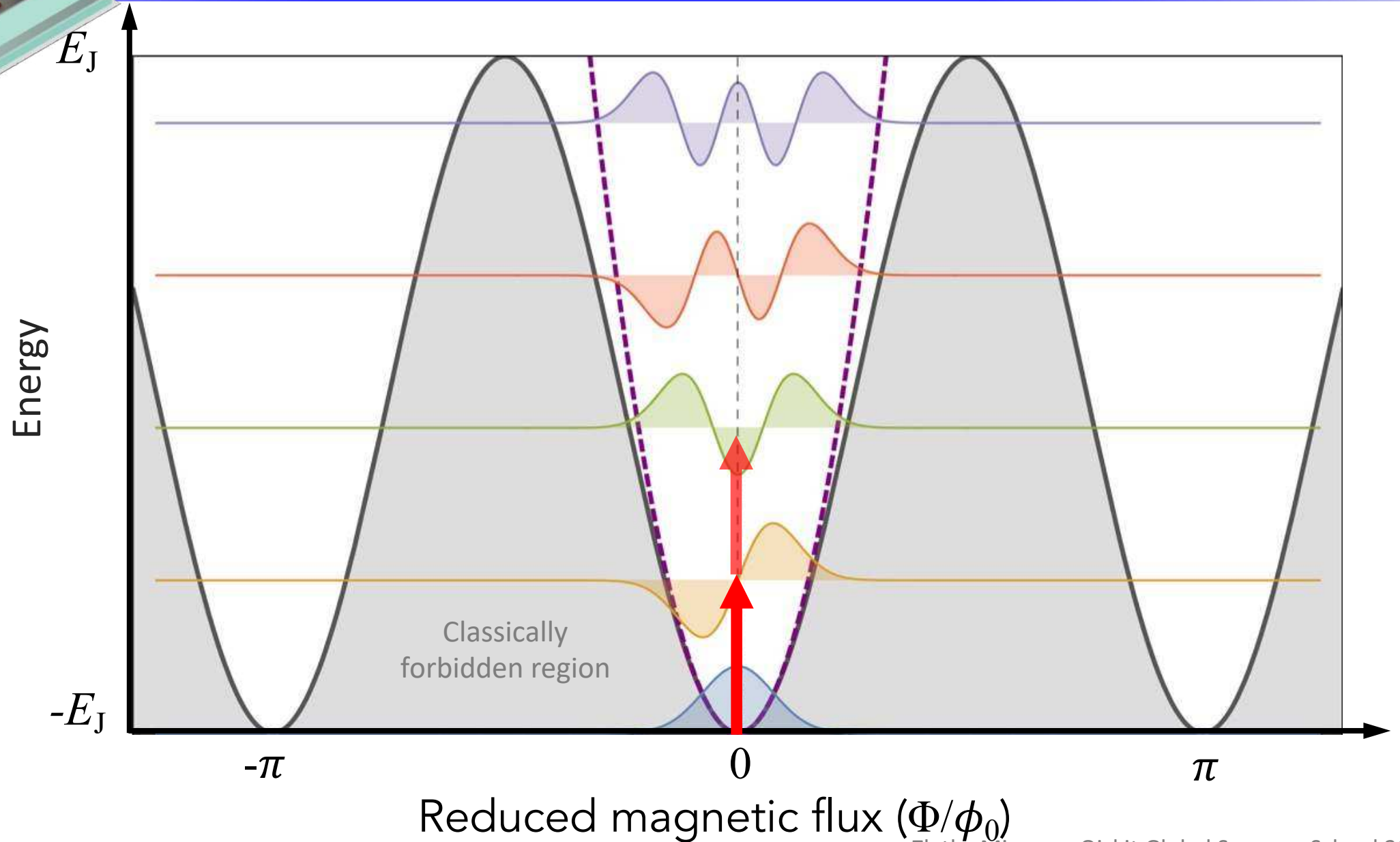
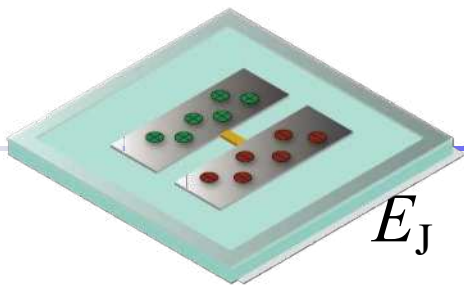
0 0.35 0.7 (mm)

0 1 2 (mm)

More complex quantum chips – see next tutorial

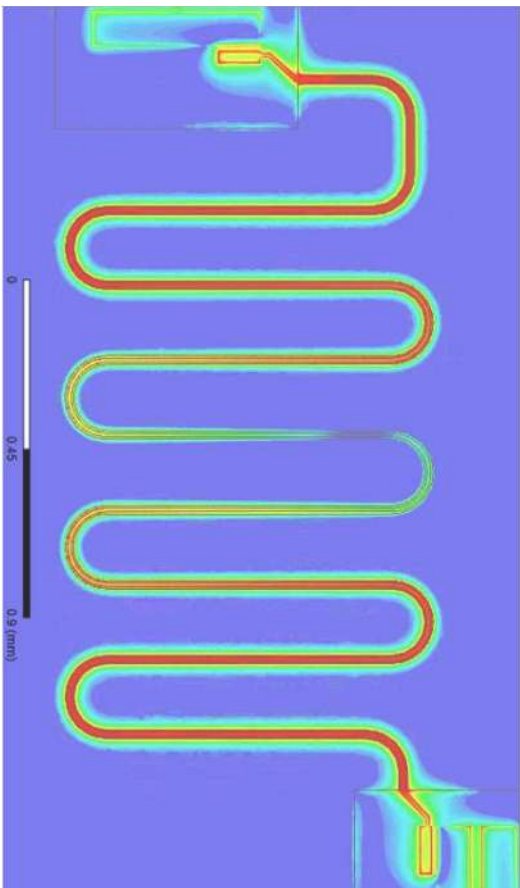


Exploring a transmon qubit

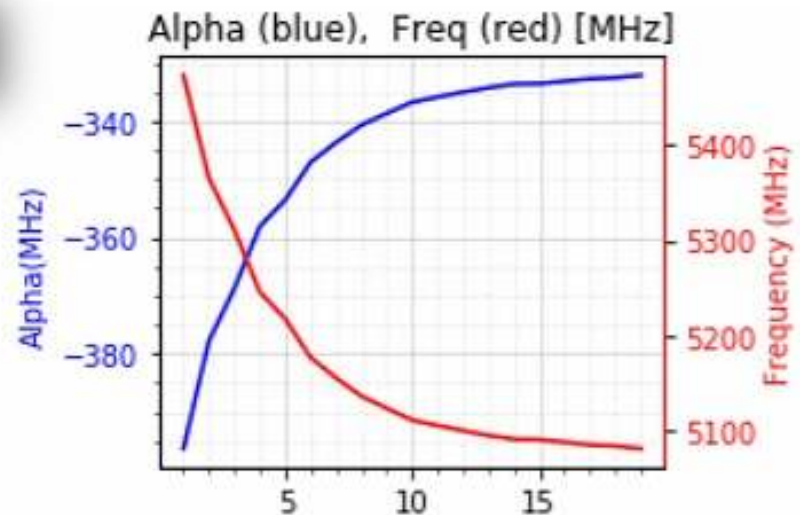


Automated analysis and reports

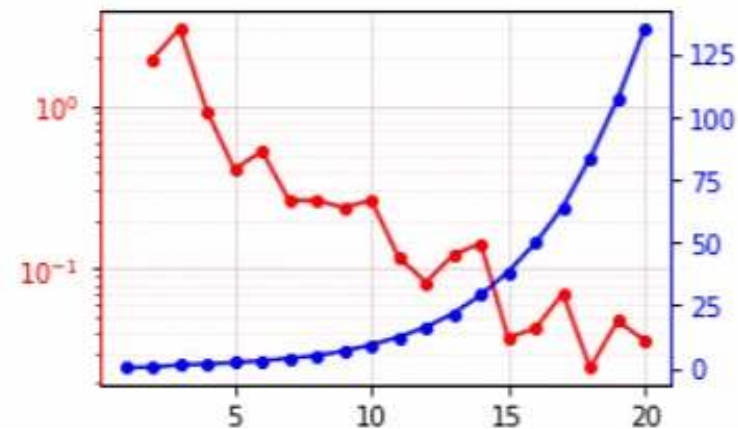
```
metal.analysis.lumped_model.analyze('Q1')
```



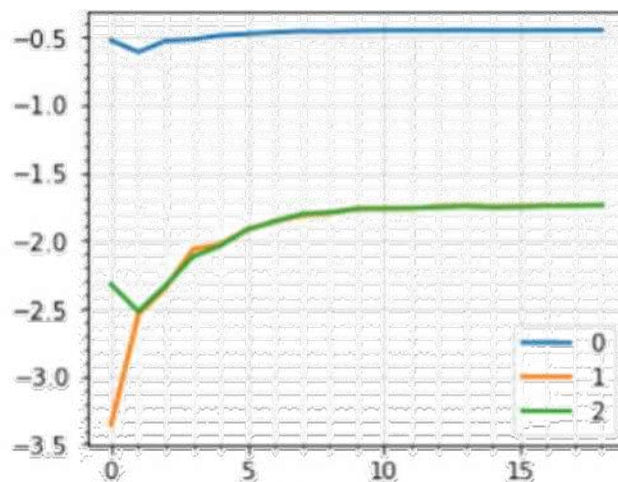
Qubit frequency & anharmonicity



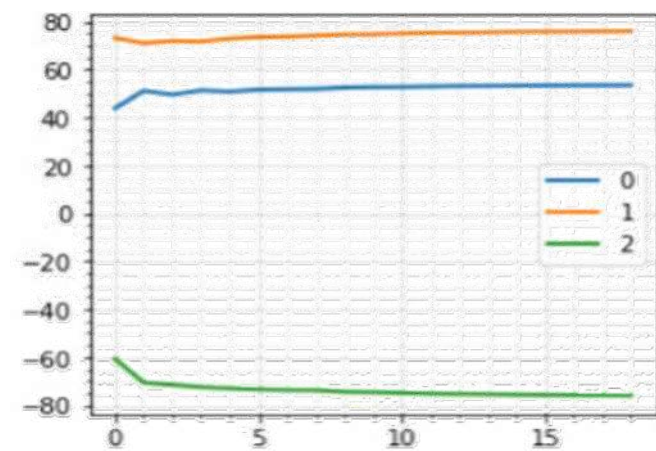
FE simulation convergence



cross-Kerr χ coupling (MHz)



Linear g coupling (MHz)



Analysis pass number

QComponents

Filter

Name	QComponent class	QComponent module	Build status	ID
Q1	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	1
Q2	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	2
Q3	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	3
Q4	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	4
cpw1	RouteMeander	qiskit_metal.components.interconnects.meandered	good	5
cpw2	RouteMeander	qiskit_metal.components.interconnects.meandered	good	6
cpw3	RouteMeander	qiskit_metal.components.interconnects.meandered	good	7
cpw4	RouteMeander	qiskit_metal.components.interconnects.meandered	good	8
OT01	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	9
OT02	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	10
OT03	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	11

Variables

Variable name	Value	Parsed value (in mm)
cpw_width	10 μ m	0.01
cpw_gap	6 μ m	0.006

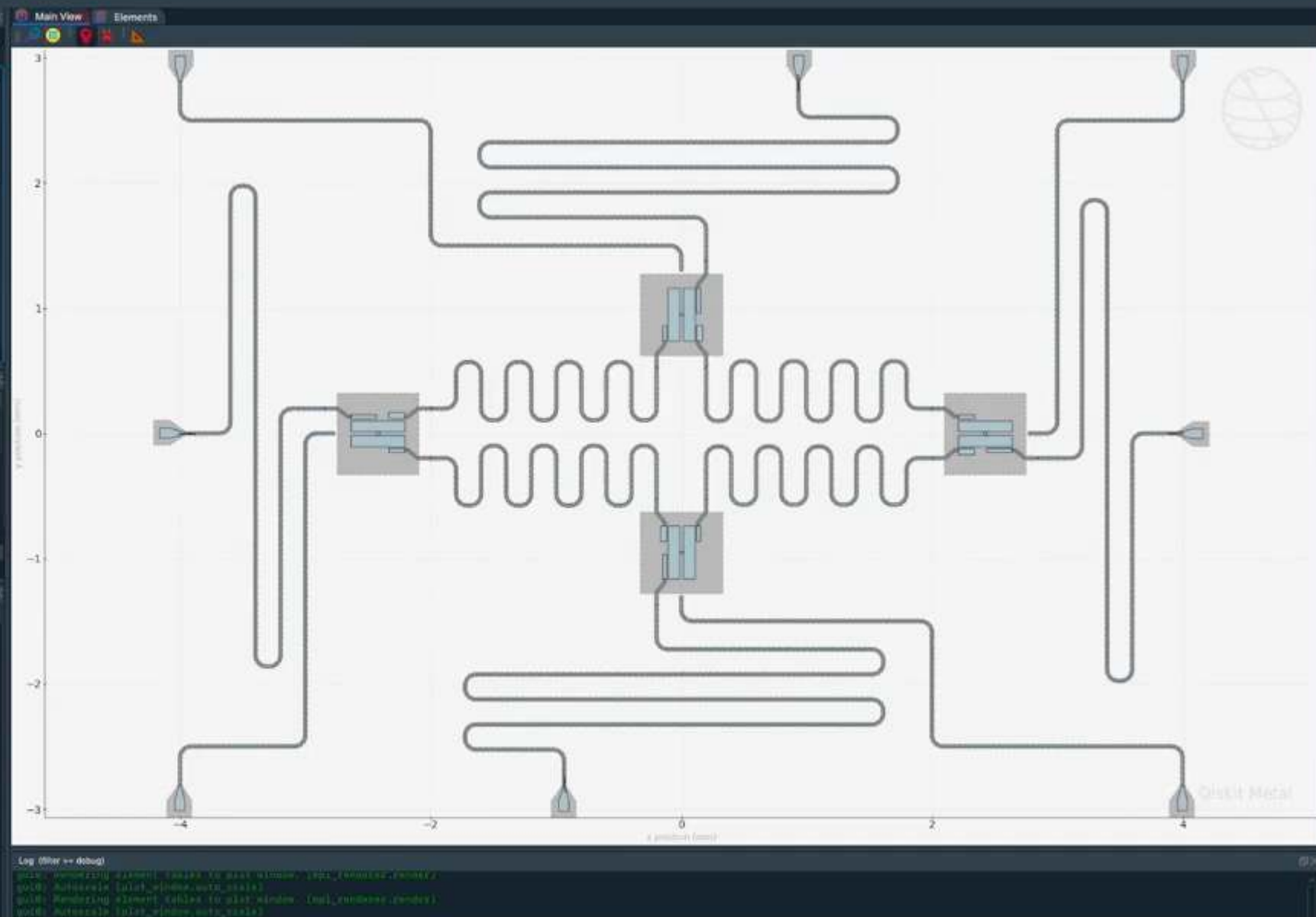
Add variable Delete variable

Library Pins Variables

Edit component

Name	Value	Parsed value
------	-------	--------------

Select a QComponent to edit
from the QComponents window



Use cases Devices made with Qiskit Metal

Korea Hackathon – Metal Projects

Project 1A: Metal - Designing a simple qubit chip

Abstract

The designing of a simple two qubit system using Qiskit Metal.

Description

Using Qiskit Metal, design a single plane, two-qubit chip using superconducting qubits, which are coupled together with a coplanar waveguide (CPW) bus resonator (capacitively), and with each qubit having a CPW readout resonator. The readout resonators should be capacitively coupled to transmission line(s) which connect to launcher pads for wire bonds. All components should be selected from those available in the library.

A simple model of the potential layout is shown below,



The project members are to determine the actual physical layout, with the below target parameters in mind.

Parameters	Target Value	Parameters	Target Value
Freq_Q1	5.5 GHz	Freq_readout_Q1	7.5 GHz
Freq_Q2	5.7 GHz	Freq_readout_Q2	7.7 GHz
Anharmonicity (Ec)	300 MHz	χ	400 kHz
		κ	2 MHz
Freq_bus	6.6 GHz		
g			

The project members can also decide on the chip size, though should consider issues such as, potential box/substrate modes, or cross talk between the different components on the chip. It can be assumed wirebonds are available to be added where desired.

Members

Korea Hackathon – Metal Projects

Project 1B: Metal - Designing a simple qubit chip

Abstract

The designing of a simple two qubit system using Qiskit Metal.

Description

Using Qiskit Metal, design a single plane, two-qubit chip using superconducting qubits, which are coupled together with a coplanar waveguide (CPW) bus resonator, and with each qubit having a CPW readout resonator. The readout resonators should be coupled to transmission line(s) which connect to launcher pads for wire bonds. The qubits should be newly written components, as should the capacitive couplers. Only the CPW transmission line components can be from the available library.

The project members are to determine the actual physical layout, with the below target parameters in mind.

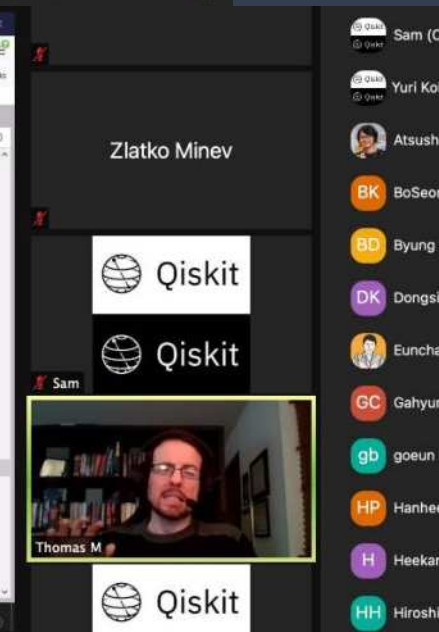
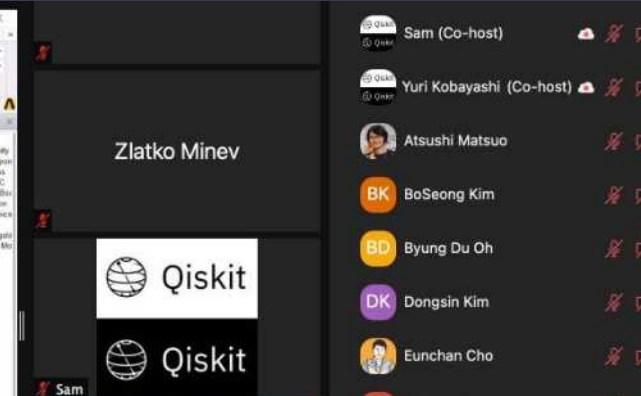
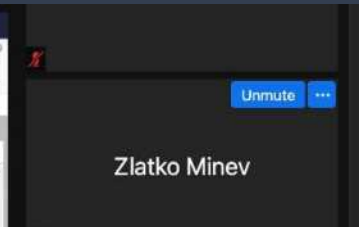
Parameters	Target Value	Parameters	Target Value
Freq_Q1	5.5 GHz	Freq_readout_Q1	7.5 GHz
Freq_Q2	5.7 GHz	Freq_readout_Q2	7.7 GHz
Anharmonicity (Ec)	300 MHz	χ	400 kHz
		κ	2 MHz
Freq_bus	6.6 GHz		
g			

The project members can also decide on the chip size, though should consider issues such as, potential box/substrate modes, or cross talk between the different components on the chip. It can be assumed wirebonds are available to be added where desired.

Members

Deliverables

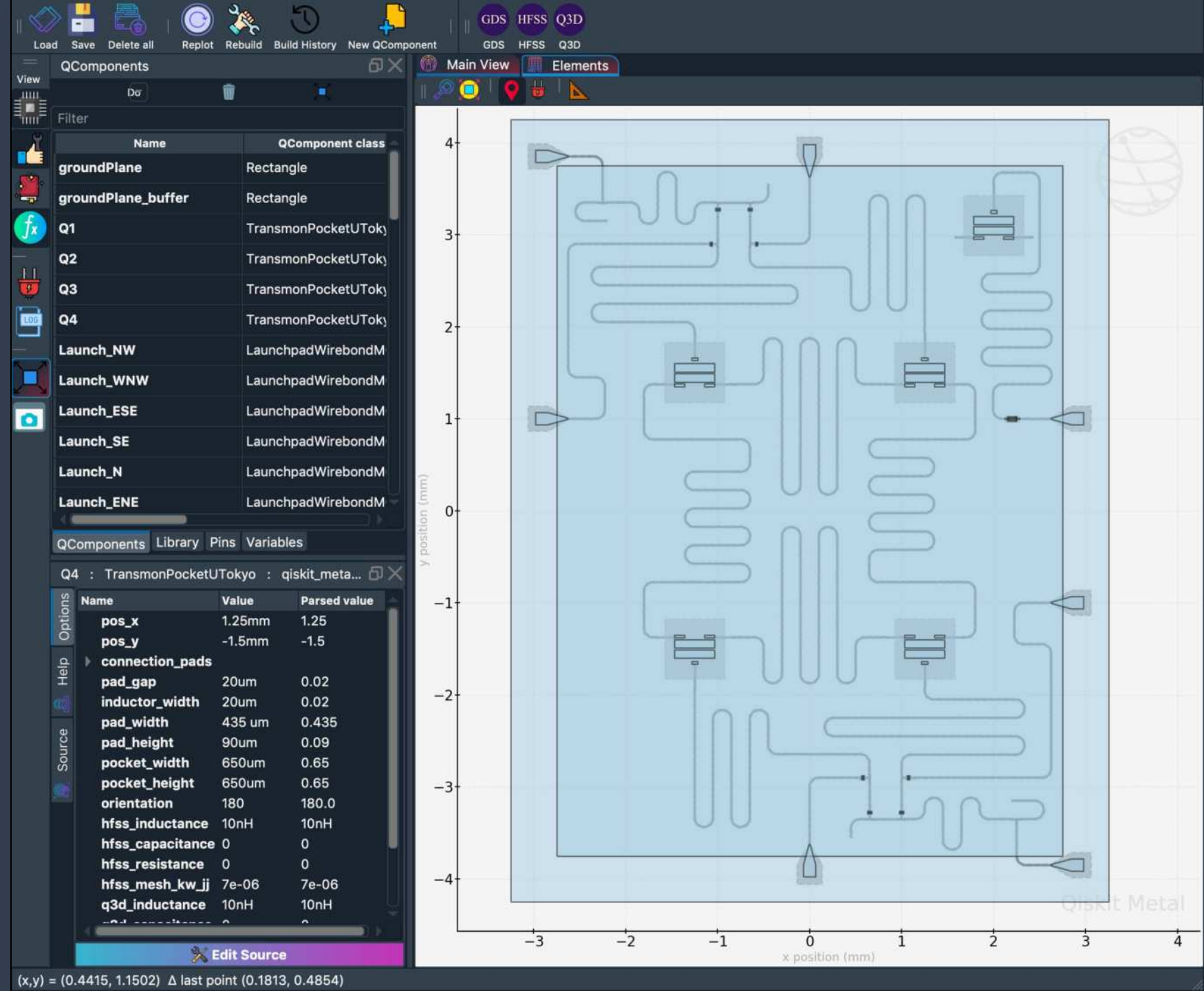
- The design notebook.
- Documentation showing the successful simulation/analysis of the chip design meeting the desired parameters.



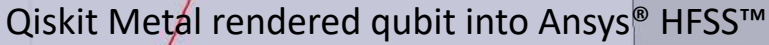
BIG BIG THANKS TO Yuri, Samanvay,
Hanhee, Junye & team!!
Thomas, Dennis, Metal coaches, ...

Devices made with Metal

IBM 5Q Tsuru U Tokyo

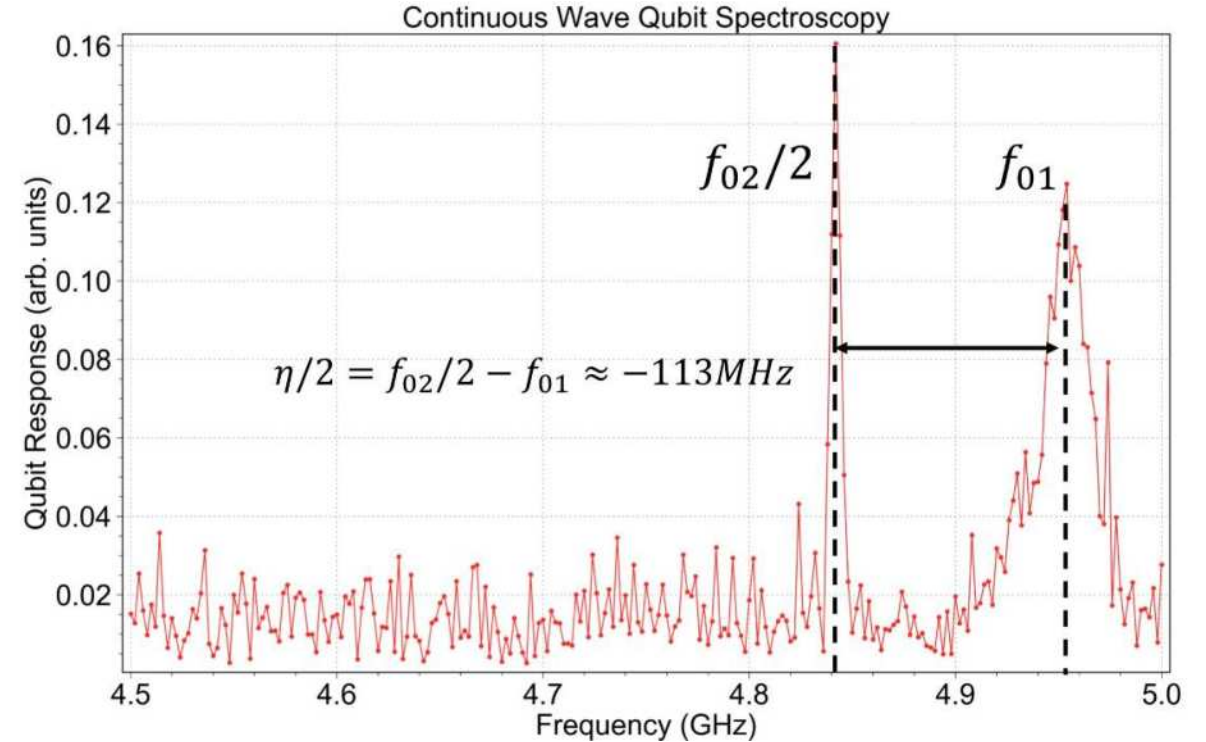
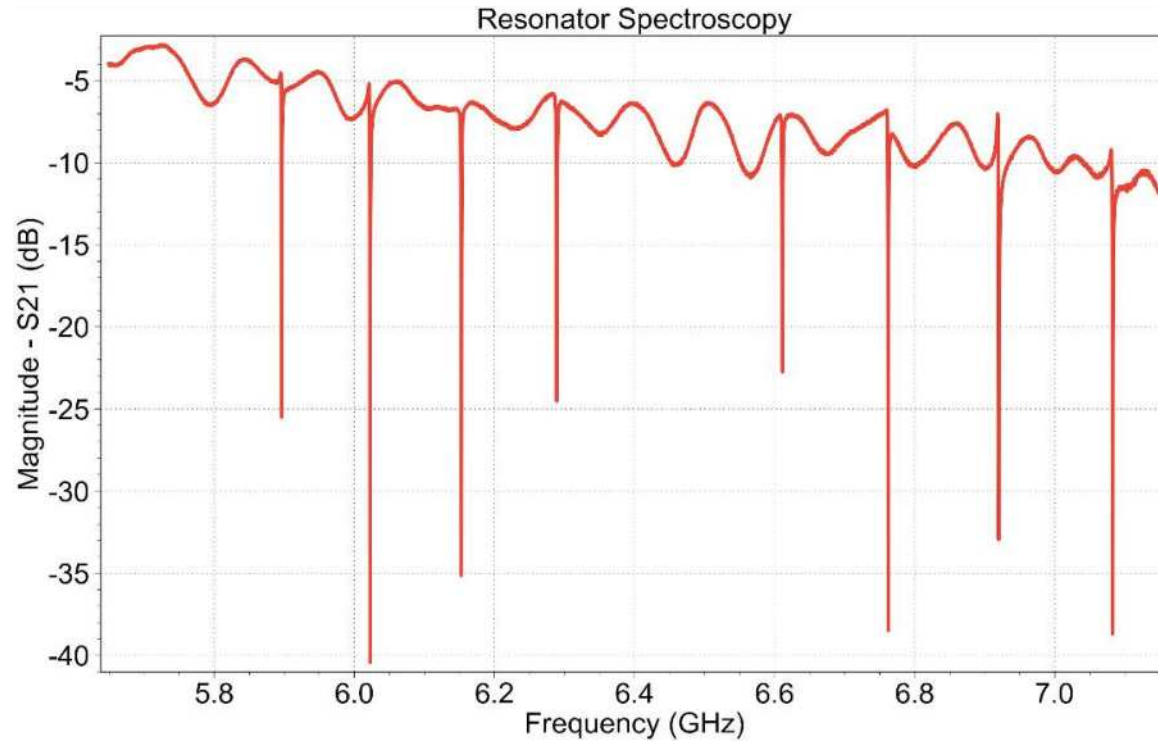


The screenshot displays the QCAD application interface for PCB layout design. The top menu bar includes File, Edit, View, Tools, and Windows. The left sidebar contains the QComponents panel, which lists various components such as P1_C, P2_C, Feedline, and several STG components. The main workspace shows a PCB layout with a central vertical feedline and four horizontal microstrip lines branching off it. The bottom status bar indicates the current layer is 'Top' and the unit is 'mm'.



Experimental results of the Chalmers device designed with Qiskit Metal

Data courtesy of *C. Warren* & Amr Osman (Bylander, Chalmers)



0.0.3

Search Docs

Home

Installing Qiskit Metal

Frequently Asked Questions

Roadmap

Qiskit Metal Workflow

Quantization Methods Overview

Contributor Guide

Contributing to Qiskit Metal

Where Things Are

Reporting Bugs and Requesting Enhancements

Contributing Code

Contributing to Documentation

Tutorials

Overview

Components

Analysis

Renderers

Video Recordings

Circuit Example Library

Qubits

Resonators

Composite Bi-Partite Systems

Qubit Couplers

Input-Output Coupling

Small Quantum Chips

Design Flow

Libraries

All Quantum Devices

API References

Overview

QDesigns

Docs > Qiskit Metal | Quantum Device Design & Analysis (Q-EDA) 0.0.3

Qiskit Metal | Quantum Device Design & Analysis (Q-EDA) 0.0.3

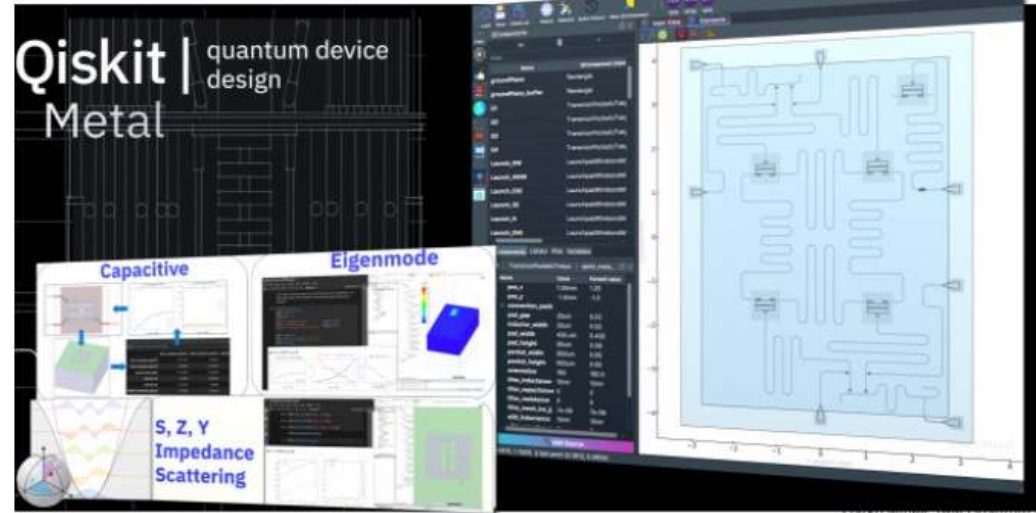
• ATTENTION

This is an alpha version of Qiskit Metal, the code is still under development. Please let us know about anything you might want us to add or elaborate upon in the Slack channel [#metal](#) in the [qiskit workspace](#).

• HINT

You can open this documentation using

```
import qiskit_metal
qiskit_metal.open_docs()
```

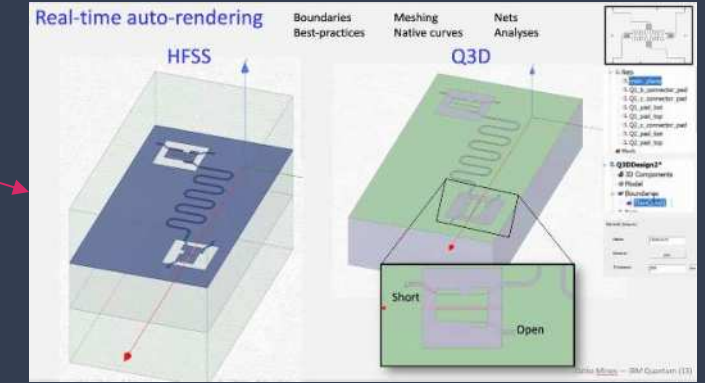
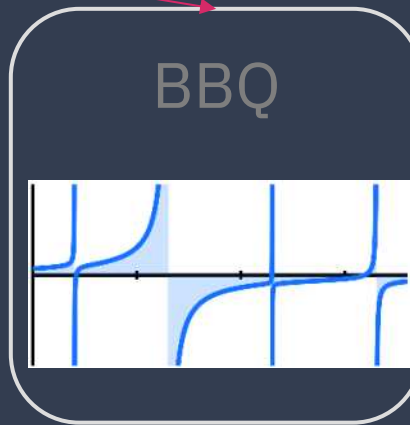
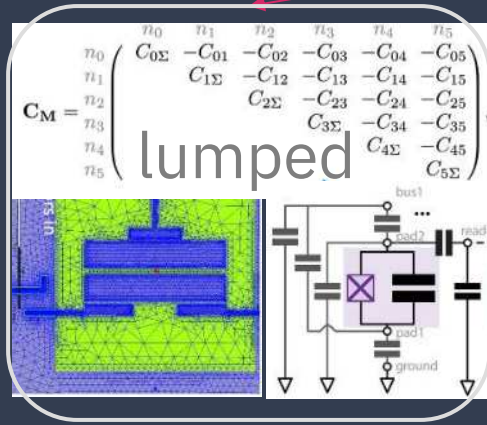
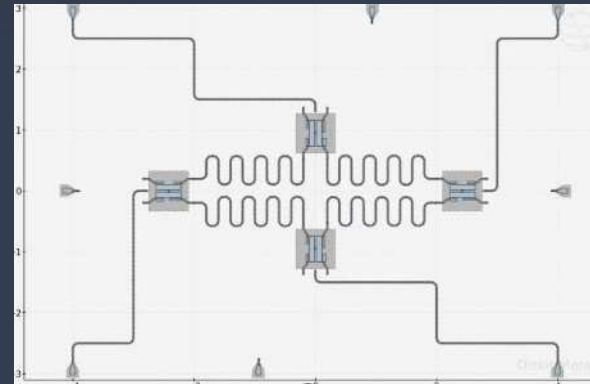
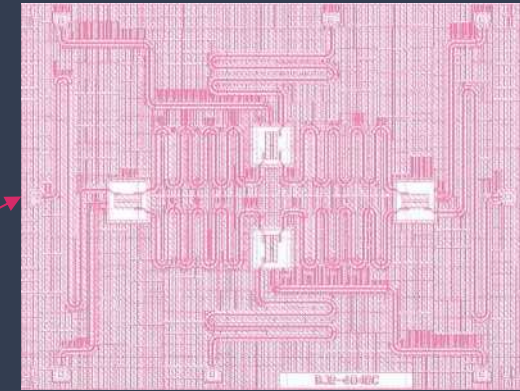
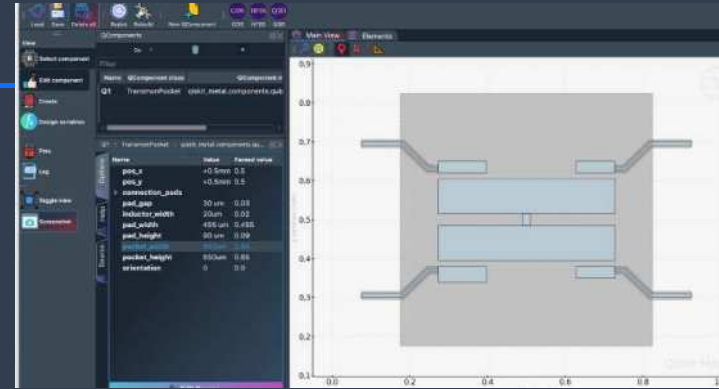
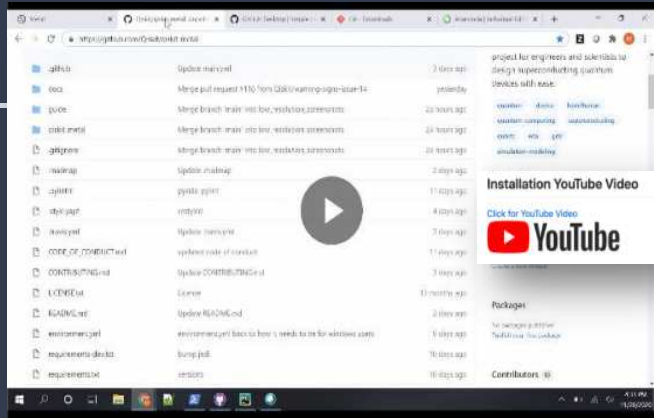


About

Qiskit Metal | for quantum device design & analysis ('Qiskit Metal') is an open-source framework (and library) for the design of superconducting quantum chips and devices. Call it quantum EDA (Q-EDA) and analysis. Qiskit Metal is:

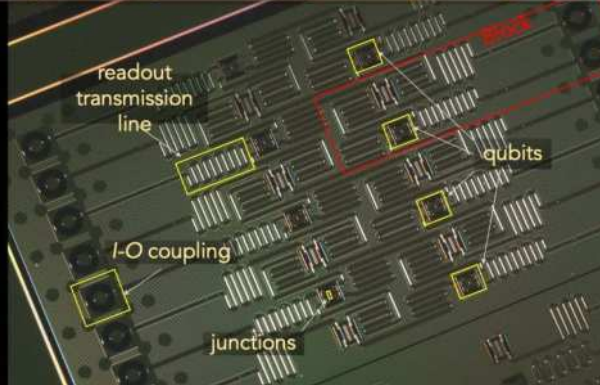
- Open source
- Community-driven

Tutorials

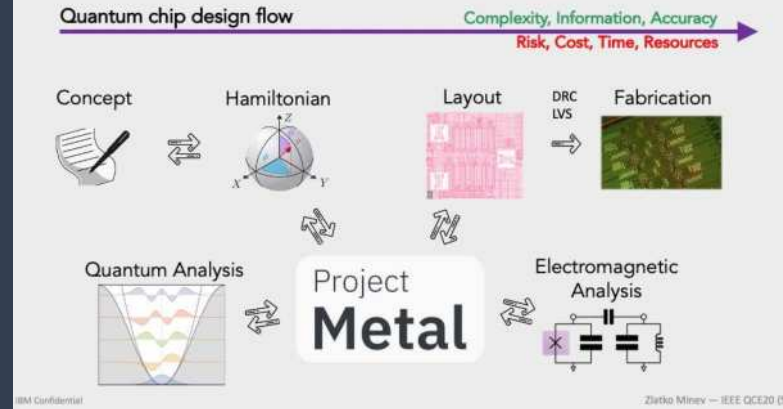


Regularity is our greatest ally in chip design

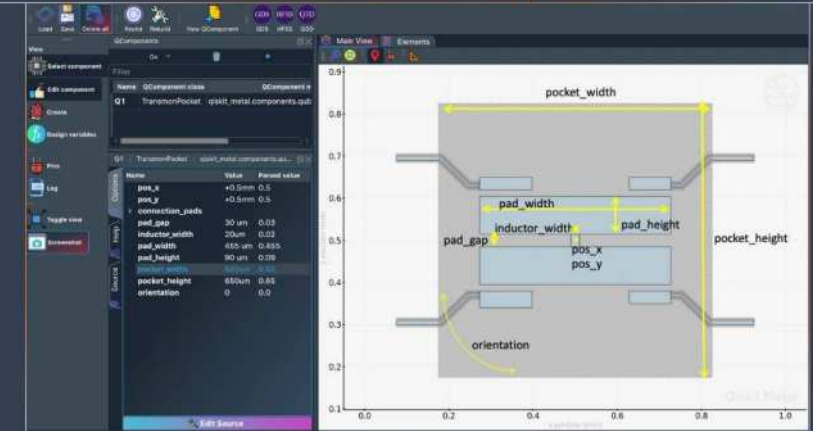
Reuse
Fine-tune
Automate
Extend



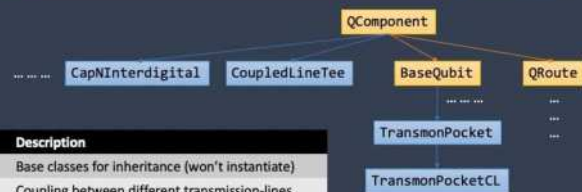
Quantum chip design flow



Fine-tune and automate parameters



Reuse and Extend the QLibrary

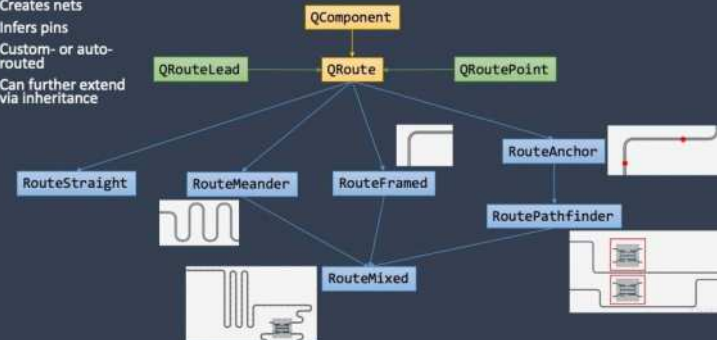


Folder	Description
Core	Base classes for inheritance (won't instantiate)
Couplers	Coupling between different transmission-lines
Lumped	Capacitive/inductive elements
Qubits	Information
Sample Shapes	Geometries sand-box
Terminations	Single pin element to terminate transmission lines
Tlines	Two pins transmission lines

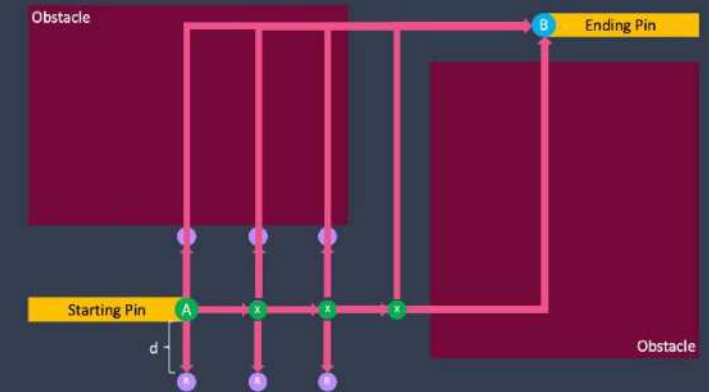
- Extensible horizontally and vertically
- Simple plug-n-play custom QComponents

Transmission Line

- Creates nets
- Infers pins
- Custom- or auto-routed
- Can further extend via inheritance



Making Connections



Tutorial 6

Qiskit | quantum device design

There's a description for every job



$$\hat{H}_{\text{tot}}$$

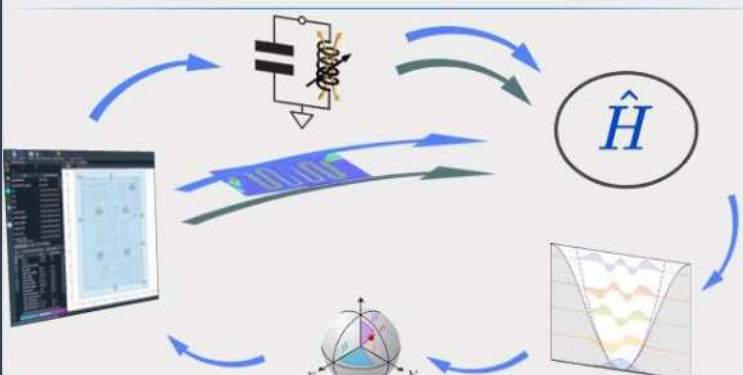
$$\hat{H}_{\text{tot}} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$

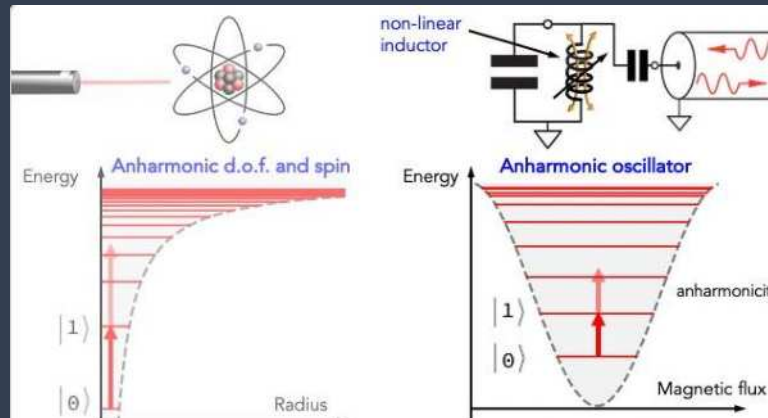
Zlatko Mineev, IBM Quantum (9)

Qiskit | quantum device design

Quantum analysis

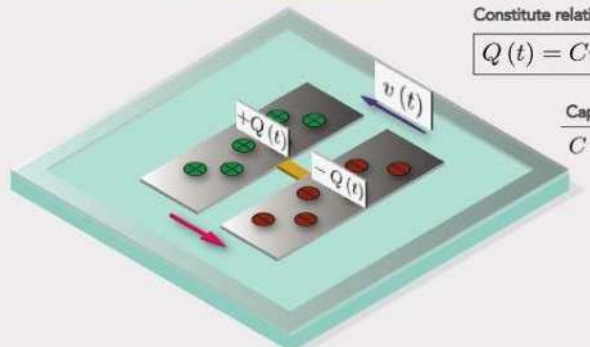


Zlatko Mineev, IBM Quantum (10)



Zlatko Mineev — Qiskit Global Summer School 2020 (26)

Charge and capacitance



Constitute relationship

$$Q(t) = Cv(t)$$

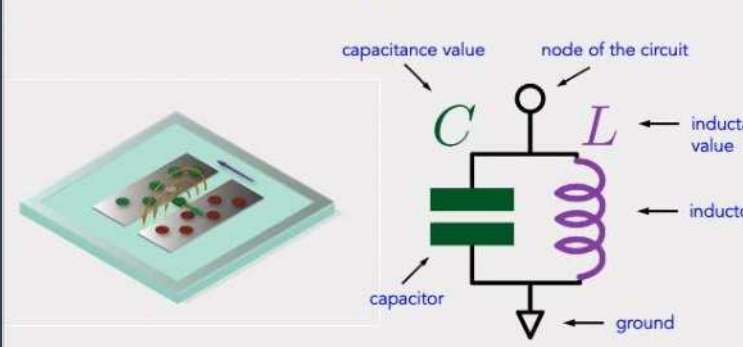
Capacitance

$$C \in (0, \infty)$$

For a good discussion, see "The Feynman Lectures on Physics Vol. II Ch. 22: AC Circuits." Caltech.

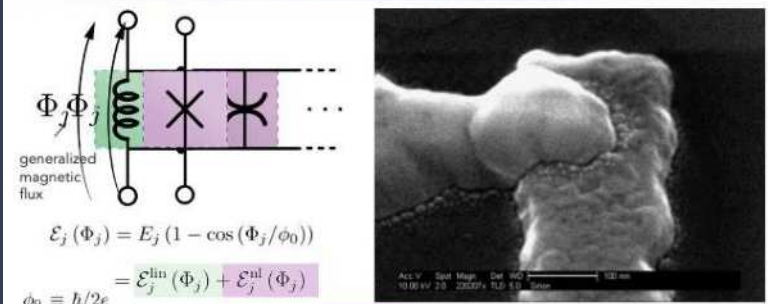
Zlatko Mineev — Qiskit Global Summer School 2020 (40)

Electromagnetic oscillator



Zlatko Mineev — Qiskit Global Summer School 2020 (44)

Josephson tunnel junction



$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

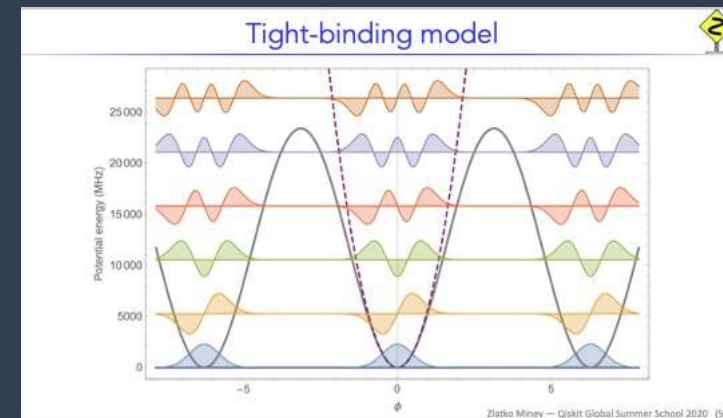
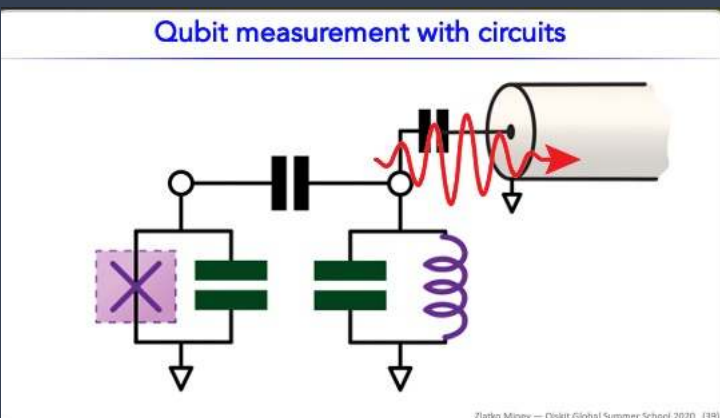
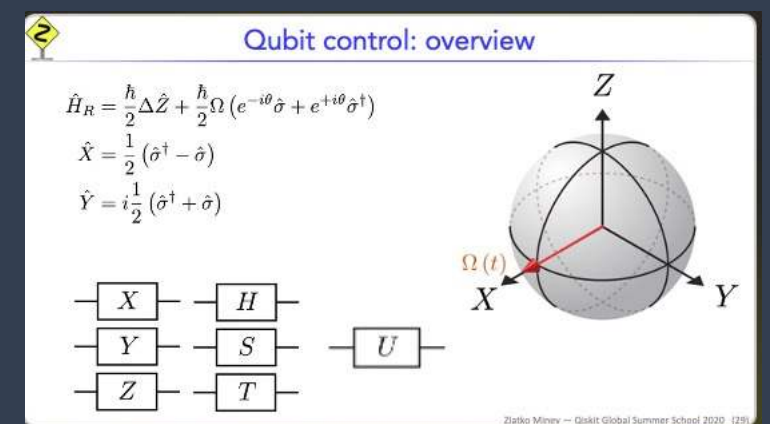
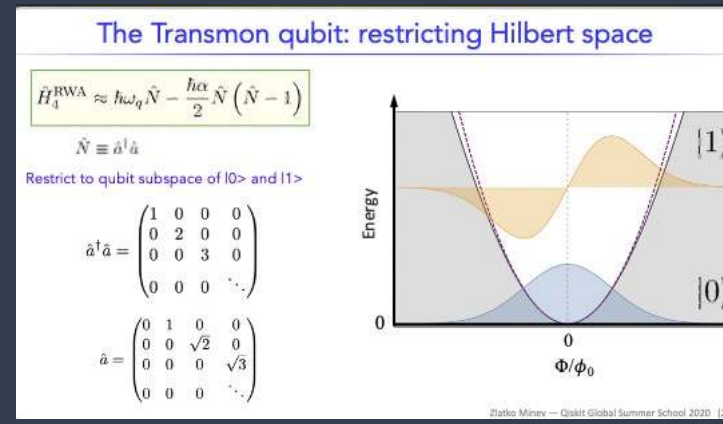
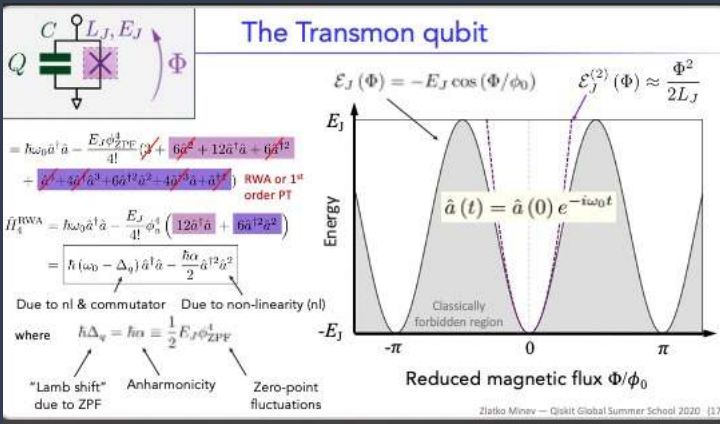
$$\phi_0 \equiv h/2e$$

$$= \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$

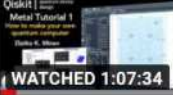
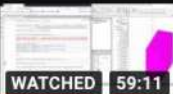
Circuit image: Mineev et al., EPR to appear (2020)

Zlatko Mineev — Qiskit Global Summer School 2020 (48)

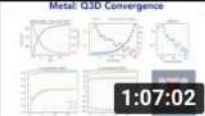
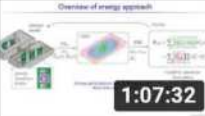
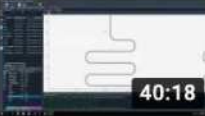


Demo notebook

Tutorials so far

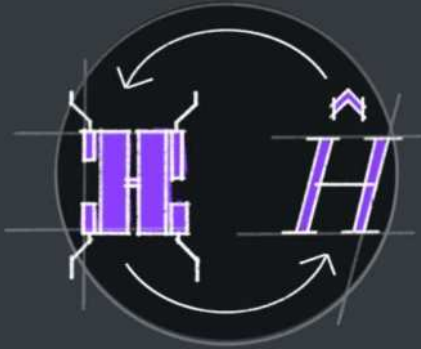
- 1  **Qiskit Metal E01 - Overview**
Qiskit 
- 2  **Qiskit Metal E02 - End to end example of Quantum Chip Design - Part 1 of 2**
Qiskit
- 3  **Qiskit Metal E03 End to end example of Quantum Chip Design - Part 2 of 2**
Qiskit
- 4  **Qiskit Metal E04 - QComponents for parametric design**
Qiskit
- 11  **Qiskit Metal E06 - Quantum Analysis 101 Prerequisites**
Qiskit
- 12  **Qiskit Metal E07 - Introduction to the transmon qubit**
Qiskit

New analysis (pre-recorded)

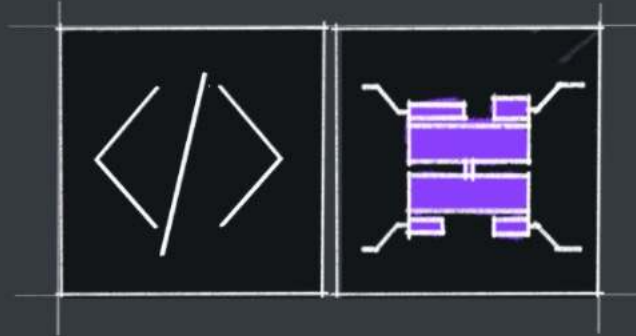
- 5  **Qiskit Metal E05.1 - Analysis - Capacitance and Frequency Control**
Qiskit
- 6  **Qiskit Metal E05.2 - Analysis - Eigenmode and Energy Participation**
Qiskit
- 7  **Qiskit Metal E05.3 - Analysis - EPR Theory**
Qiskit
- 8  **Qiskit Metal E05.4 - Analysis - Summary EPR Quantization with Code Example**
Qiskit
- 9  **Qiskit Metal E05.5 - Analysis - Finish Eigenmode Start Impedance Analysis**
Qiskit
- 10  **Qiskit Metal E05.6 - Analysis - Extracting S Parameters for a Hanging Resonator**
Qiskit

Why the Vision of Qiskit Metal

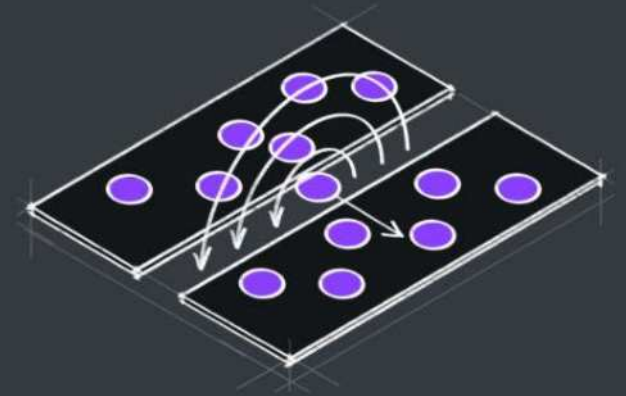
End-to-end automation



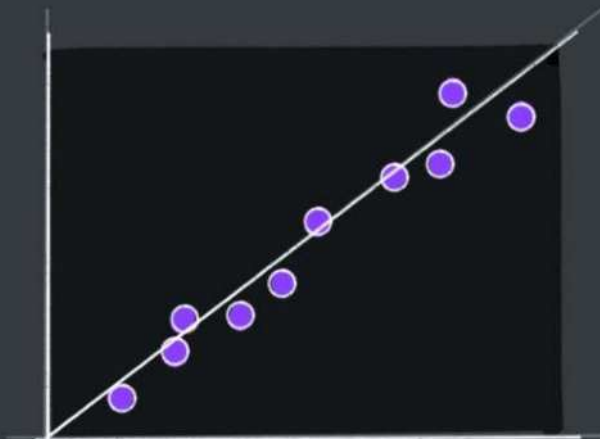
Flexible & extensible



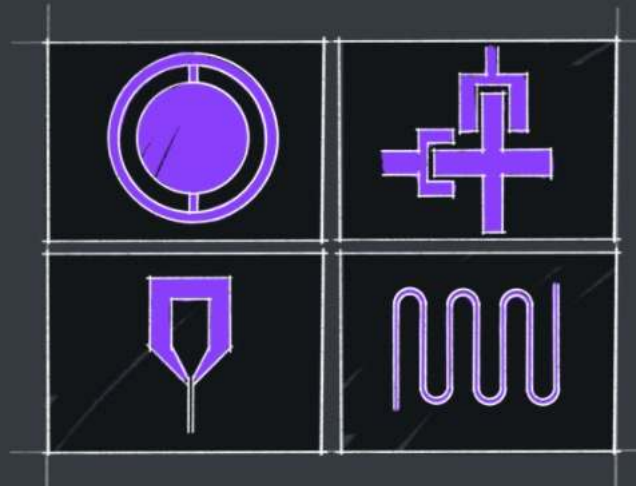
Light-weight interoperability



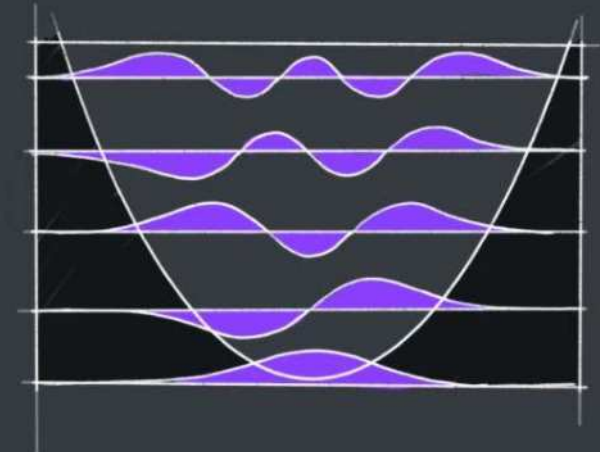
Experimentally tested



Library of components



Cutting edge resources



Building together

Call for community participation

Open source

Education

See summer school lectures 16-21 by Z. Mineev from
Introduction to Quantum Computing and Quantum Hardware
and the *Qiskit Textbook*

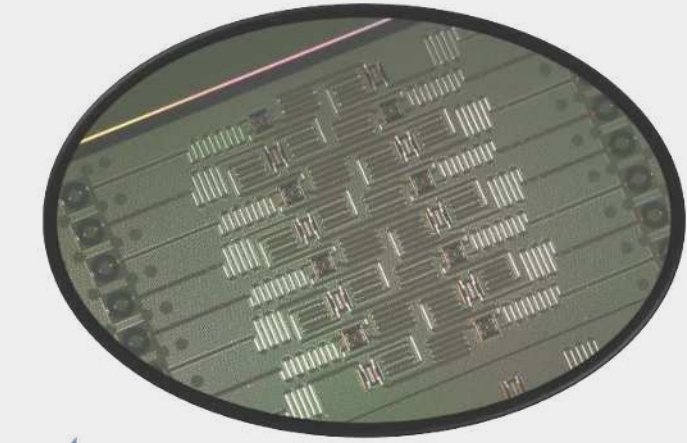
qiskit.org/metal



@zlatko_minev



Physical Devices ↔ Quantum Hamiltonian



Energy participation ratio (EPR)

Minev et al. arXiv:1902.10355 & arXiv:2010.00620
(Yale)

$\hat{H}$

Quantum Physics

[Submitted on 1 Oct 2020]

Energy-participation quantization of Josephson circuits

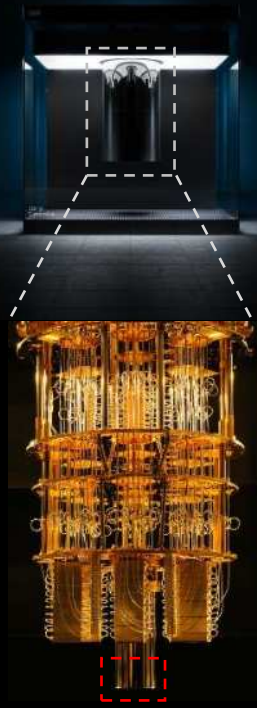
Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, Michel H. Devoret

Accepted in principle: *Nature npj qi*

See also Minev
Yale dissertation Sec. 4.1
(arXiv: 1902.10355)

Yale

What needs to be designed?



metals

junctions

geometry &
topology

qubit frequency,
anharmonicity, ...

dielectrics

many-body
Hamiltonian

qubit-qubit
coupling

readout
frequency,
coupling

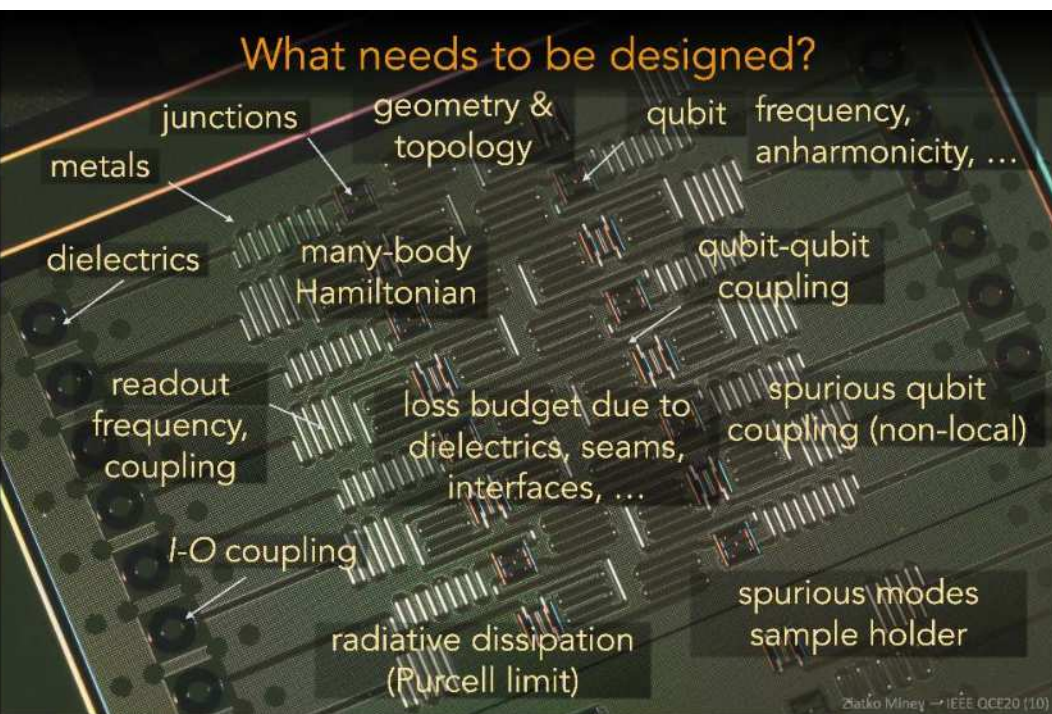
loss budget due to
dielectrics, seams,
interfaces, ...

spurious qubit
coupling (non-local)

I-O coupling

radiative dissipation
(Purcell limit)

spurious modes
sample holder



The solution of all these questions reduces to:

Where is the energy?

What *fraction* of the energy of a mode p_j (p_l) is stored in the non-linear (dissipative) element?

$$0 \leq p_j, p_l \leq 1$$

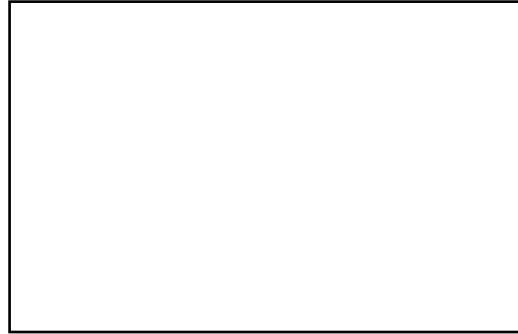

$$\hat{H}_{\text{full}}$$


$$\mathcal{D}[\sqrt{\kappa}\hat{a}]\rho$$

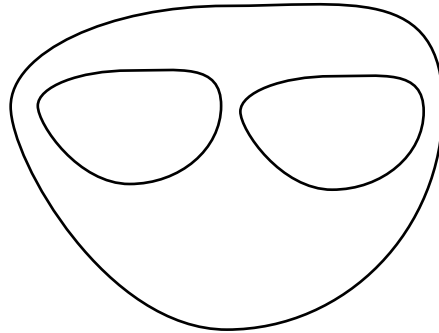
$$\hat{H}_{\text{full}}$$

Hamiltonian of a linear distributed system

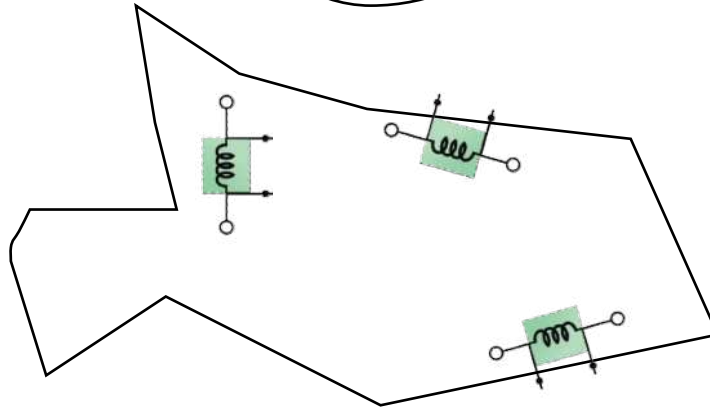
This is a box



$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$



$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$



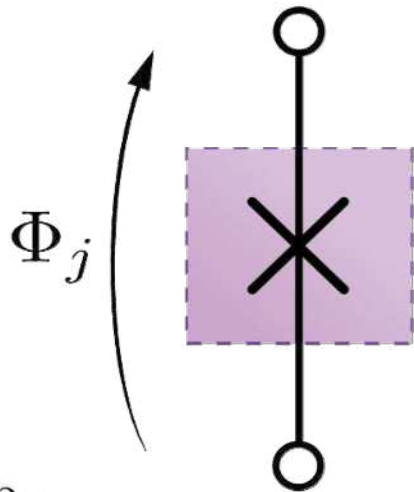
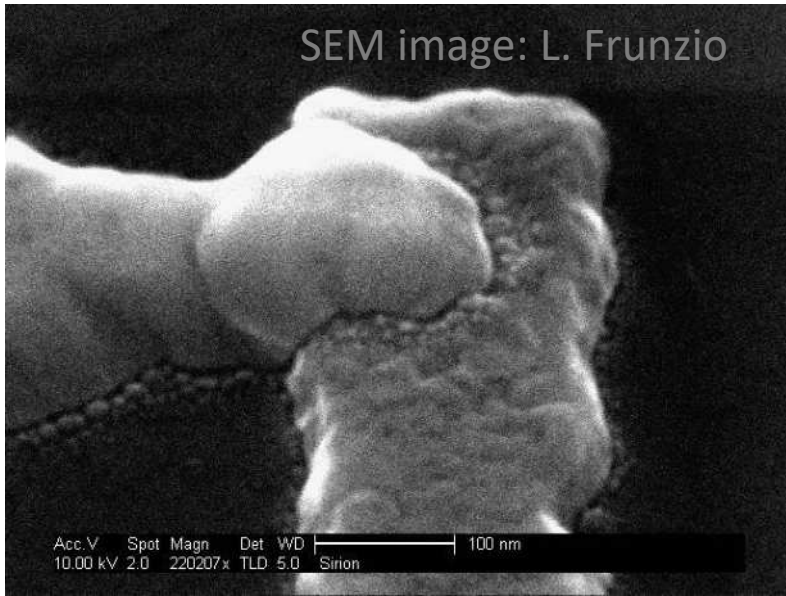
$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$

There are two kinds of physicists:

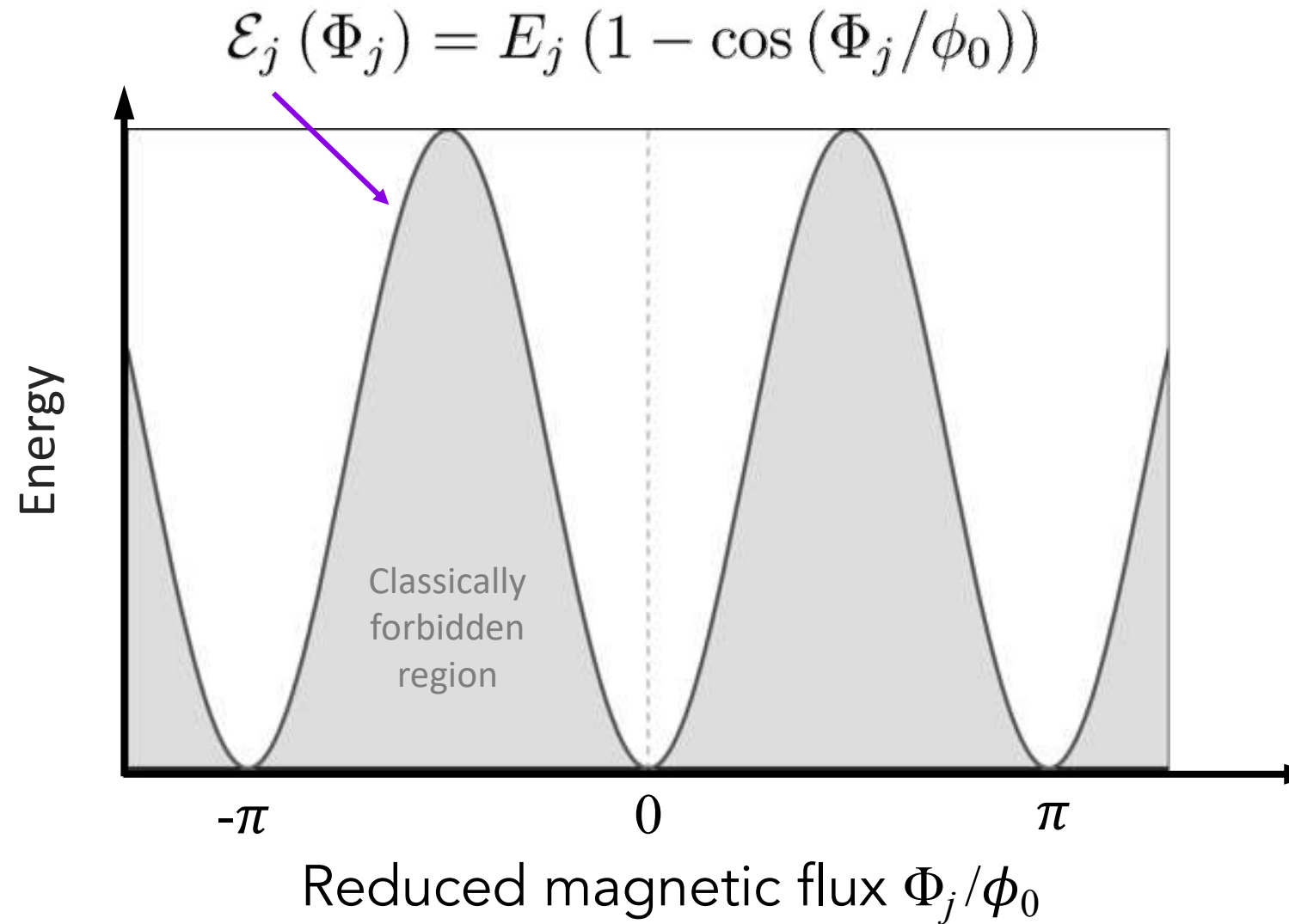
Those who believe all of physics is *spins*.

Those who believe all of physics is *oscillators*.

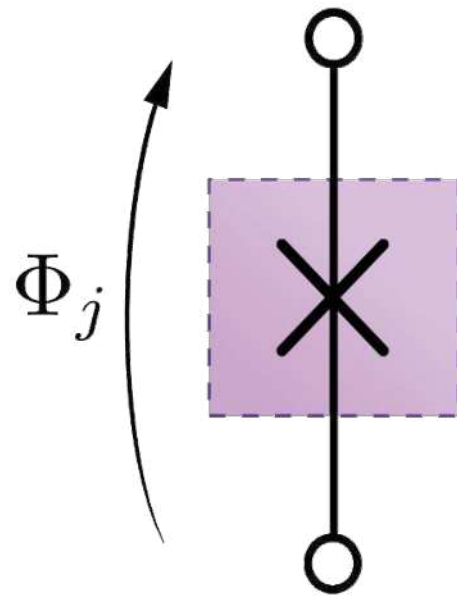
Energy of a non-linear element



$$\phi_0 \equiv \hbar/2e$$

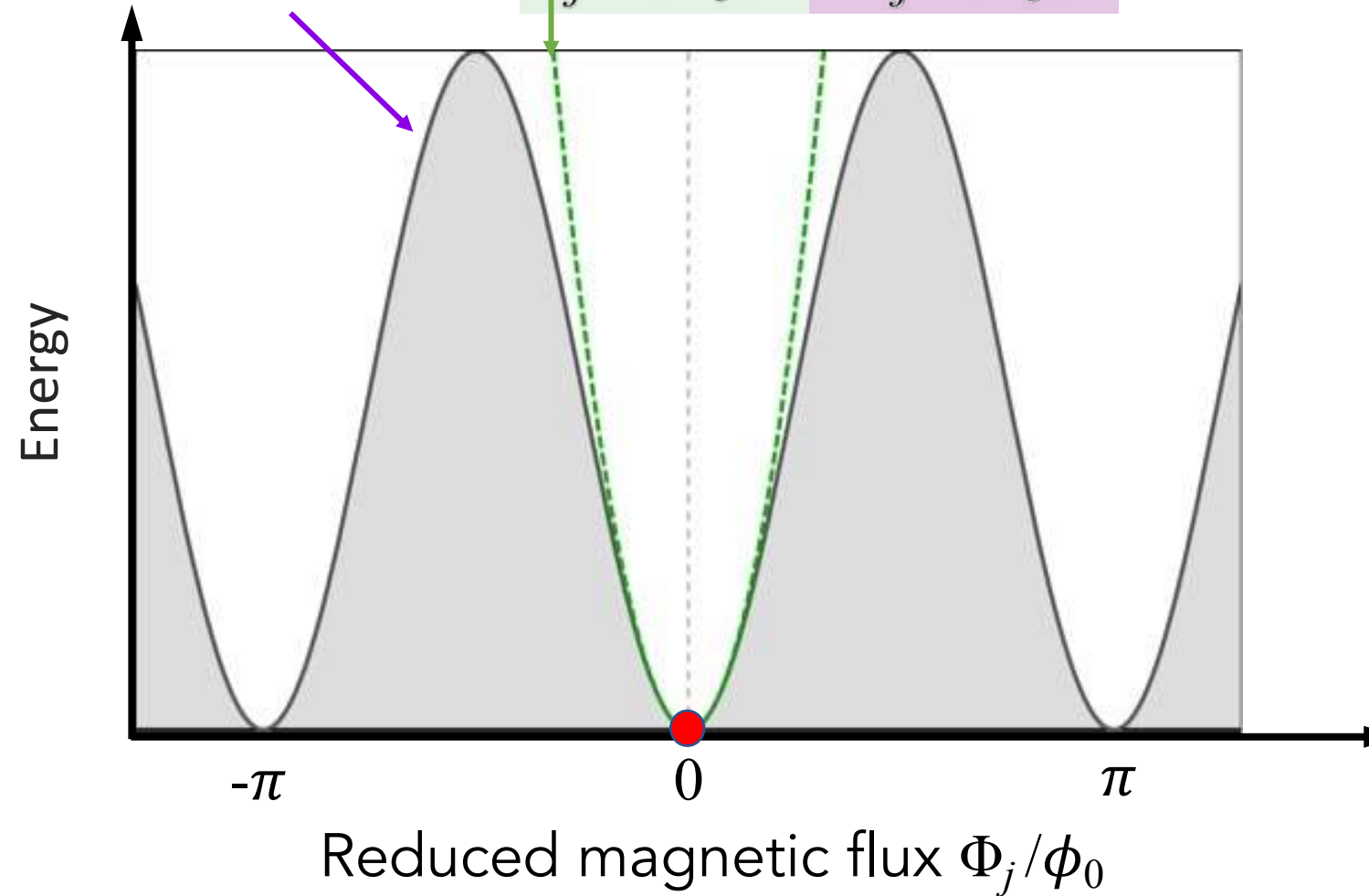


Energy of a non-linear element

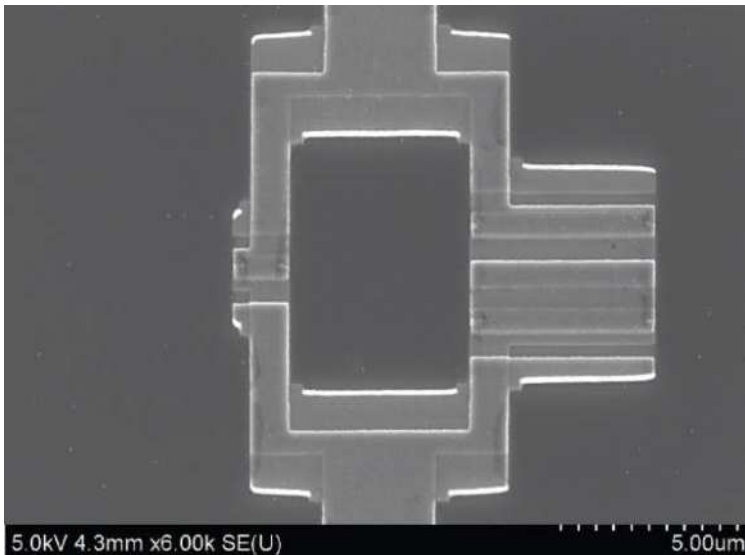
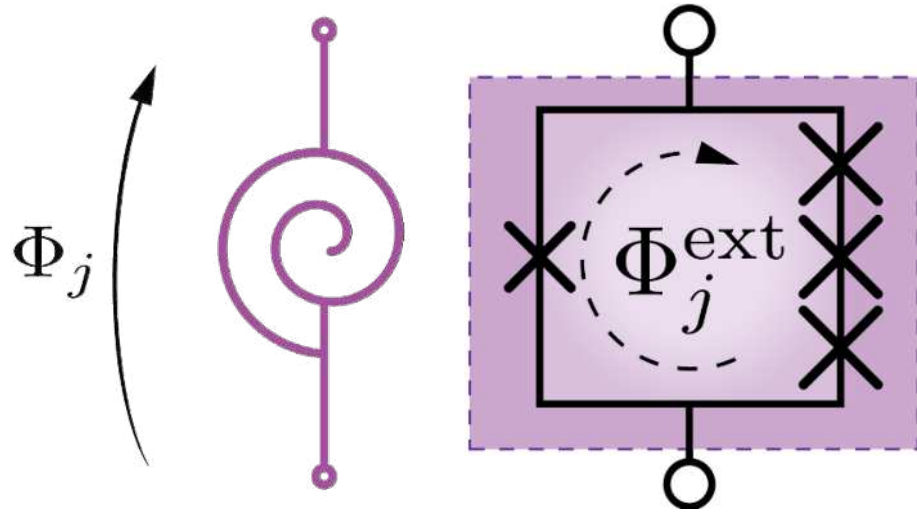


Momentarily only

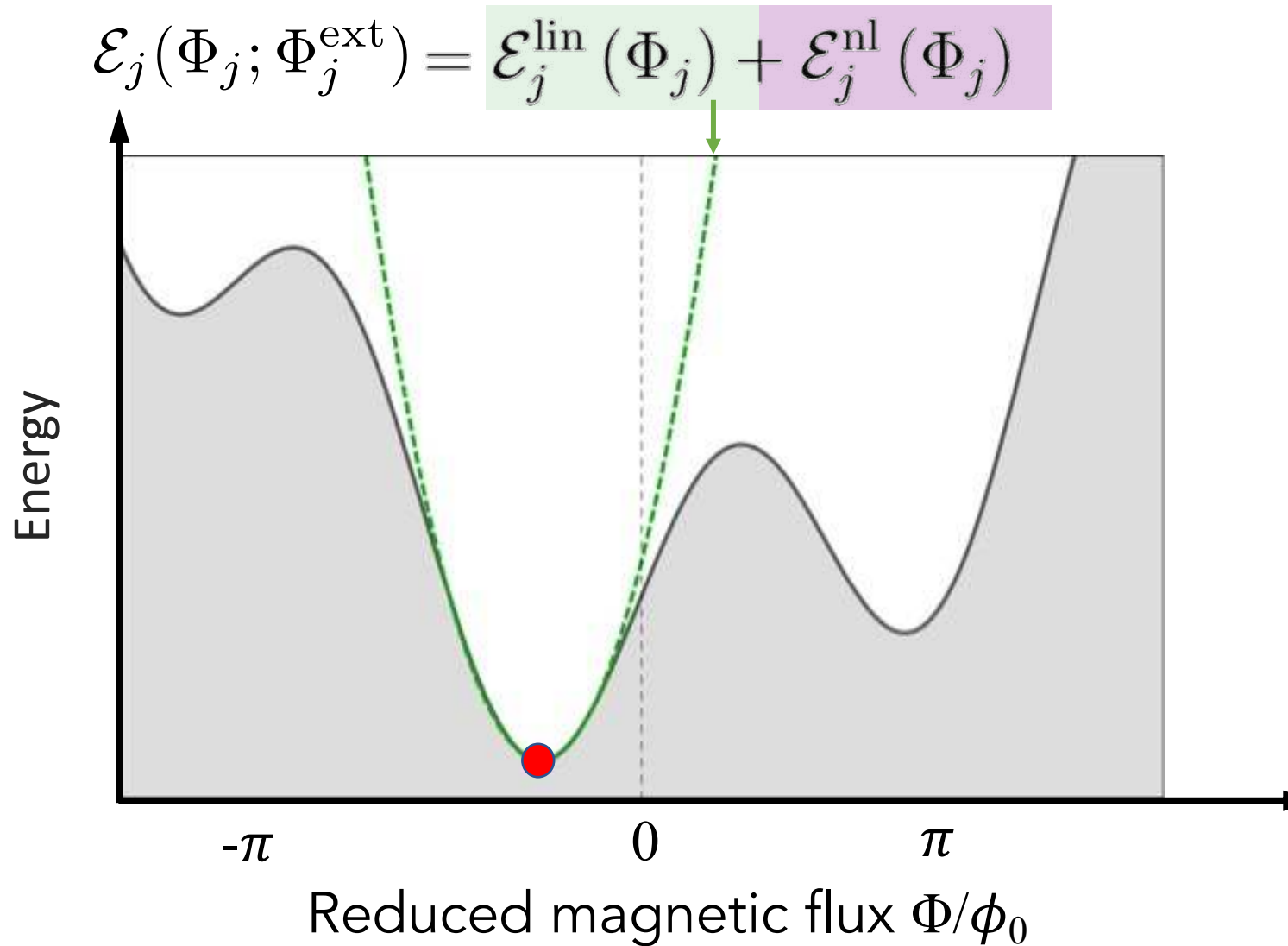
$$\mathcal{E}_j(\Phi_j) = \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$



Composite-circuit non-linear dipole

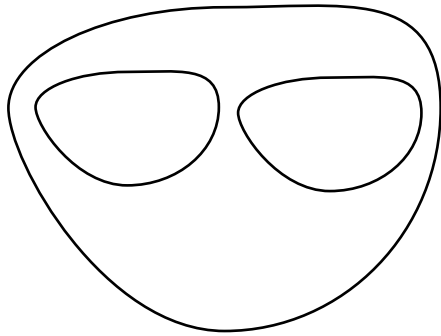


SEM image: Frattini & Sivak; see APL (2017)

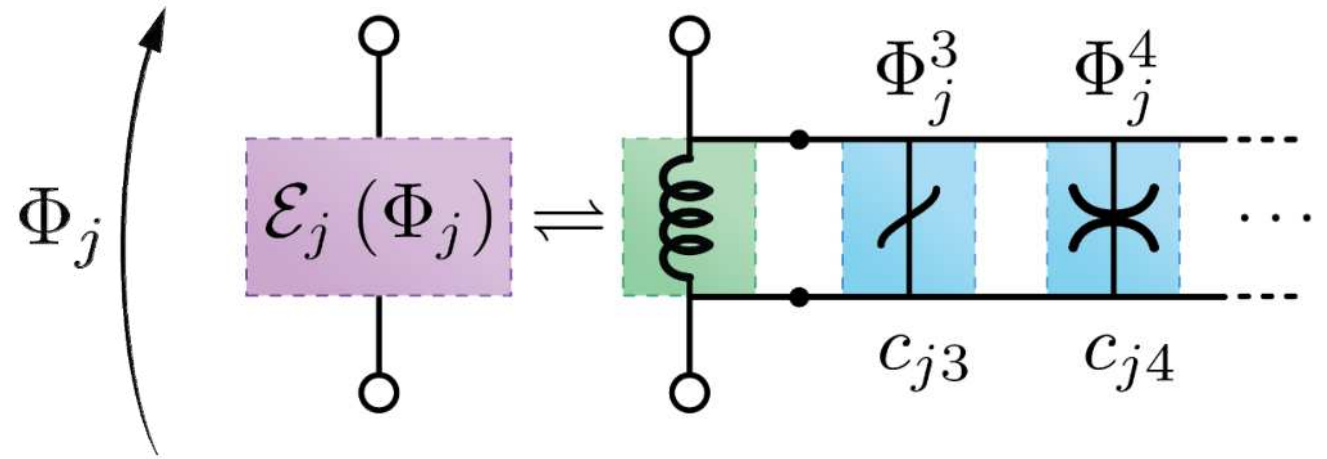


$$\phi_0 \equiv \hbar/2e$$

So far

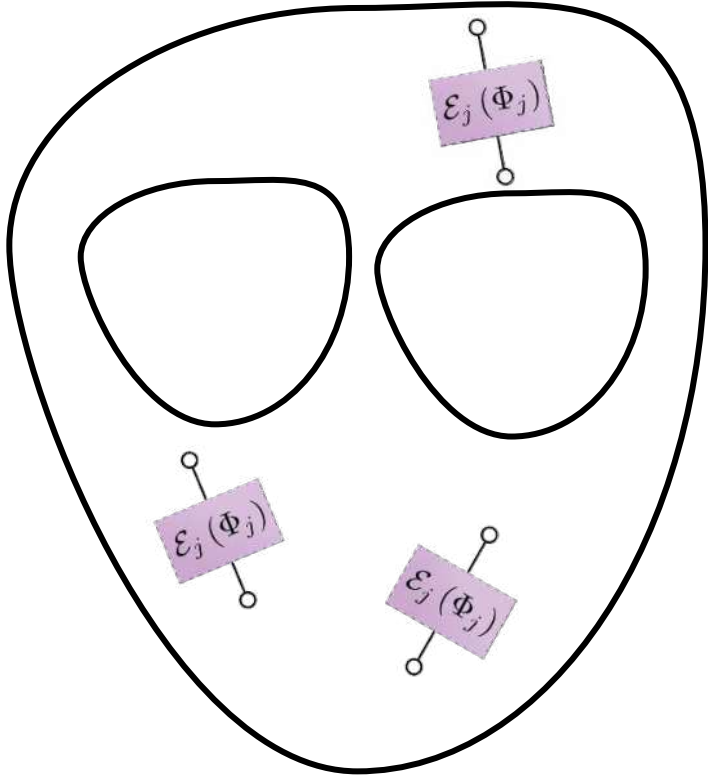


linear distributed system



non-linear dipole

Distributed system with non-linear dipoles



$$\hat{H}_{\text{full}} \left(\hat{\Phi}_{\text{bare}}, \hat{\mathbf{Q}}_{\text{bare}} \right)$$

change basis

$$\hat{H}'_{\text{full}} \left(\hat{\Phi}_{\text{eigen}}, \hat{\mathbf{Q}}_{\text{eigen}} \right)$$

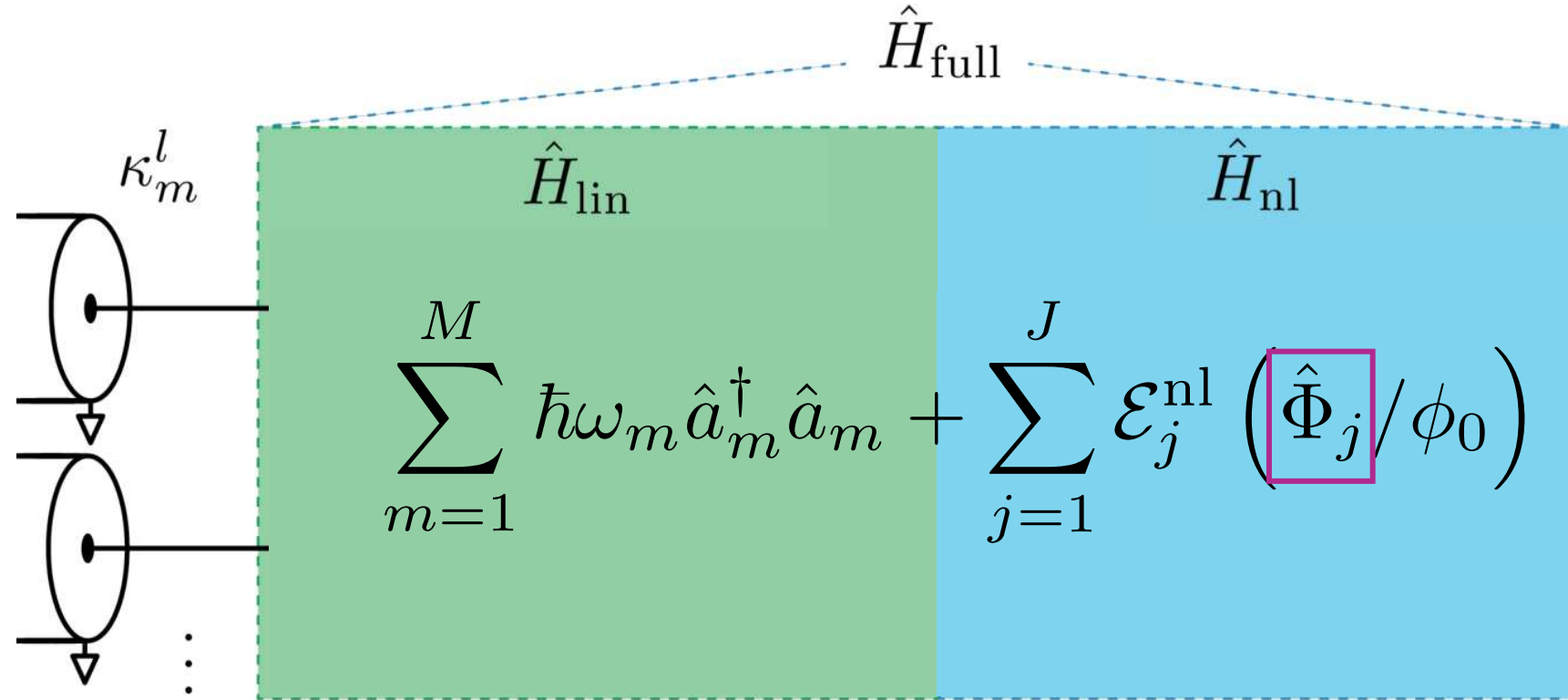
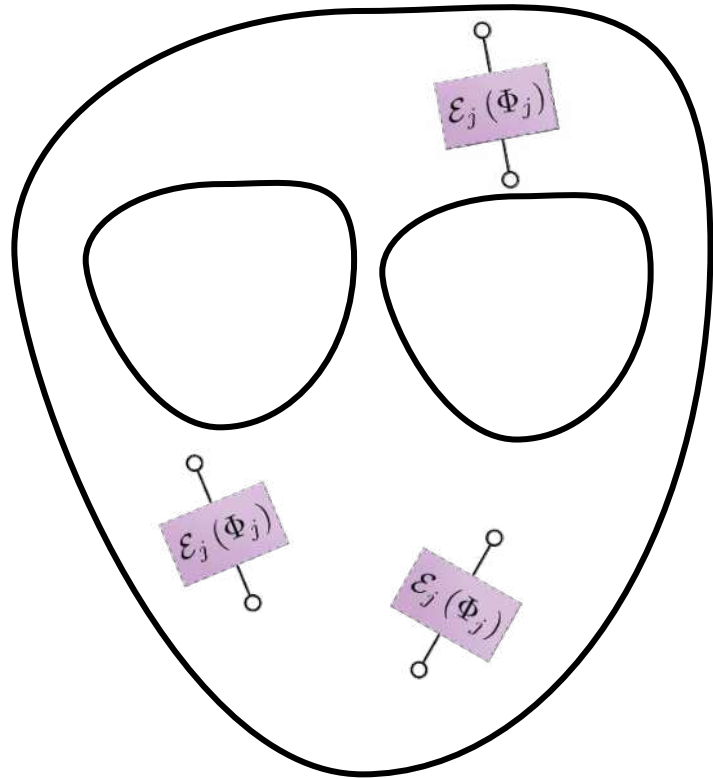
change basis

$$\hat{H}''_{\text{full}} \left(\hat{\mathbf{a}}_m, \hat{\mathbf{a}}_m^{\dagger} \right)$$

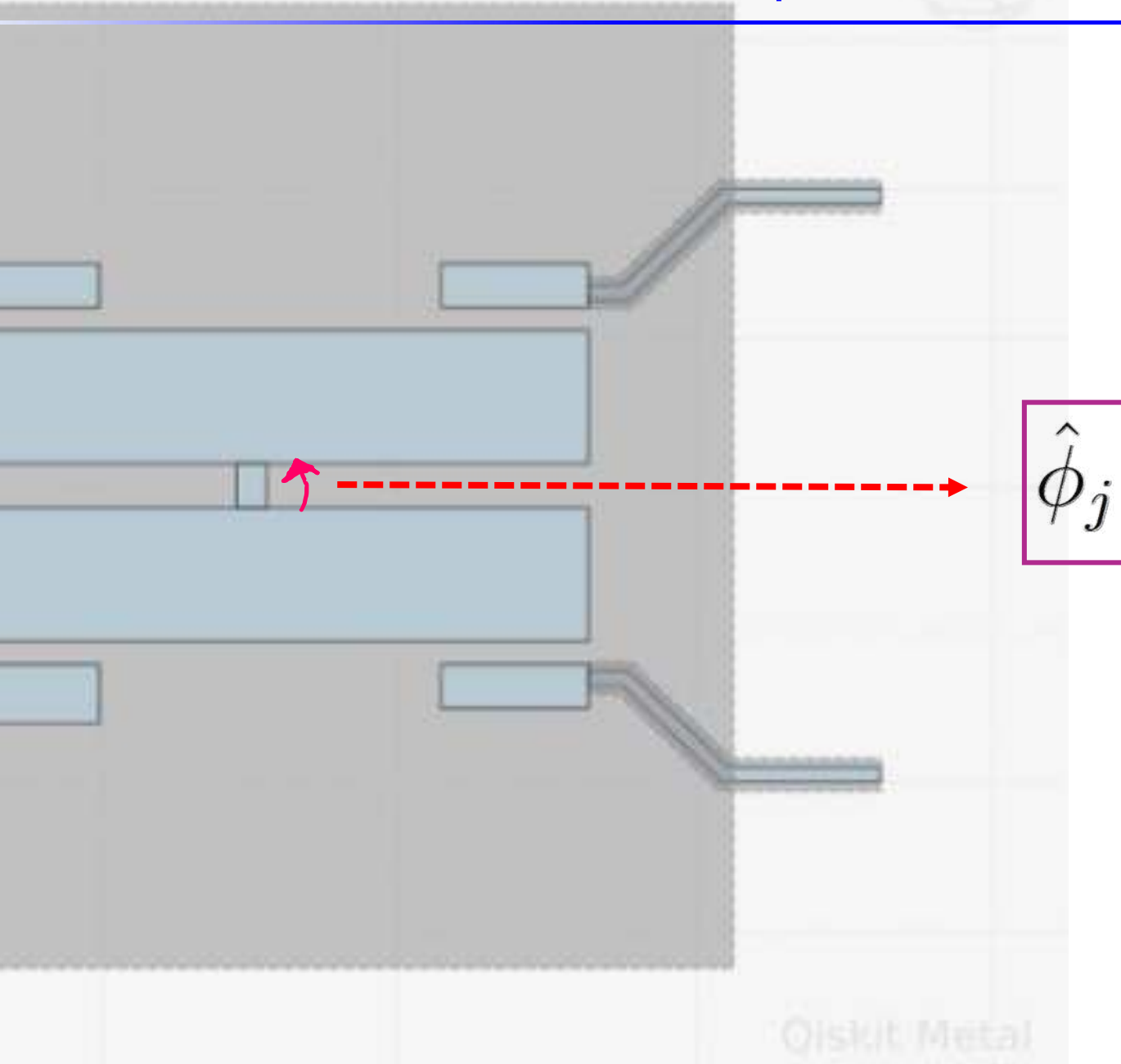
no approximation here

$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^{\dagger} \hat{a}_m + \sum_{j=1}^J \mathcal{E}_j^{\text{nl}} \left(\hat{\Phi}_j / \phi_0 \right)$$

Distributed system with non-linear dipoles

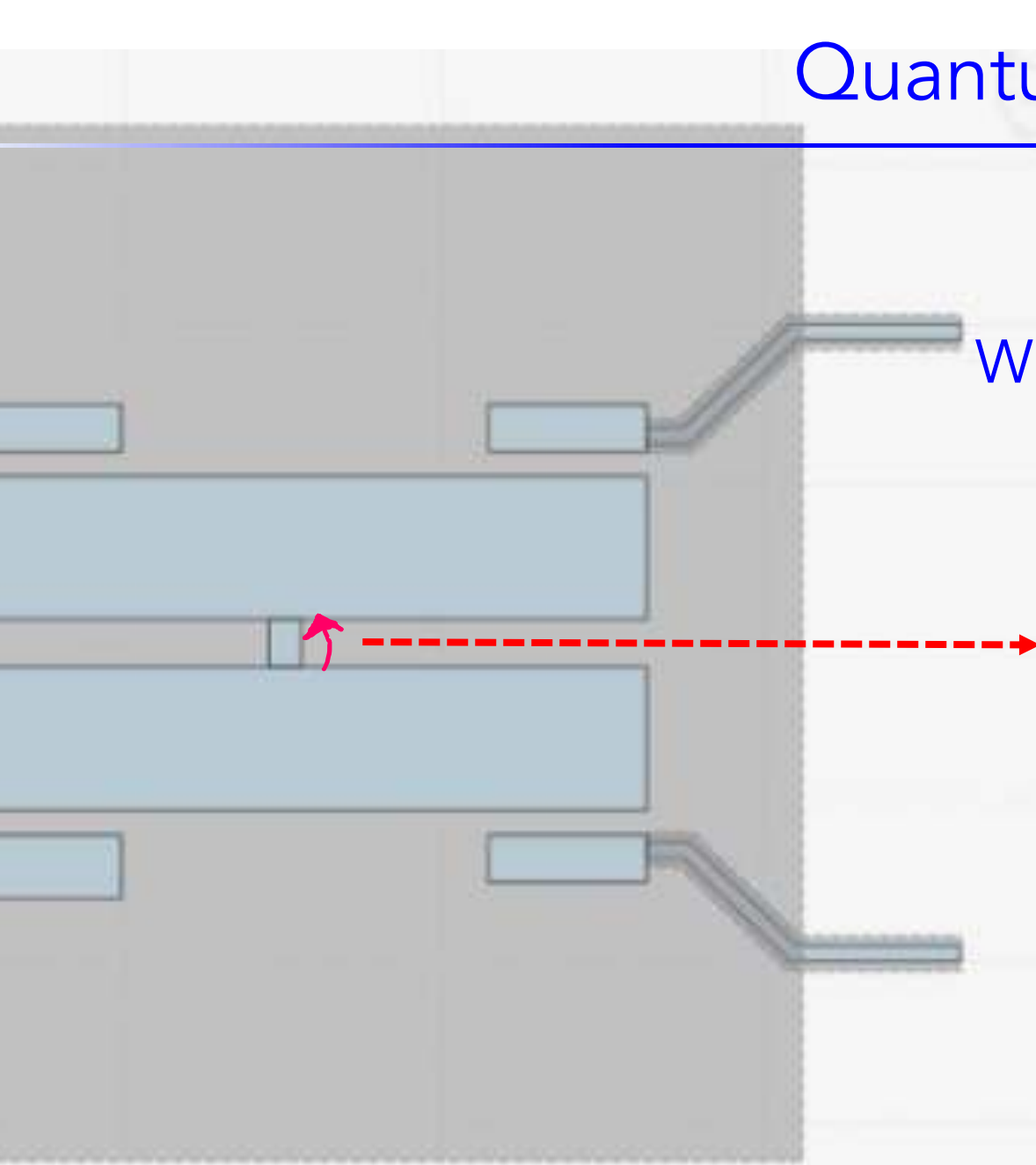


Quantum flux operator for the non-linear dipole



Quantum operator

What is zero-point fluctuation of the phase of junction j due to mode m ?



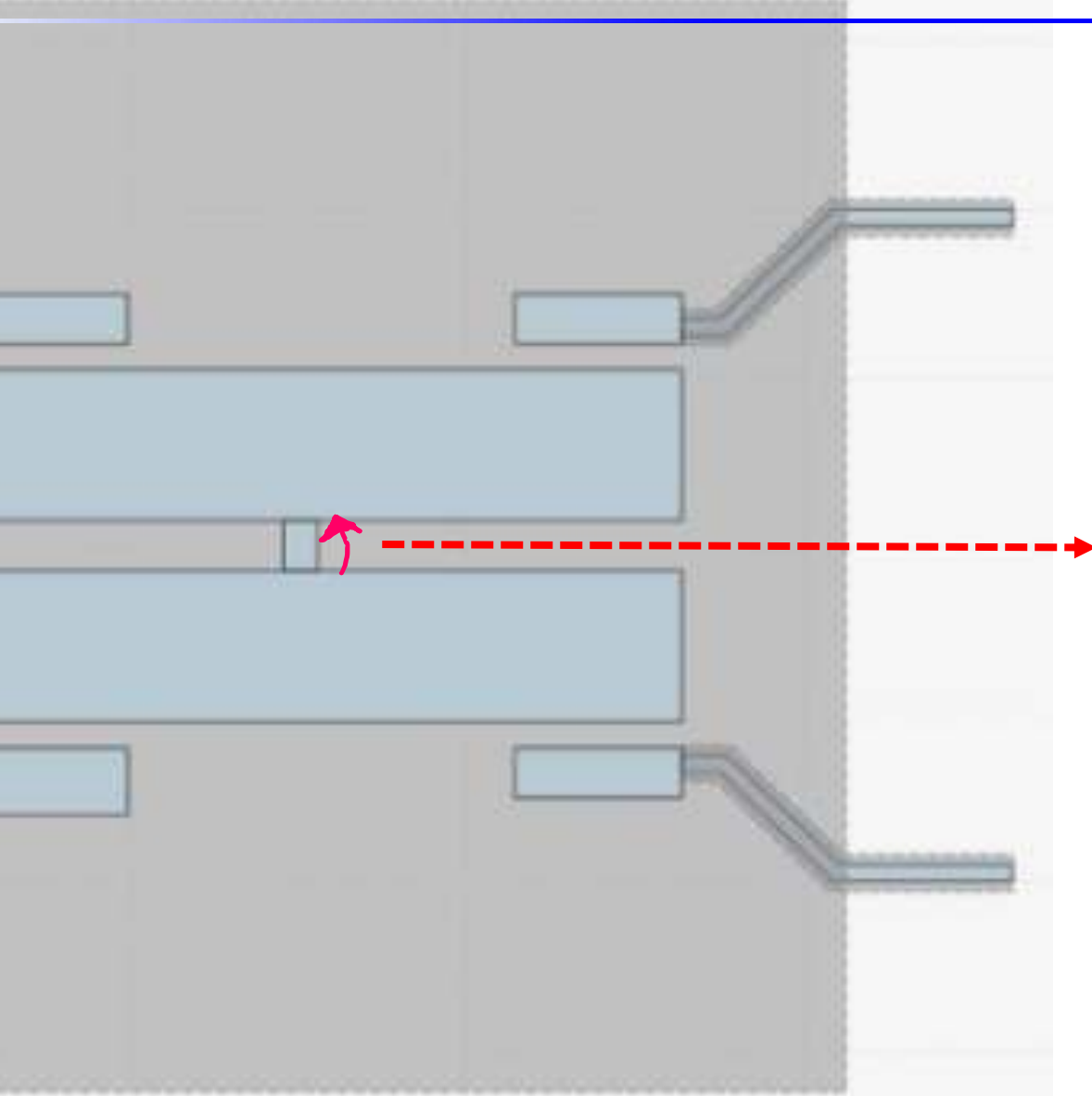
The diagram on the left shows a cross-sectional view of a quantum device with several horizontal blue bars representing gates or electrodes. A red arrow points from a small gap between two bars to the equation on the right.

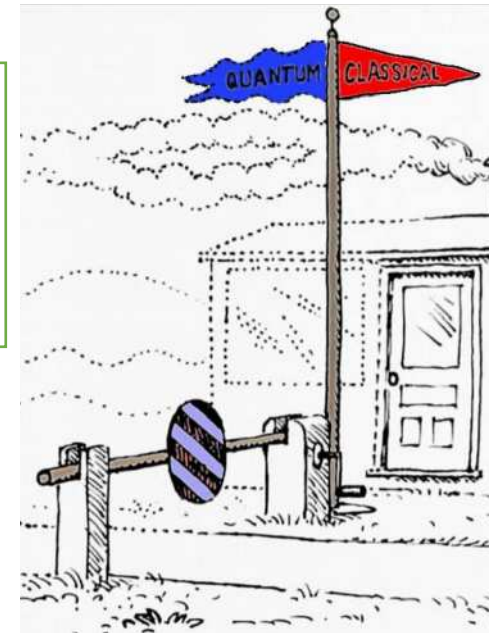
$$\hat{\phi}_j = \sum_{m=1}^M \phi_{mj} (\hat{a}_m^\dagger + \hat{a}_m)$$

This equation is a basis change!!

EPR bridge from classical to quantum

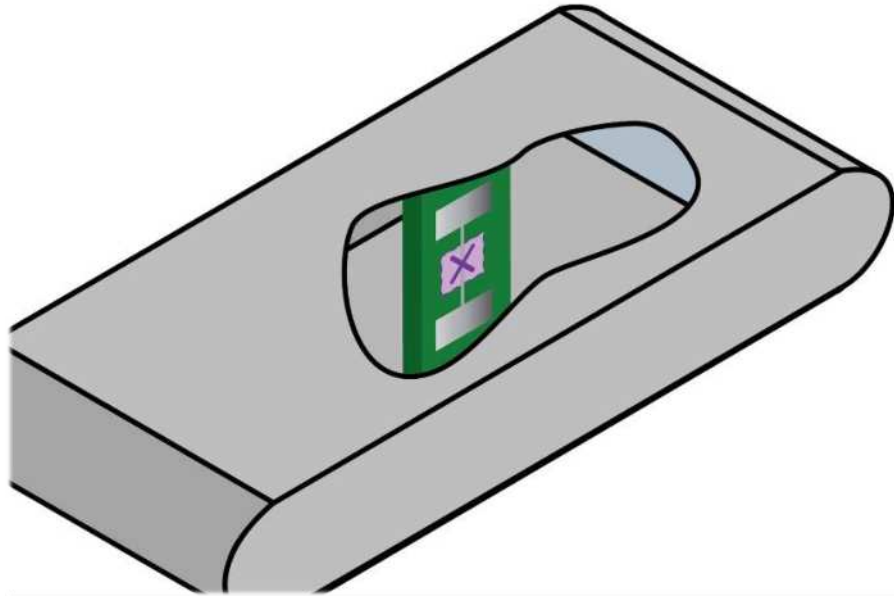
What fraction of the energy of mode m is stored in junction j ?


$$\phi_{mj} = p_{mj} \frac{\omega_m}{2E_j}$$



for $j > 1$, root requires sign bit $s_{mj} = \pm 1$

Qubit-cavity system example



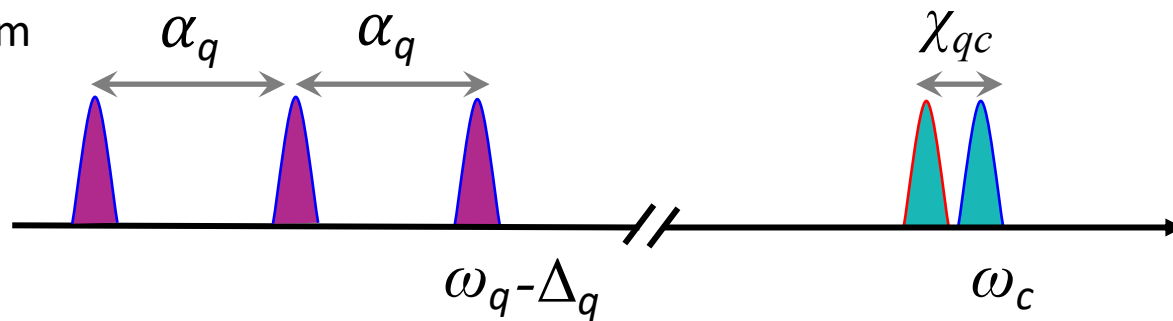
Qubit/cavity anharmonicity

$$\alpha_{q/c} = p_{q/c}^2 \frac{\hbar \omega_{q/c}^2}{8E_J}$$

Qubit-cavity dispersive shift

$$\chi_{qc} = p_q p_c \frac{\hbar \omega_q \omega_c}{4E_J}$$

Transition spectrum

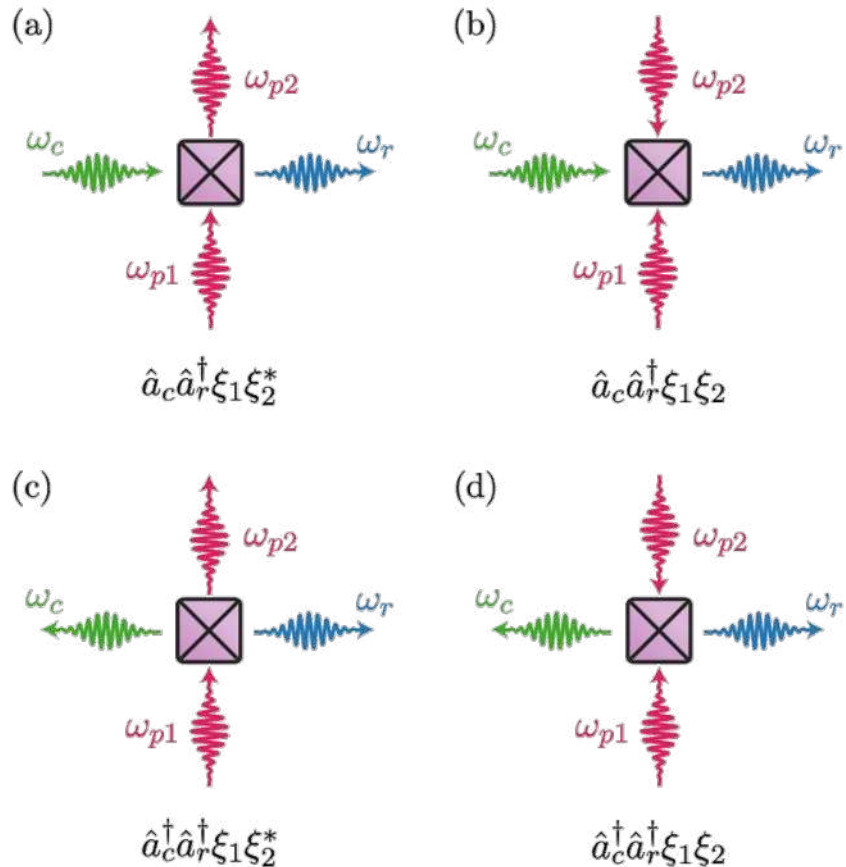


Qubit Lamb shift

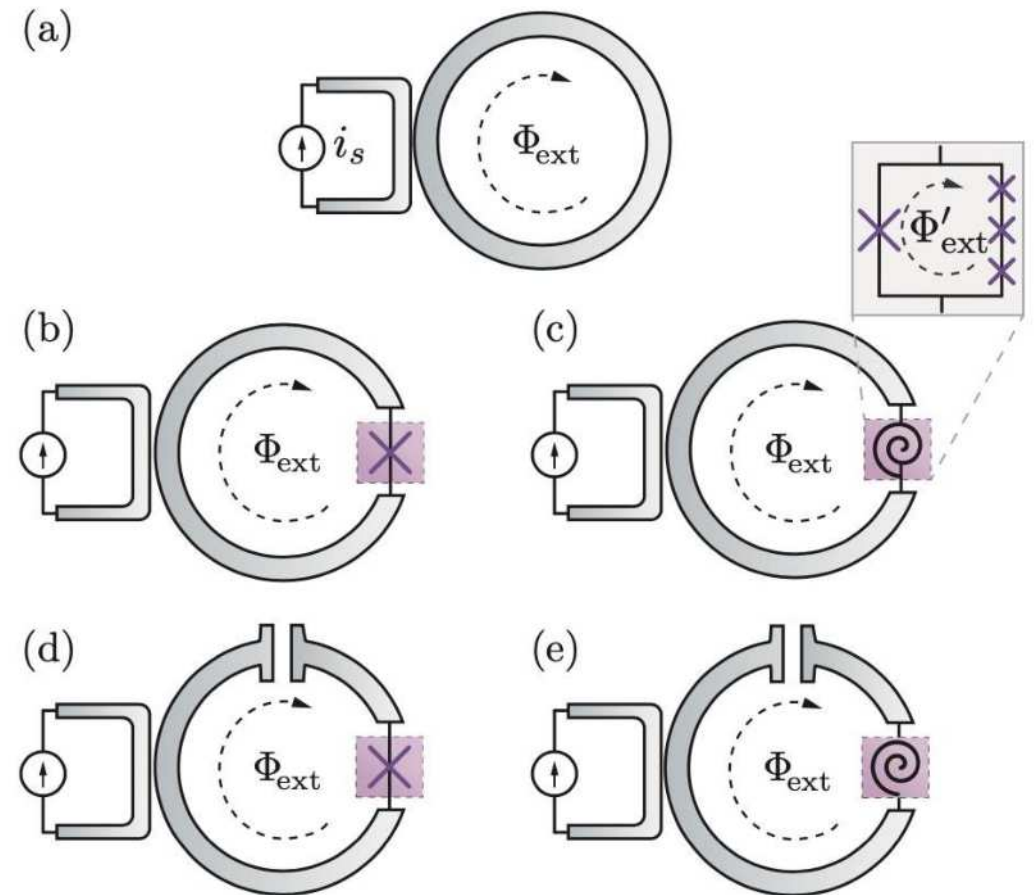
$$\Delta_q = \alpha_q - \frac{1}{2} \chi_{qc}$$

Special features

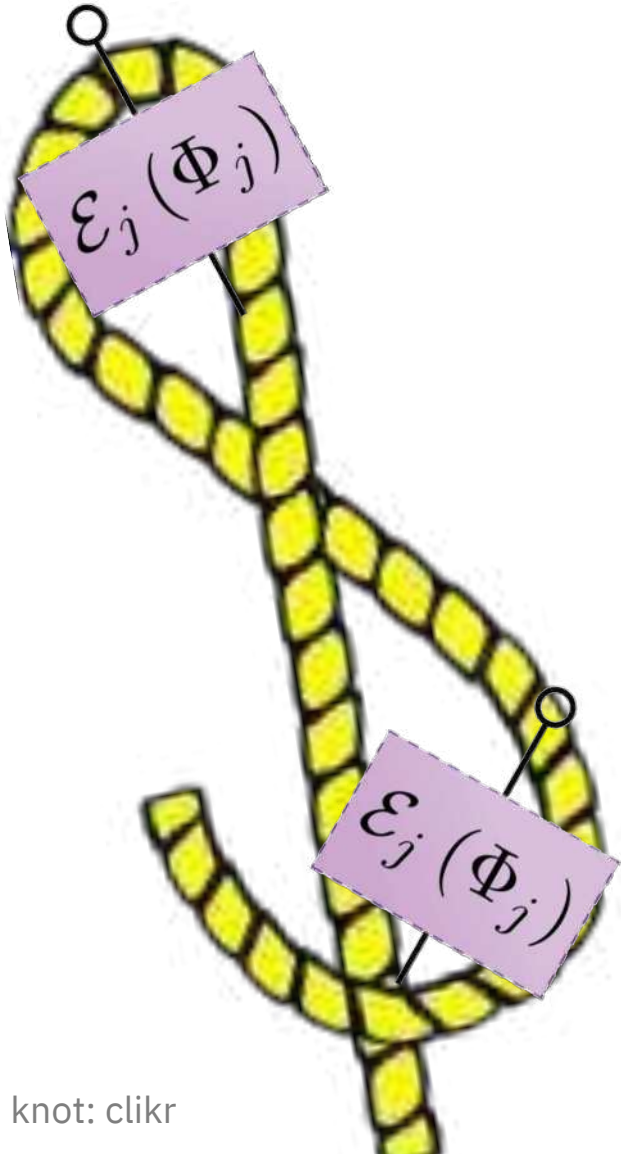
Reconstruct full Hamiltonian



Bias conditions



EPR monogamy & universal relations



I. A dipole brings
1 unit of EPR
to be diluted in the modes

$$\sum_{m=1}^M p_{mj} = 1$$

II. A mode accepts max
1 unit of EPR

$$0 \leq \sum_{j=1}^J p_{mj} \leq 1$$

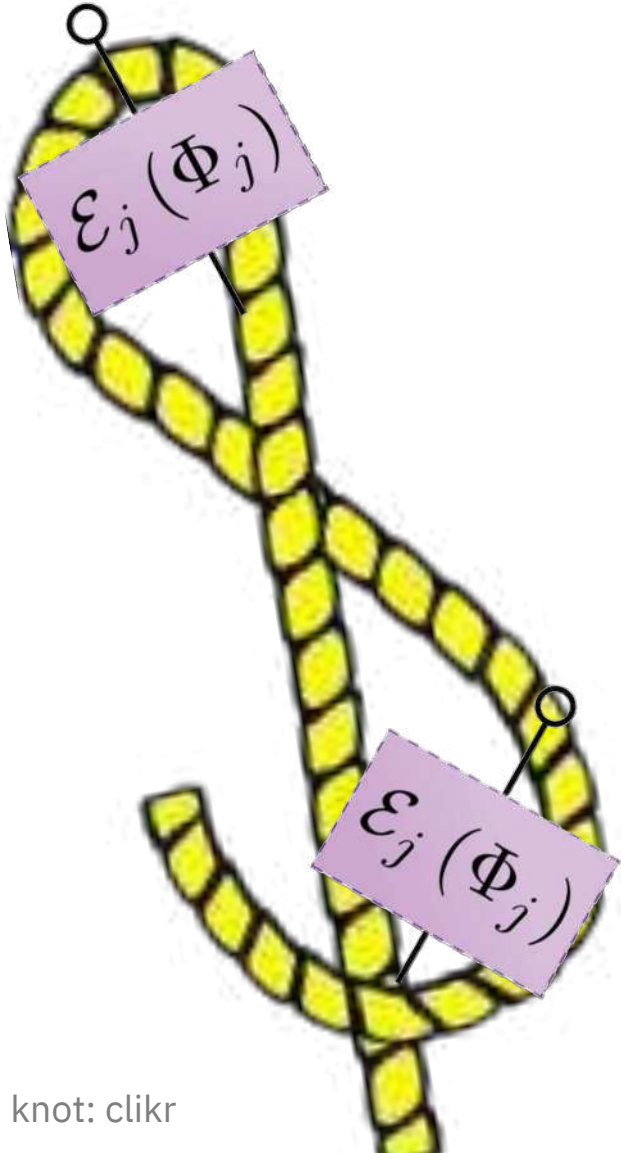
III. A dipole's EPRs are
orthogonal

$$\sum_{m=1}^M s_{mj} s_{mj'} \sqrt{p_{mj} p_{mj'}} = 0$$

* EPR is bounded

$$0 \leq p_{mj} \leq 1$$

EPR monogamy & universal relations



=>

zero-point quantum
fluctuations are
not independent

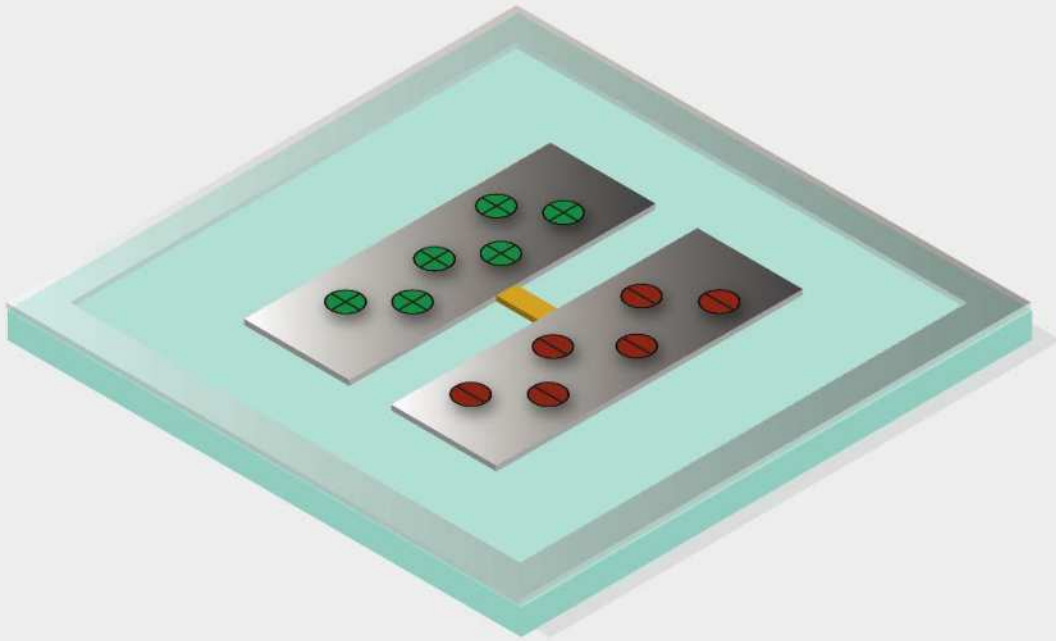
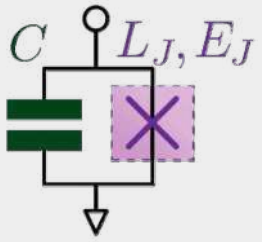
=>

design restrictions

EPR & dissipation

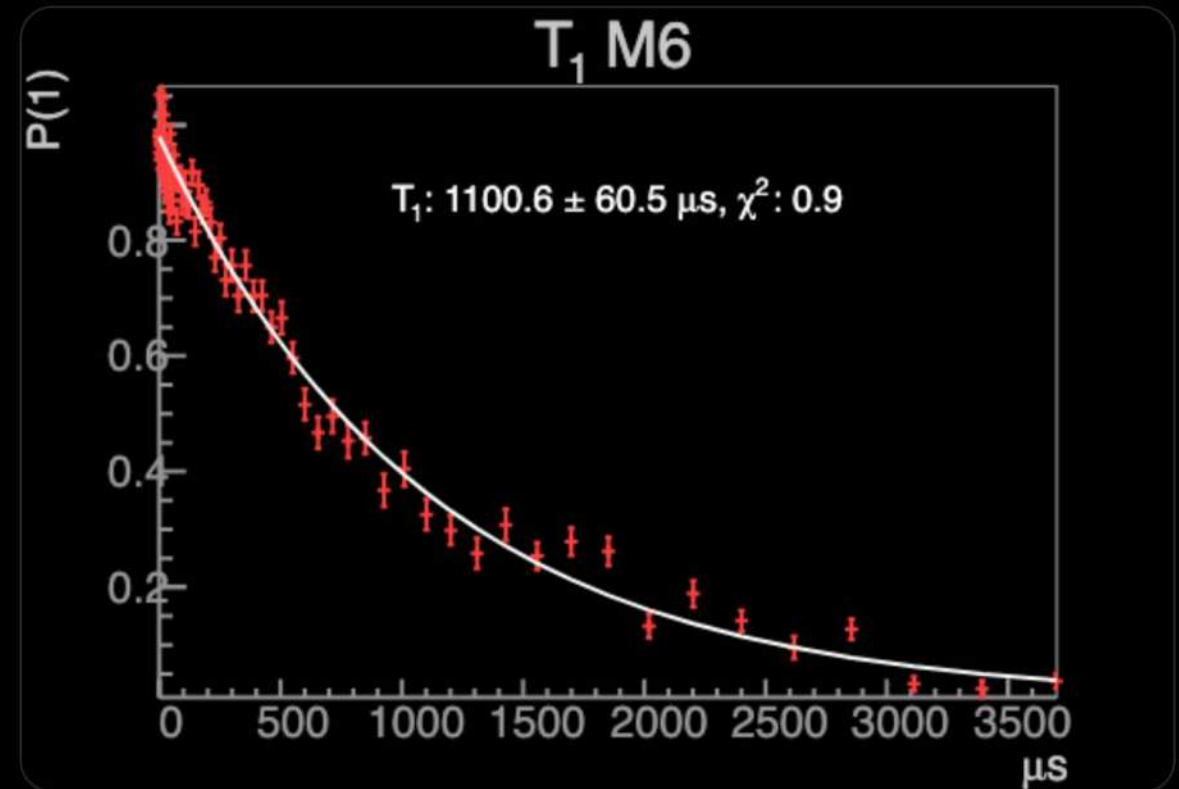
$$\mathcal{D}[\sqrt{\kappa}\hat{a}]\rho$$

Transmon coherence



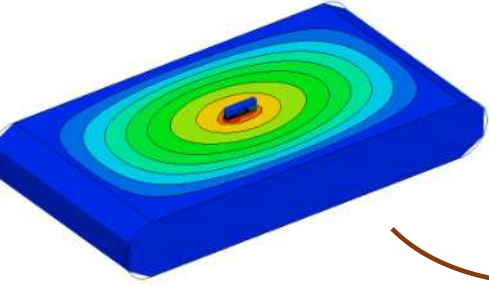
Jay Gambetta
@jaygambetta

Breaking through the millisecond barrier with our single junction transmon [@IBMResearch](#)



7:57 AM · May 20, 2021 · Twitter Web App

Lossy energy-participation ratios (EPRs)



Capacitive



bulk

$$p_{ml}^{\text{cap}} = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{1}{4} \Re \int_{V_l} \vec{E}_{\text{max}}^* \overleftrightarrow{\epsilon} \vec{E}_{\text{max}} dv ,$$

surface

$$p_{ml}^{\text{cap,surf}} = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{t_l \epsilon_l}{4} \Re \int_{\text{surf}_l} \left| \vec{E}_{\text{max}} \right|^2 ds ,$$

Inductive



bulk

$$p_{ml}^{\text{ind,surf}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 \mu_l}{4} \Re \int_{\text{surf}_l} \left| \vec{H}_{\text{max},\parallel} \right|^2 ds ,$$

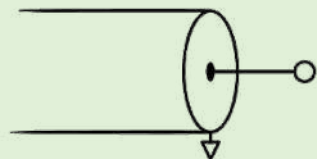
surface

$$p_{ml}^{\text{ind,bulk}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{1}{4} \Re \int_{V_l} \vec{H}_{\text{max}}^* \overleftrightarrow{\mu} \vec{H}_{\text{max}} dv .$$

seam

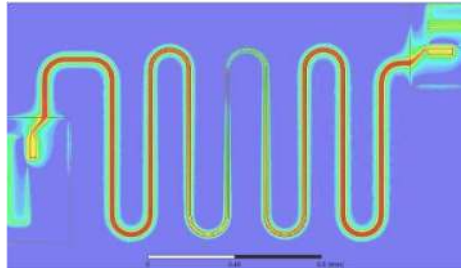
$$p_{ml}^{\text{ind,seam}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 t_l \mu_l}{4} \Re \int_{\text{seam}_l} \left| \vec{H}_{\text{max},\perp} \right|^2 dl ,$$

Radiative



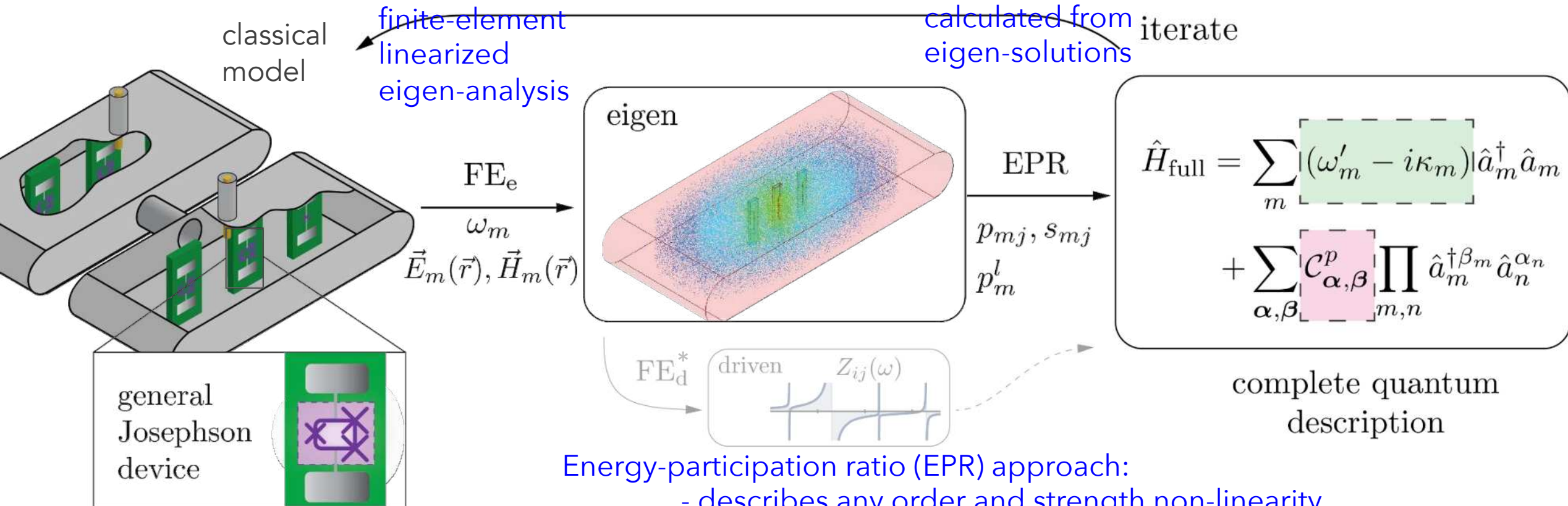
external

$$Q_{mp} = \frac{\omega_m \mathcal{E}_m(0)}{\frac{1}{2} R I_{mp}^2} .$$



Pozar (1990), Martinis *et al.* (2005), Gao (2008), Werner, *et al.* (2011), Zmuidzinas (2012), Megrant *et al.* (2012), Geerlings (2013), Quintana *et al.* (2014), Mineev *et al.* (2013), Brecht *et al.* (2015), Wang *et al.* (2015), Dial *et al.* (2016), ...

Overview of energy-participation approach



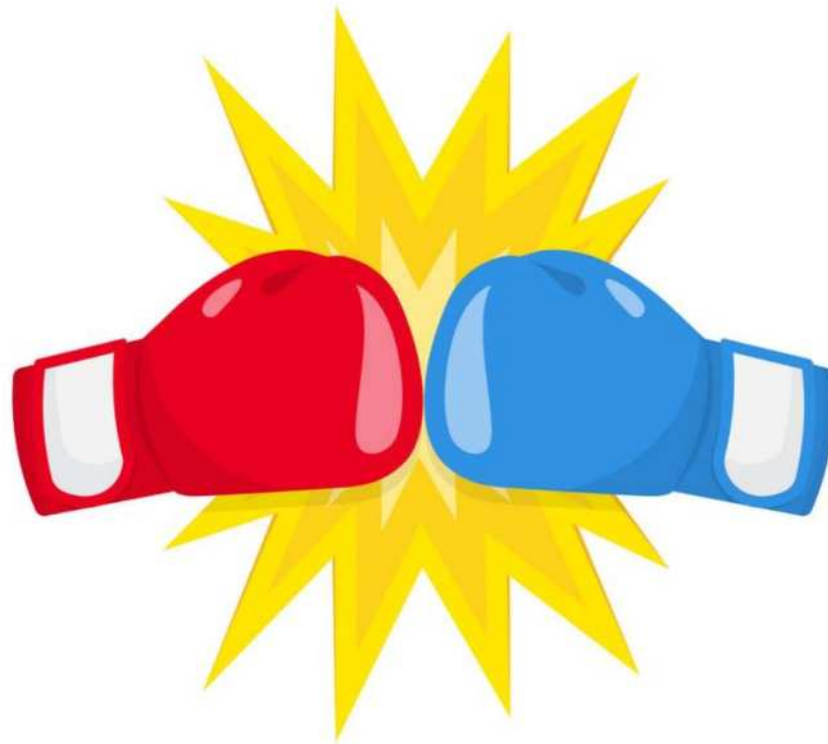
Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes) *** Jens
- fully automated in Qiskit Metal & in pyEPR

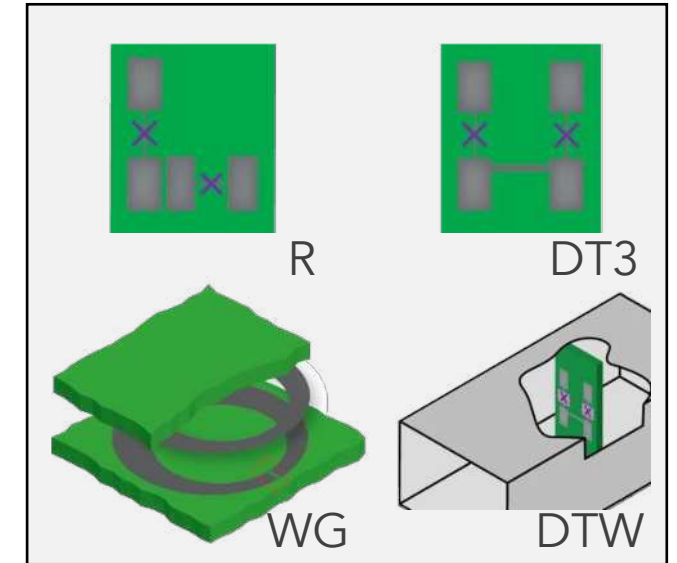
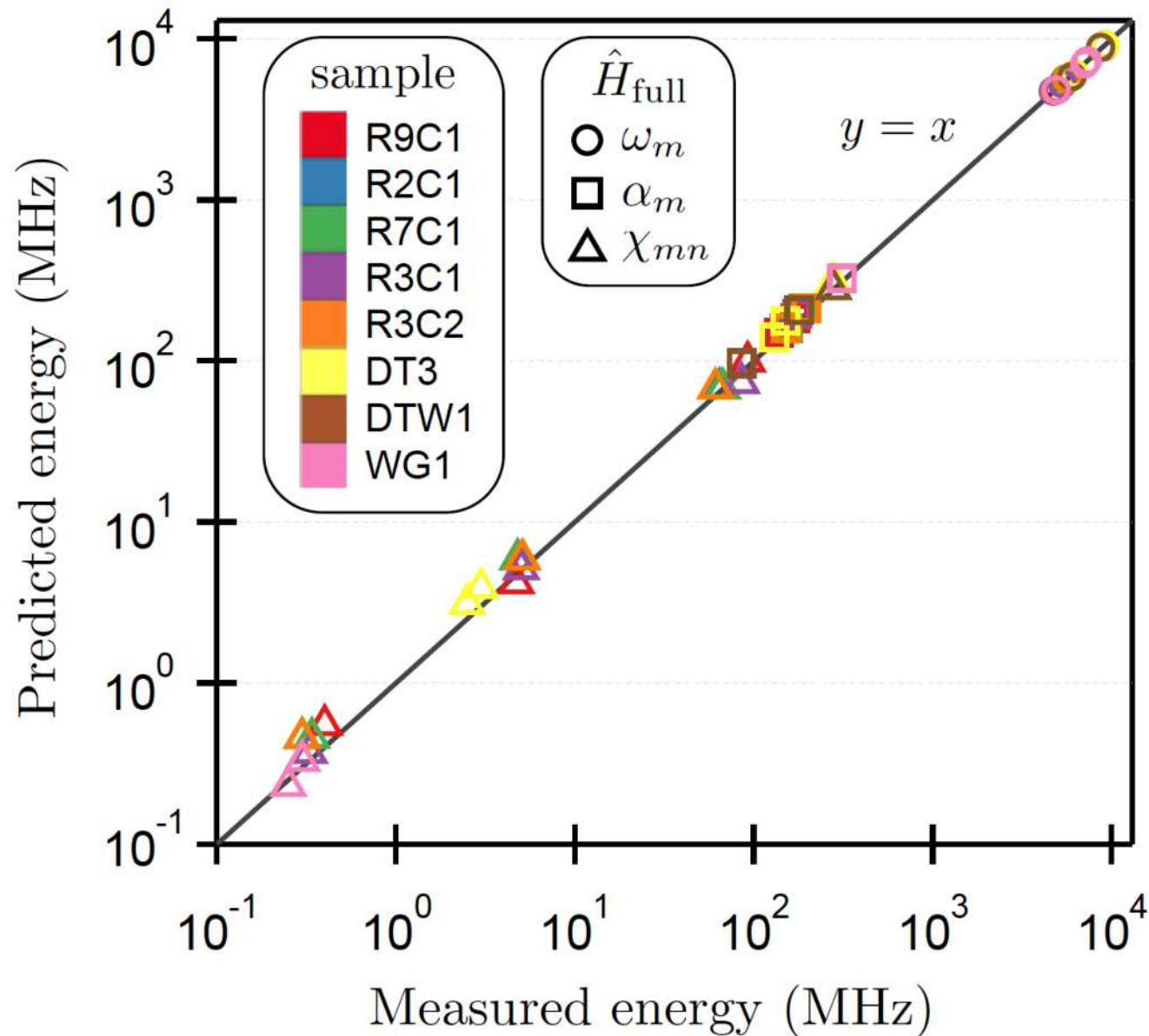
Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017), ...

EPR theory vs. experiment



Theory vs. experiment: agreement over 5 orders of magnitude



R: Minev *et al.* (2018)

WG: Minev *et al.* (2013, 2016)

DT3, DTW: Minev *et al.* (2019)

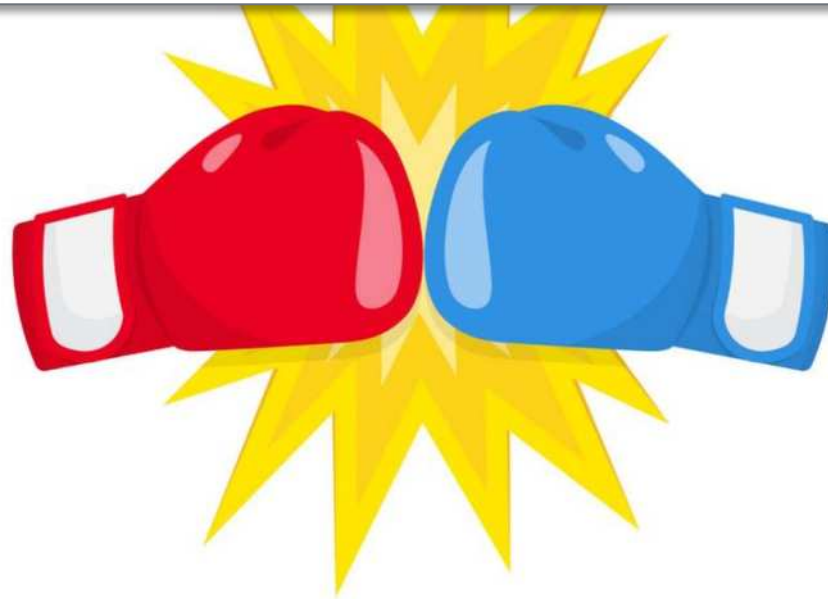
planar devices & comparison to other methods

Quantum Physics

[Submitted on 2 Feb 2021]

Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits

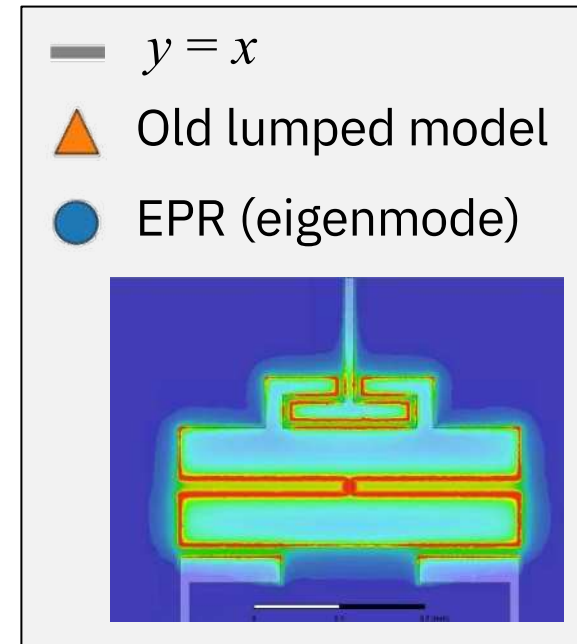
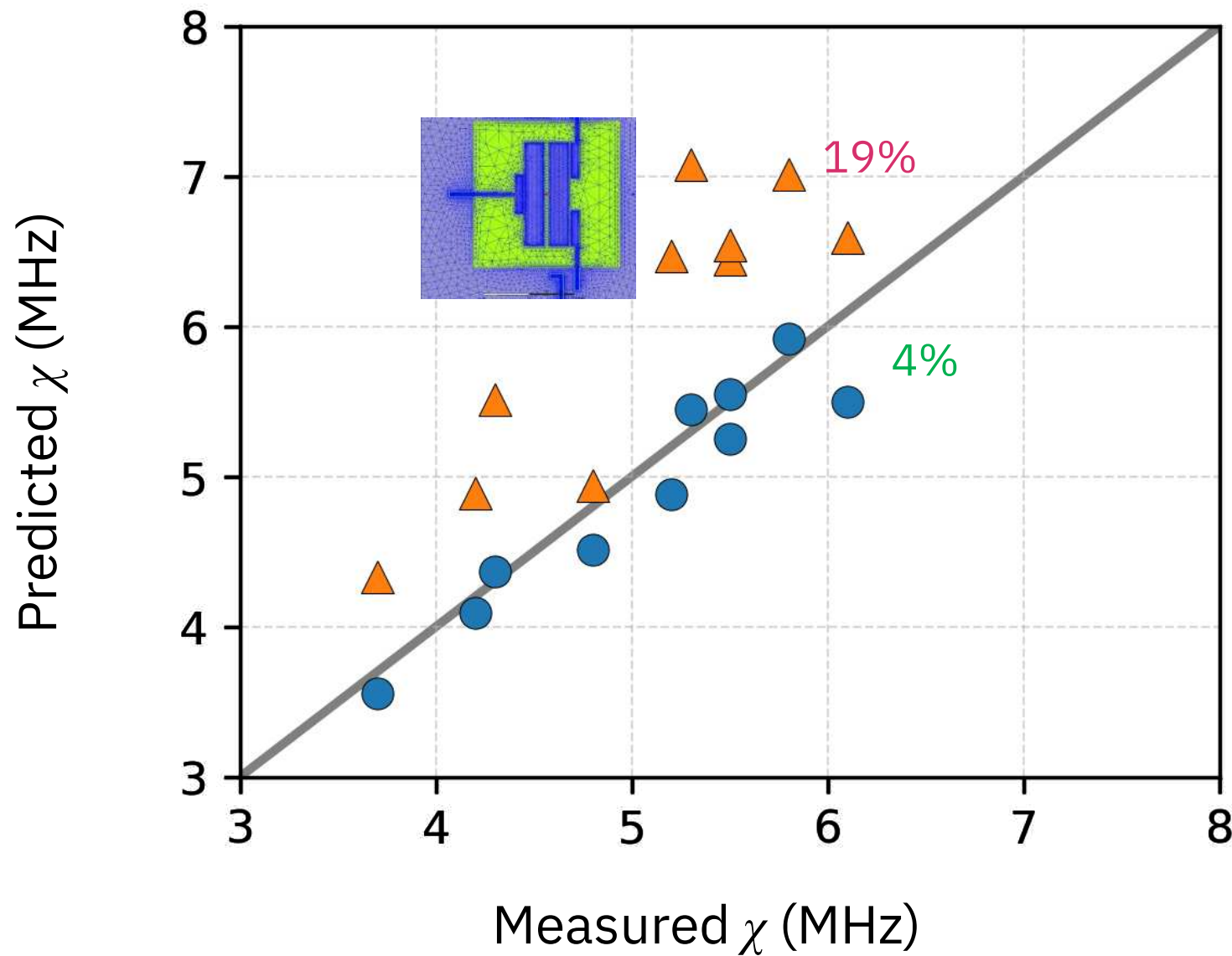
Antonio D. Corcoles, Maika Takita, Ken Inoue, Scott Lekuch, Zlatko K. Minev, Jerry M. Chow, Jay M. Gambetta



Automated with

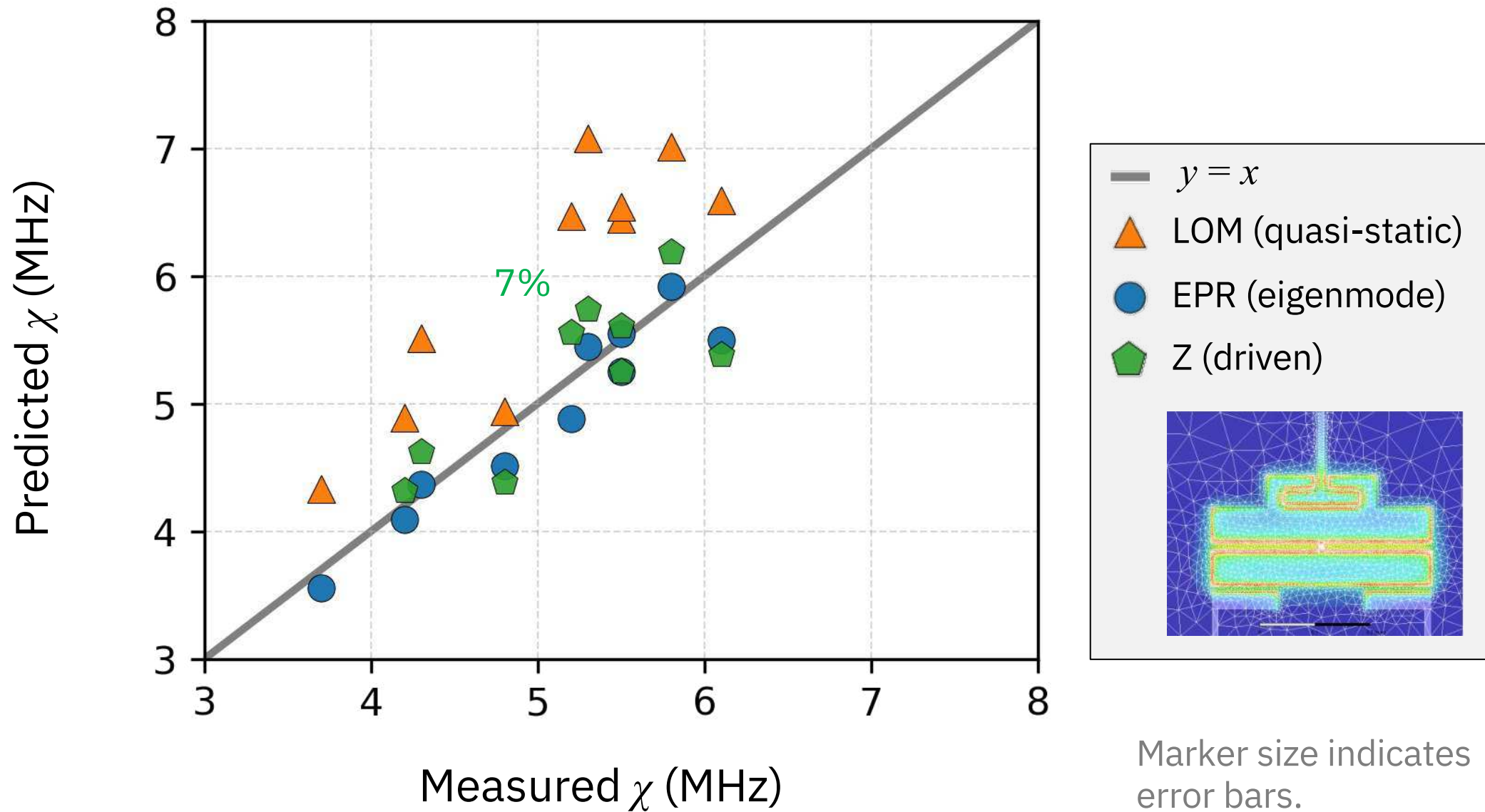
Qiskit | quantum device
design

Measured vs. predicted: qubit-readout cross-Kerr



Marker size indicates error bars.

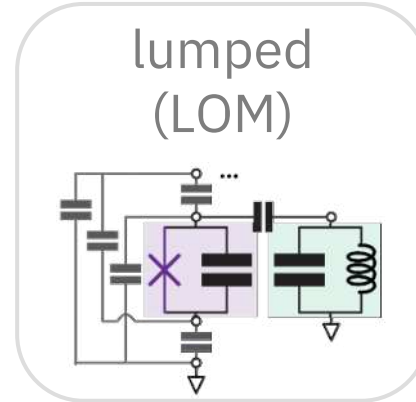
Measured vs. predicted: qubit-readout cross-Kerr



Quantization approaches



Quasi-static



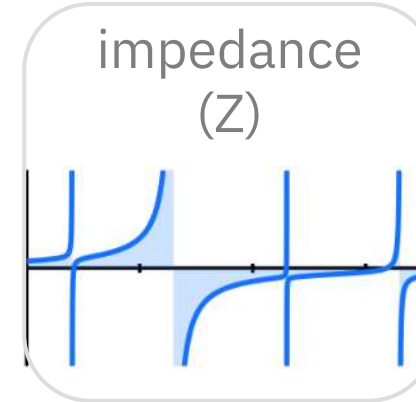
19% / 10%

higher

fastest and
cheapest

low

Full-wave

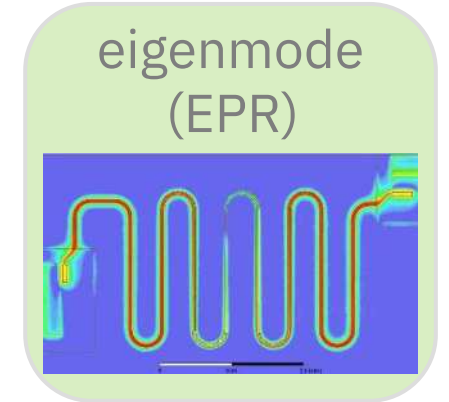


7%

low

moderate

high



4%

low

moderate

highest*

pyEPR 🍺!

Agreement
for presented results

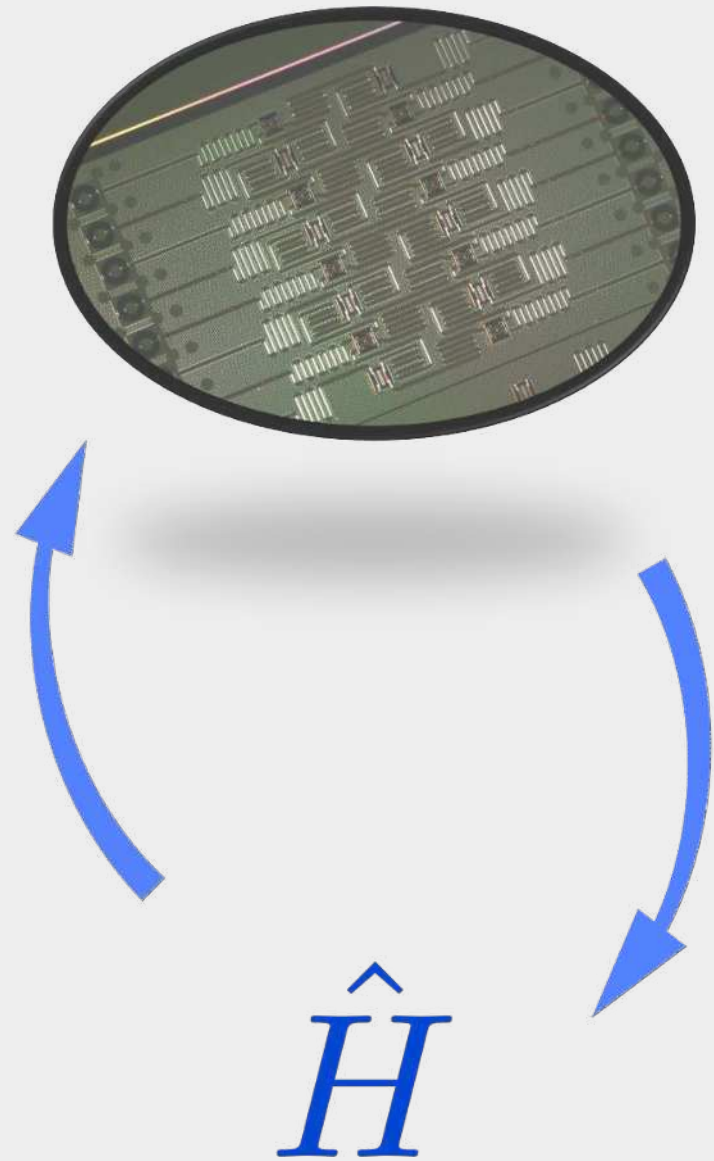
External inputs

Speed

Generality

* includes dissipative params

Physical Devices ↔ Quantum Hamiltonian

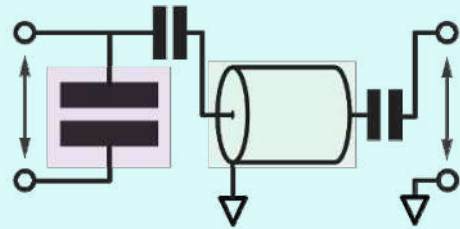


cQED with quasi-lumped models (LOM)

Minev et al. arXiv:2103.10344
(IBM)

Modular quasi-lumped models

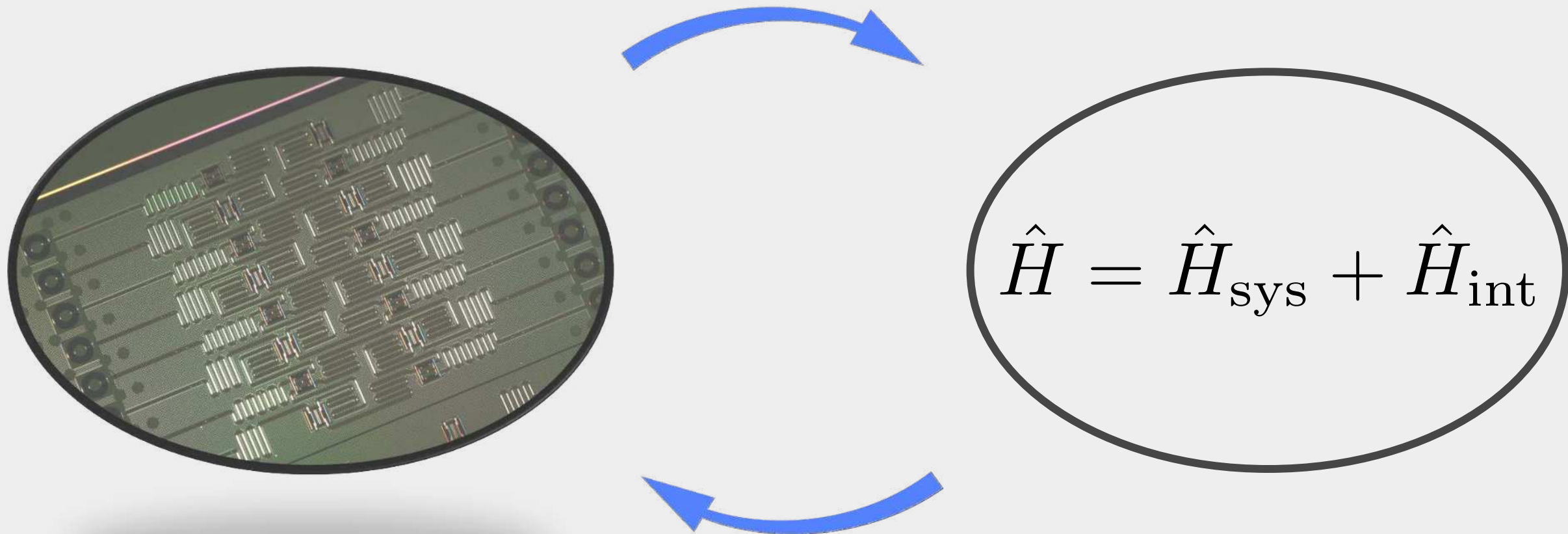
quasi-lumped



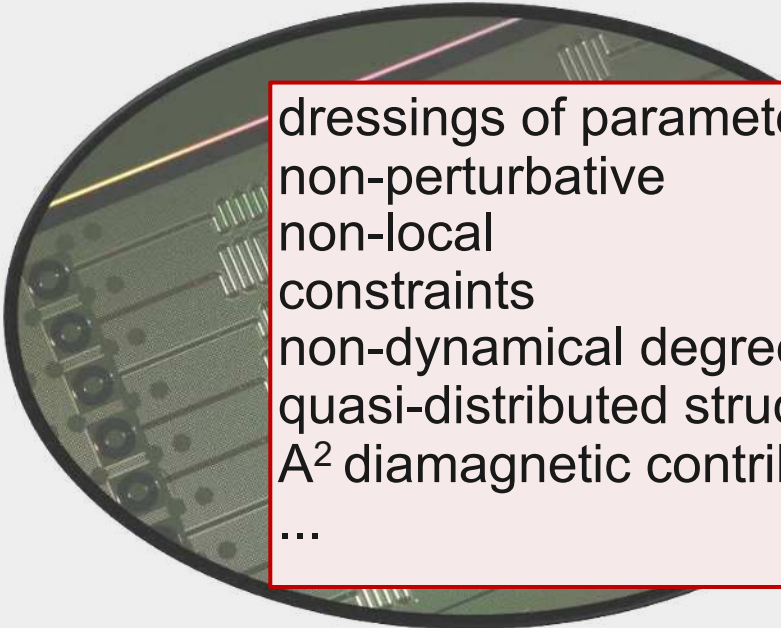
Desire:
fast
modular
precise

$\hat{H}$

Physical Hardware ↔ Modular Systems



Physical Hardware ↔ Modular Systems



dressings of parameters due to couplings
non-perturbative
non-local
constraints
non-dynamical degrees of freedom (Ding et al.)
quasi-distributed structures
 A^2 diamagnetic contributions (Malekakhlagh et al.)
...

$$\hat{H}_{\text{sys}} = \sum_n \hat{H}_n$$

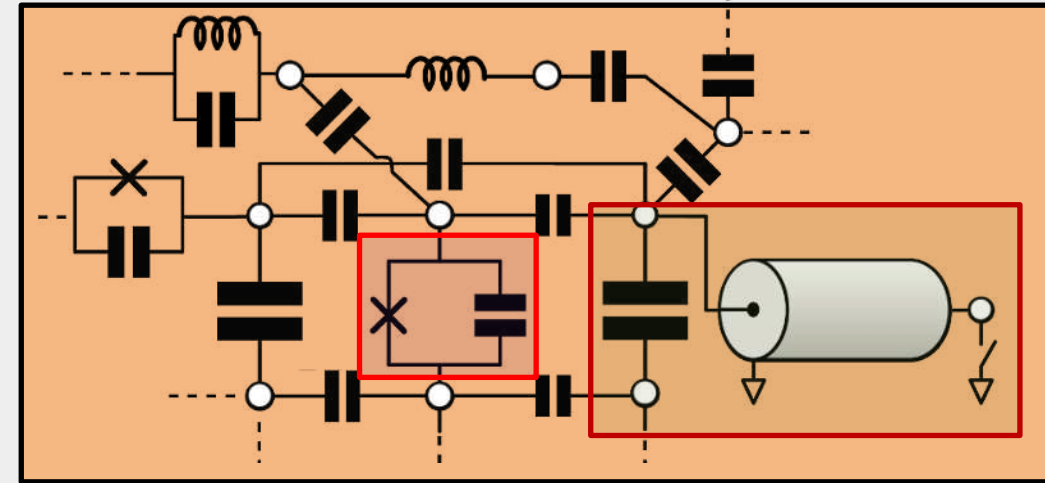
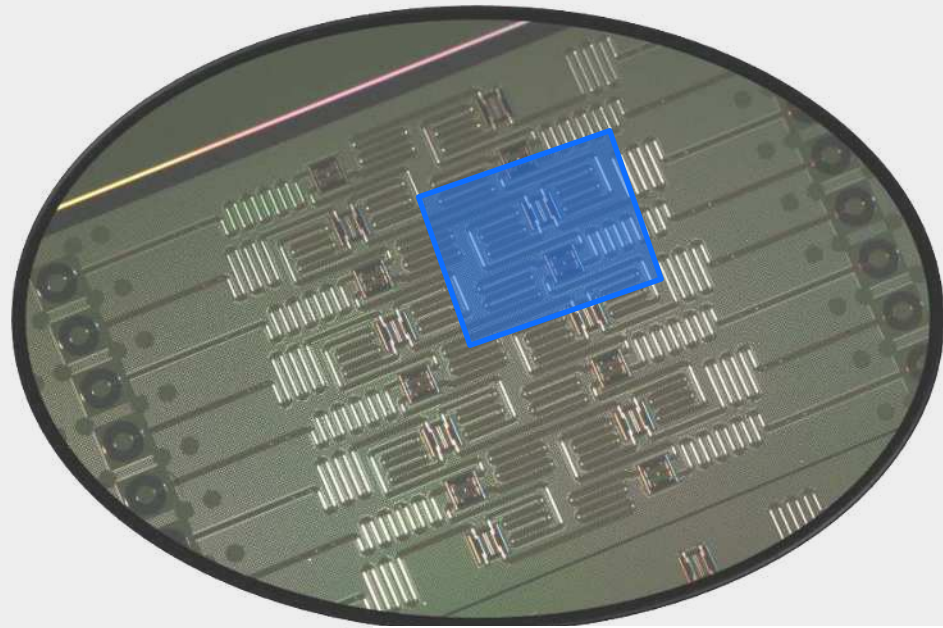
small,
known
building blocks

$$\hat{H}_{\text{int}} = \sum_n \sum_m \hbar g_{nm} \hat{A}_n \hat{B}_m$$

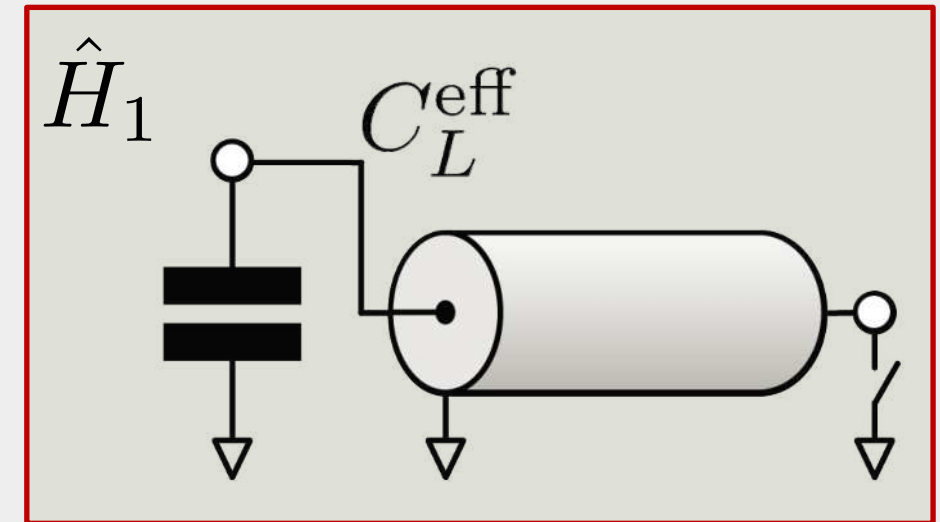
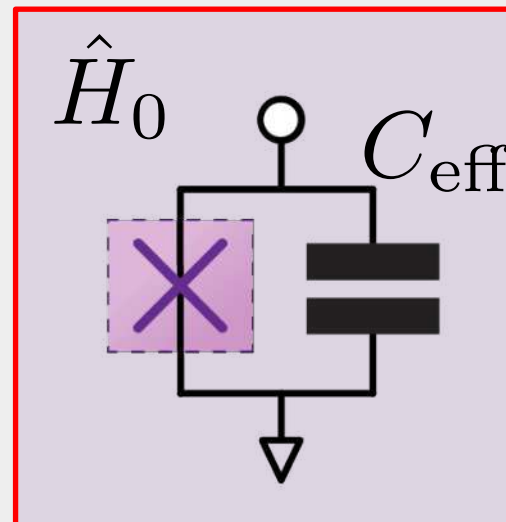
$$[\hat{H}_n, \hat{H}_m] = 0$$

$$[\hat{A}_n, \hat{B}_m] = 0$$

Method of this talk: Complex network to simple building blocks



$$\hat{H}_{01} \quad \Bigg| \quad \hat{Q}_0 \hat{Q}_1 / C_{01}^{\text{eff}} + \hat{\Phi}_0 \hat{\Phi}_1 / L_{01}^{\text{eff}}$$



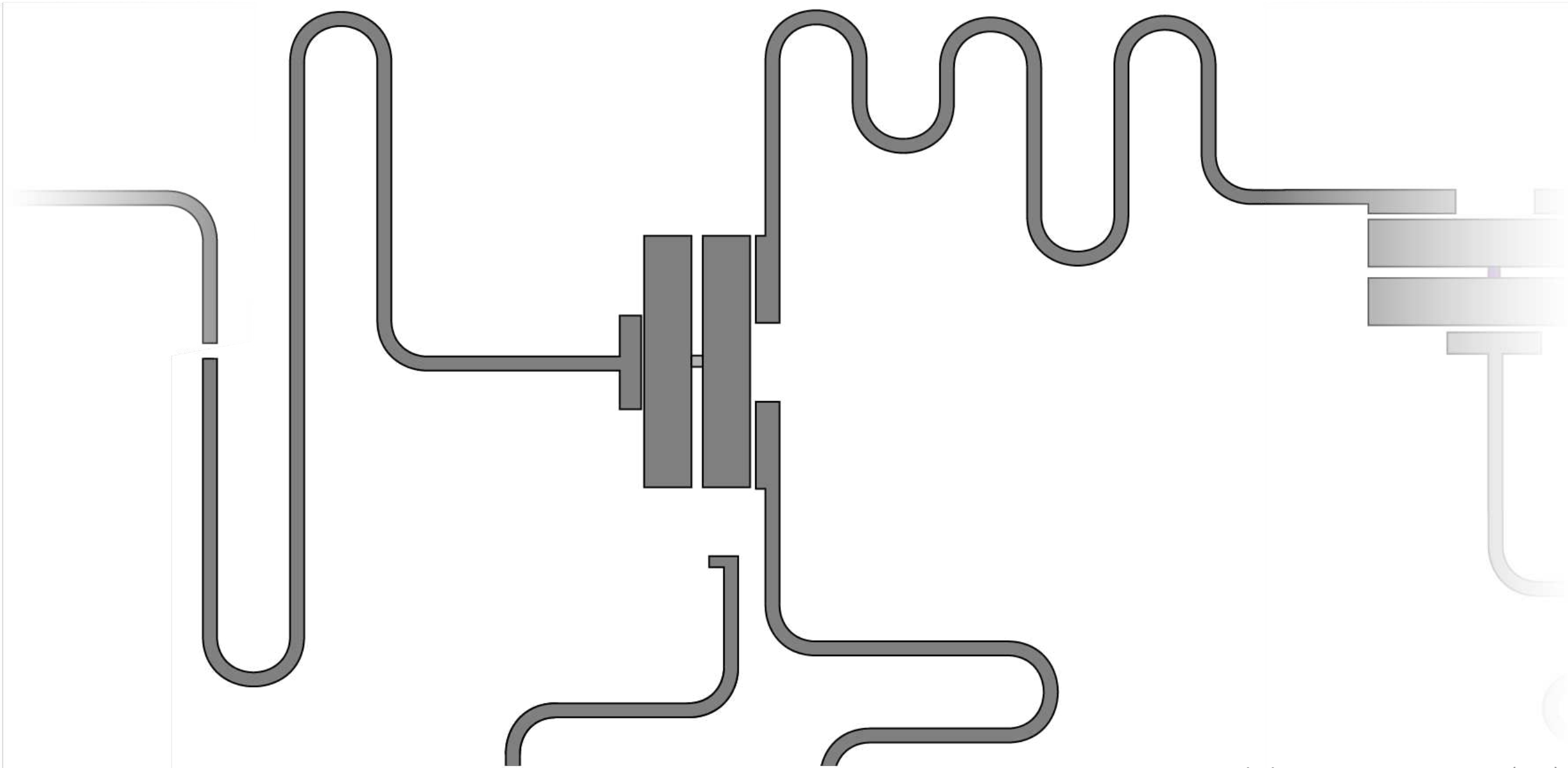
Jens Koch
Peter Groszkowski

- transmon (fixed | tunable)
- fluxonium
- 3-jct. flux qubit
- 0- π qubit
- cos(2 ϕ) qubit^{NEW}
- oscillator | Kerr oscillator^{NEW}

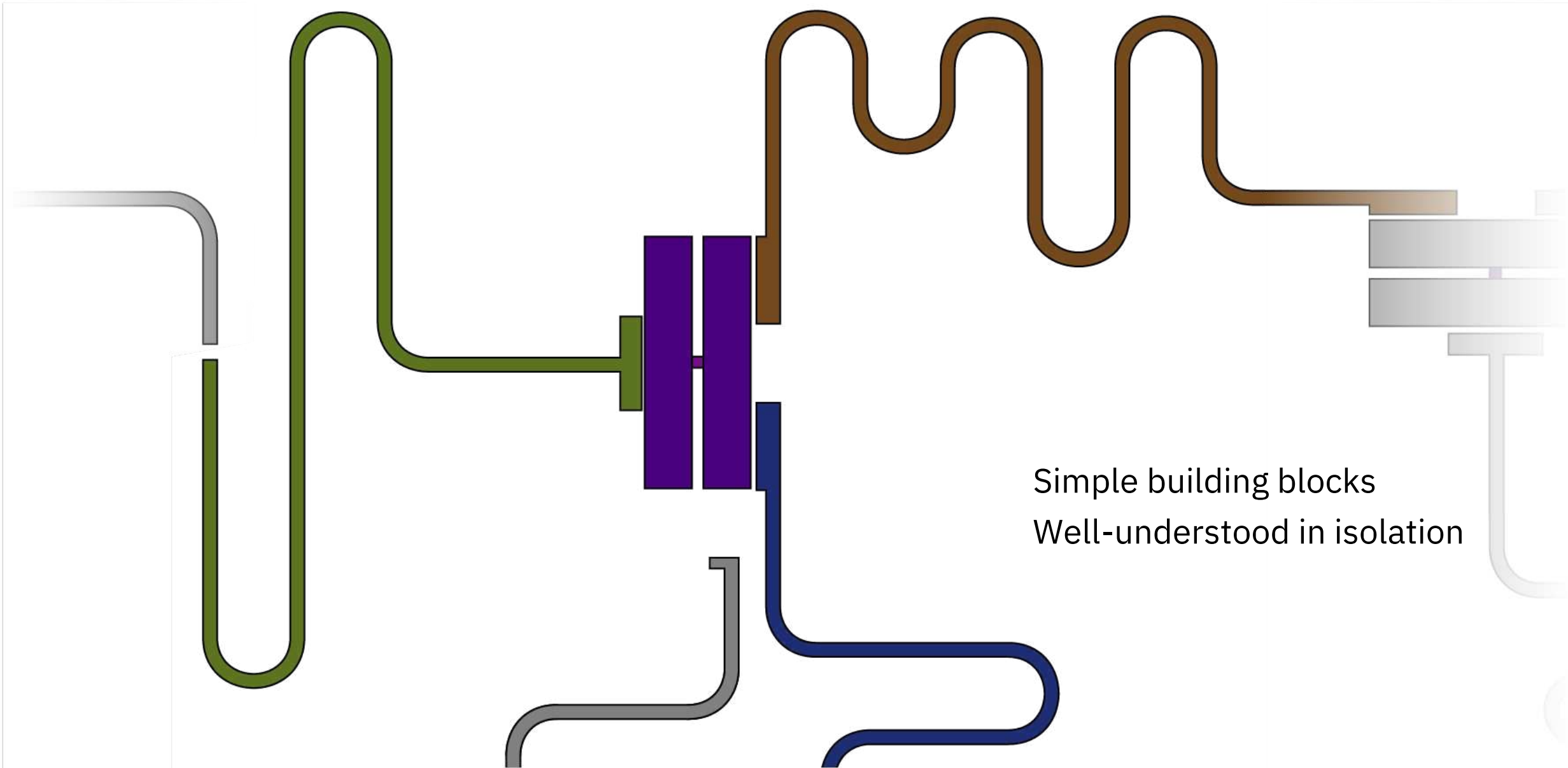
method

example

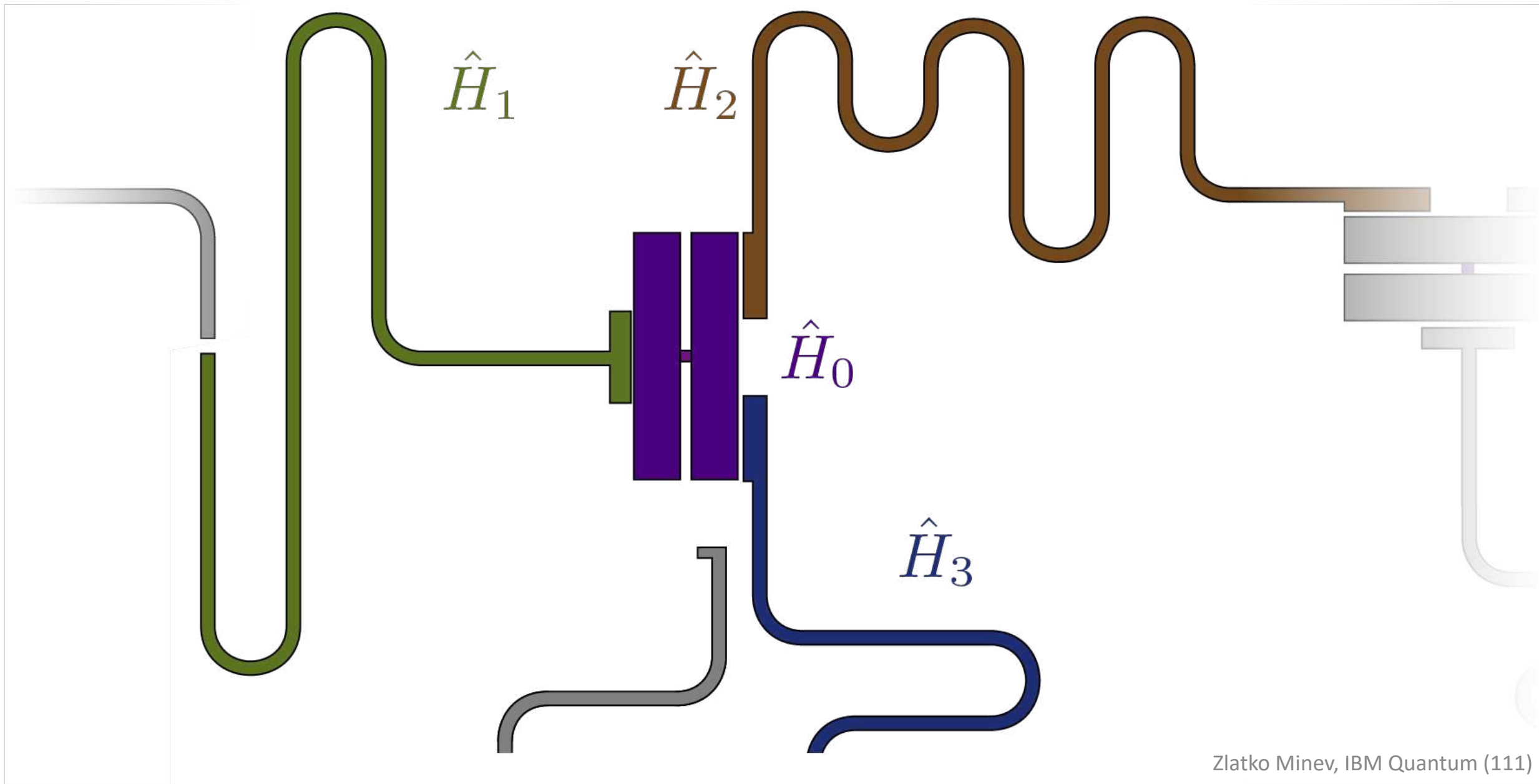
Physical layout zoom in



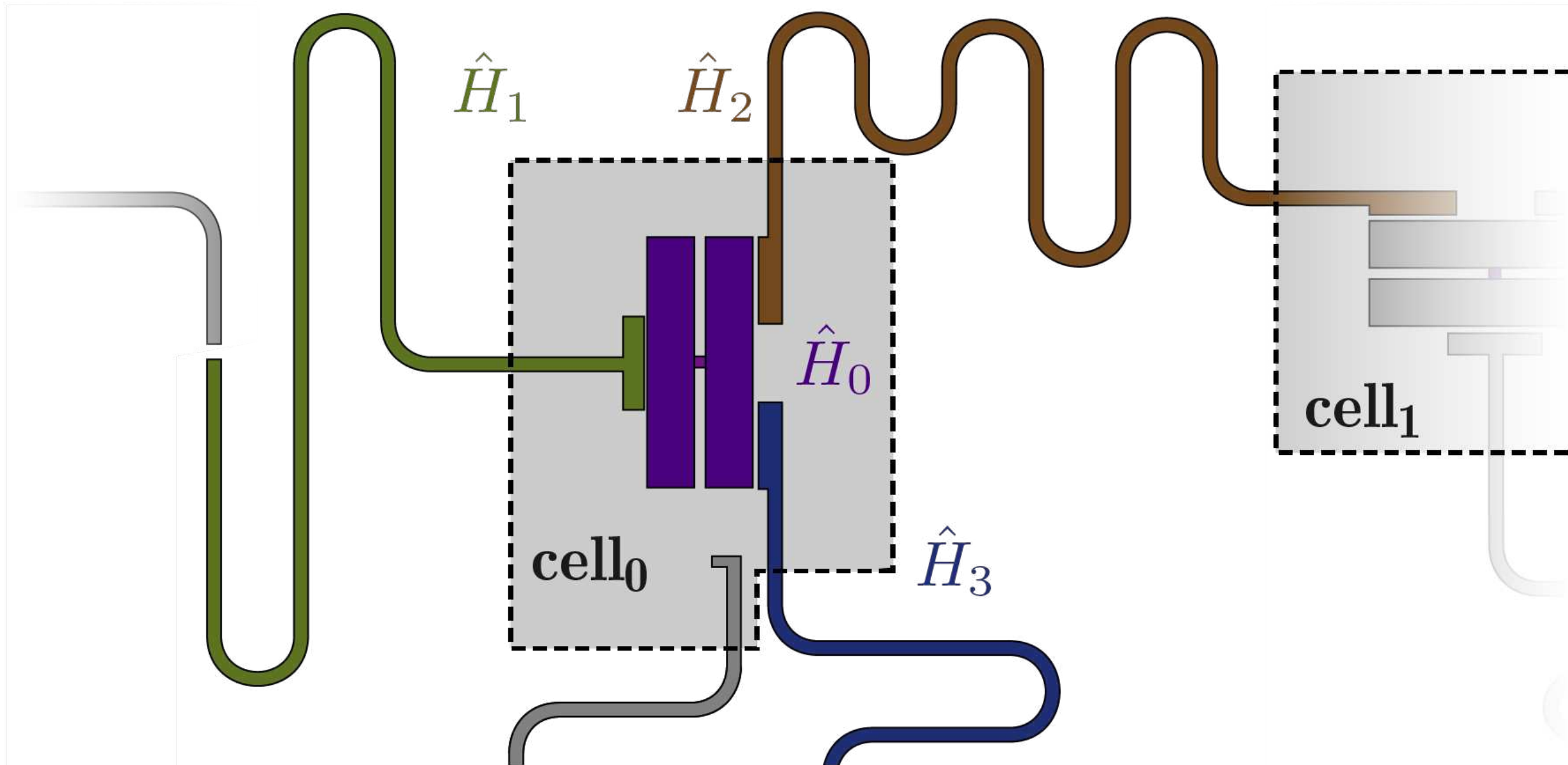
Identifying systems



Dressed subsystem Hamiltonians

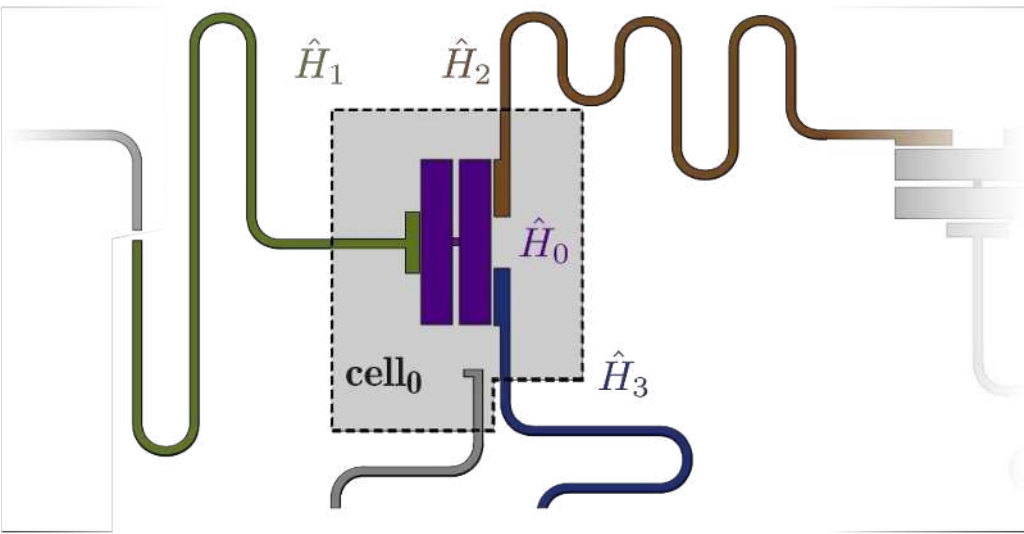


Subsystem & cell partitions



Simulating a cell

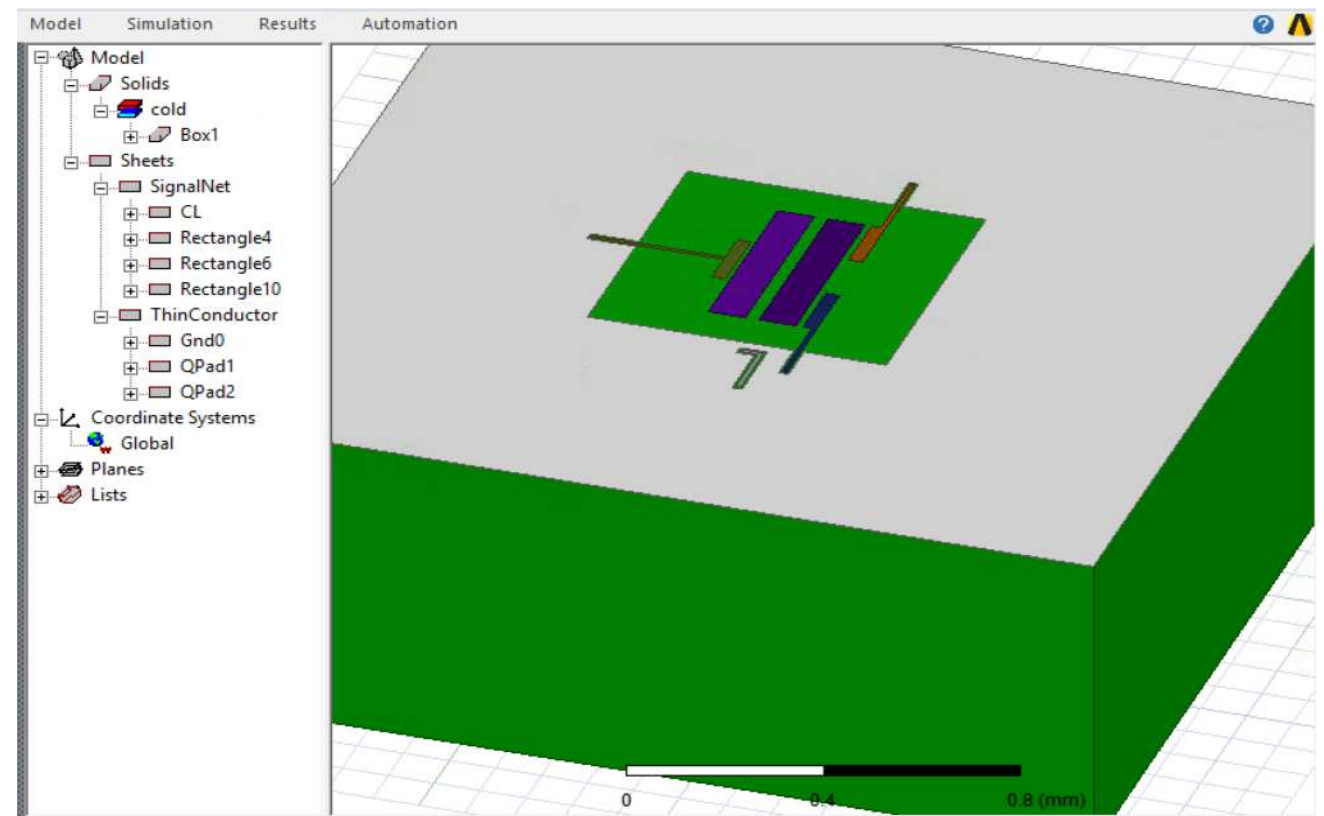
device layout



Automated with
Qiskit | quantum device
design



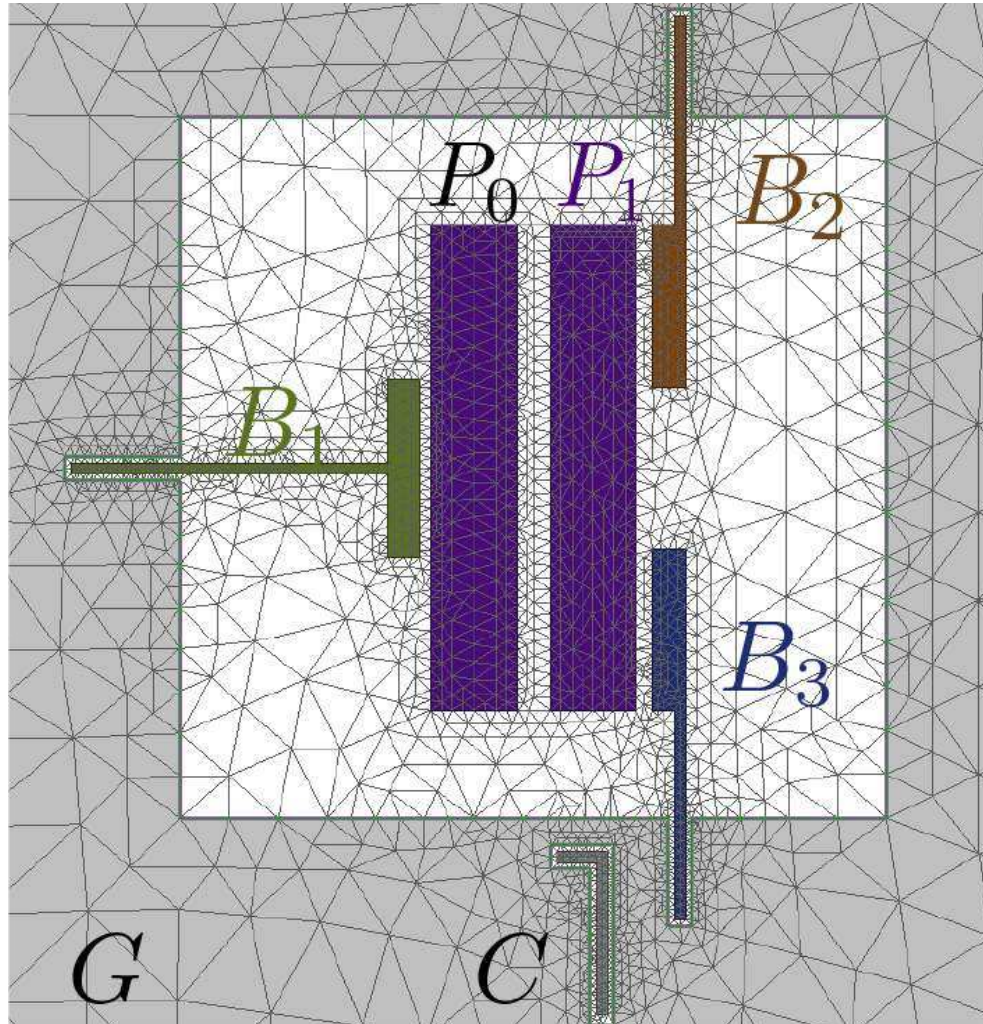
extractor model



Extracting cell parameters

netlist & simulation

Maxwell capacitance matrix



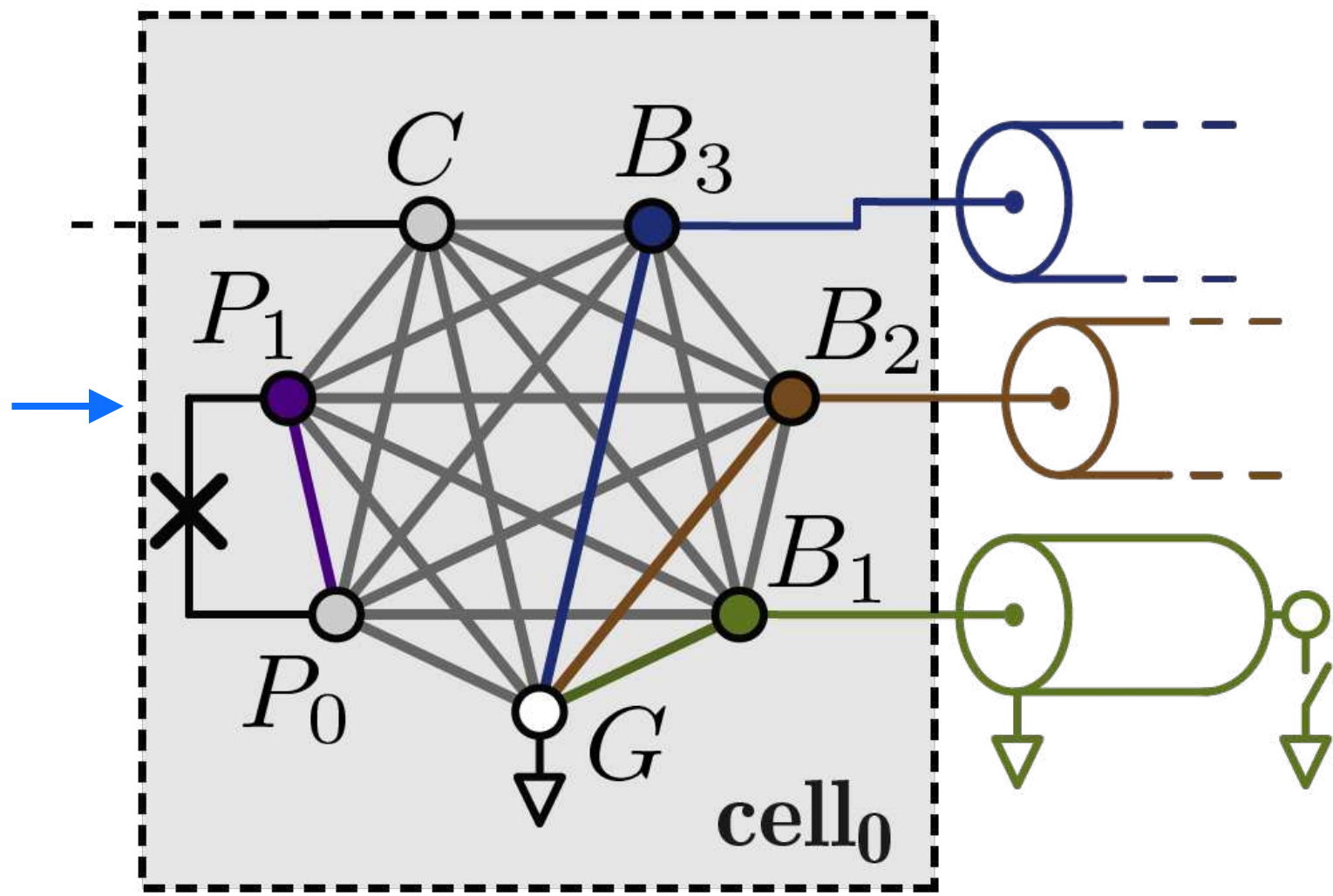
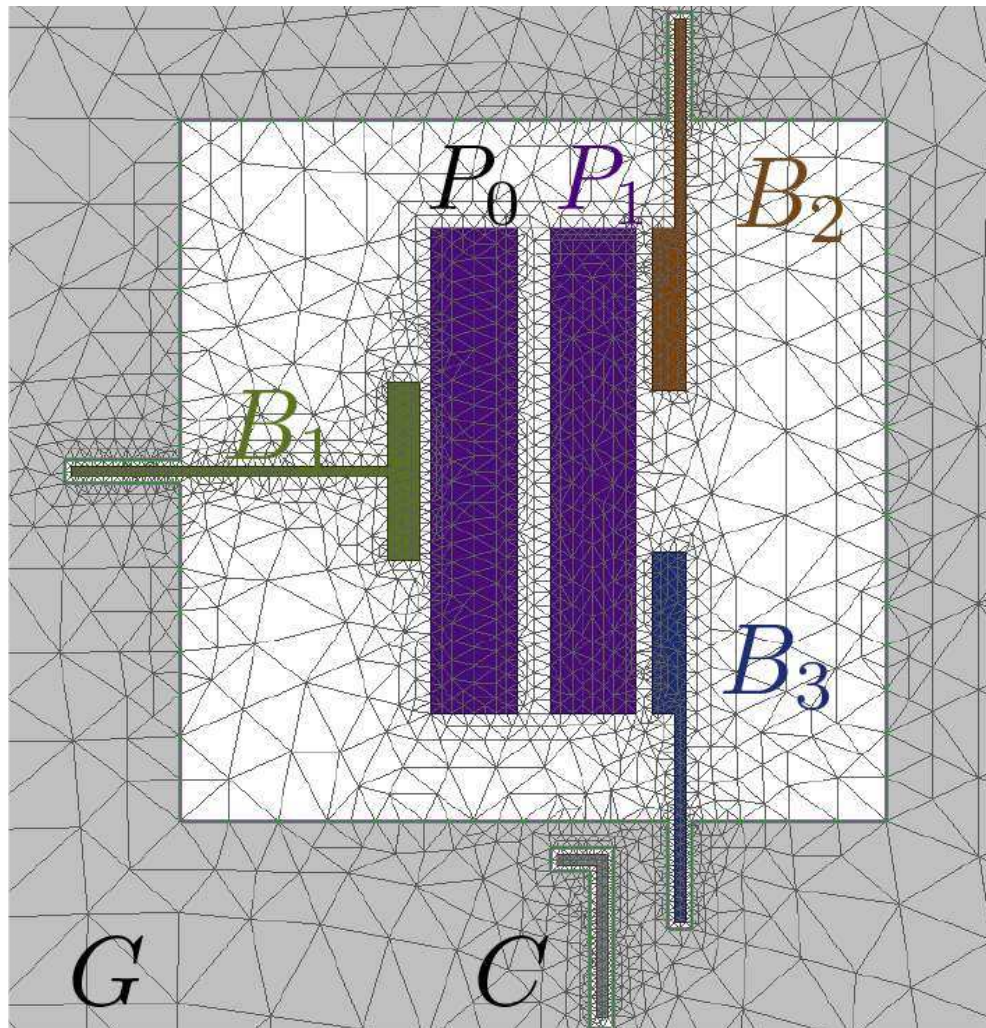
$$\begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{matrix} \begin{pmatrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix}$$

Automated with

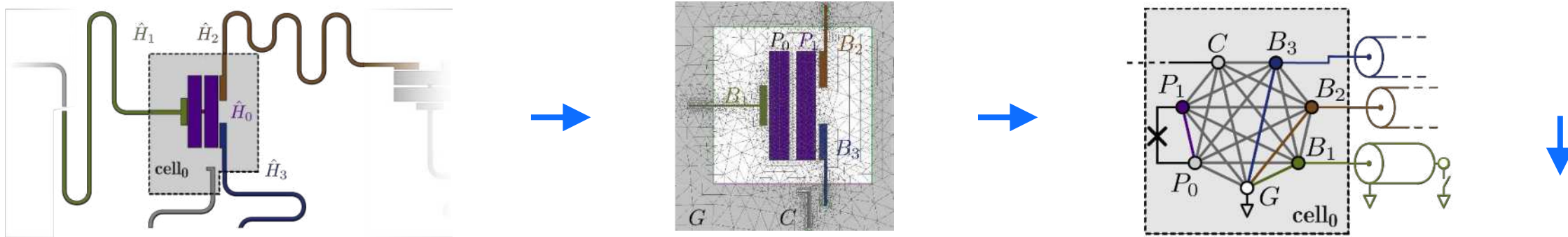
Qiskit | quantum device
design

Cell & composite-system network

full network model



Reduction to dressed subsystems



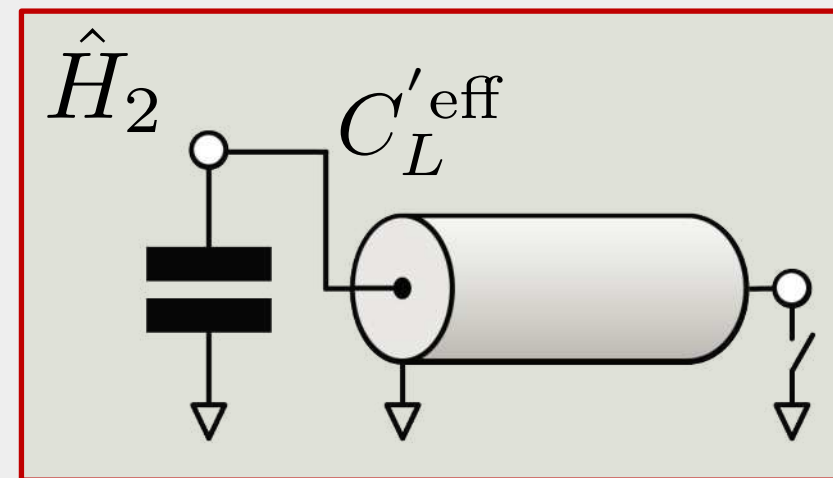
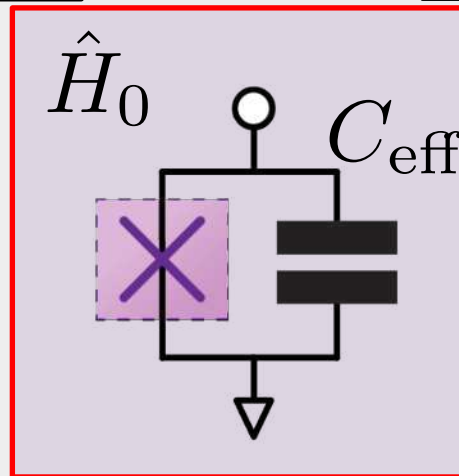
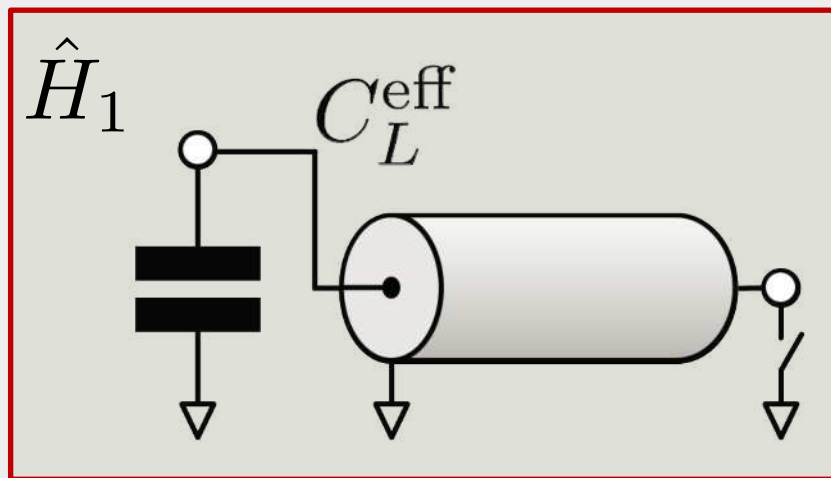
...

$$\hat{H}_{01}$$

$$\hat{H}_{12}$$

$$\hat{H}_{02}$$

...



...

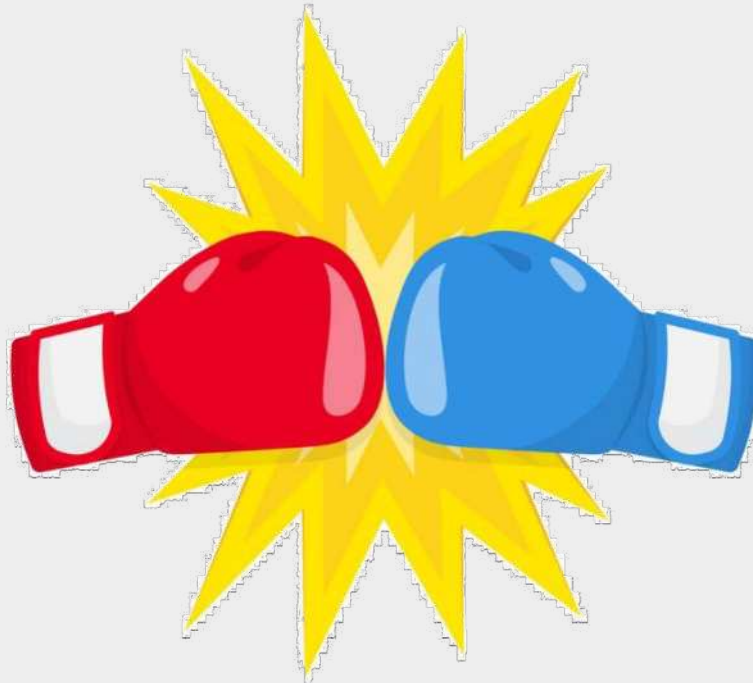
LOM theory vs. experiment

Quantum Physics

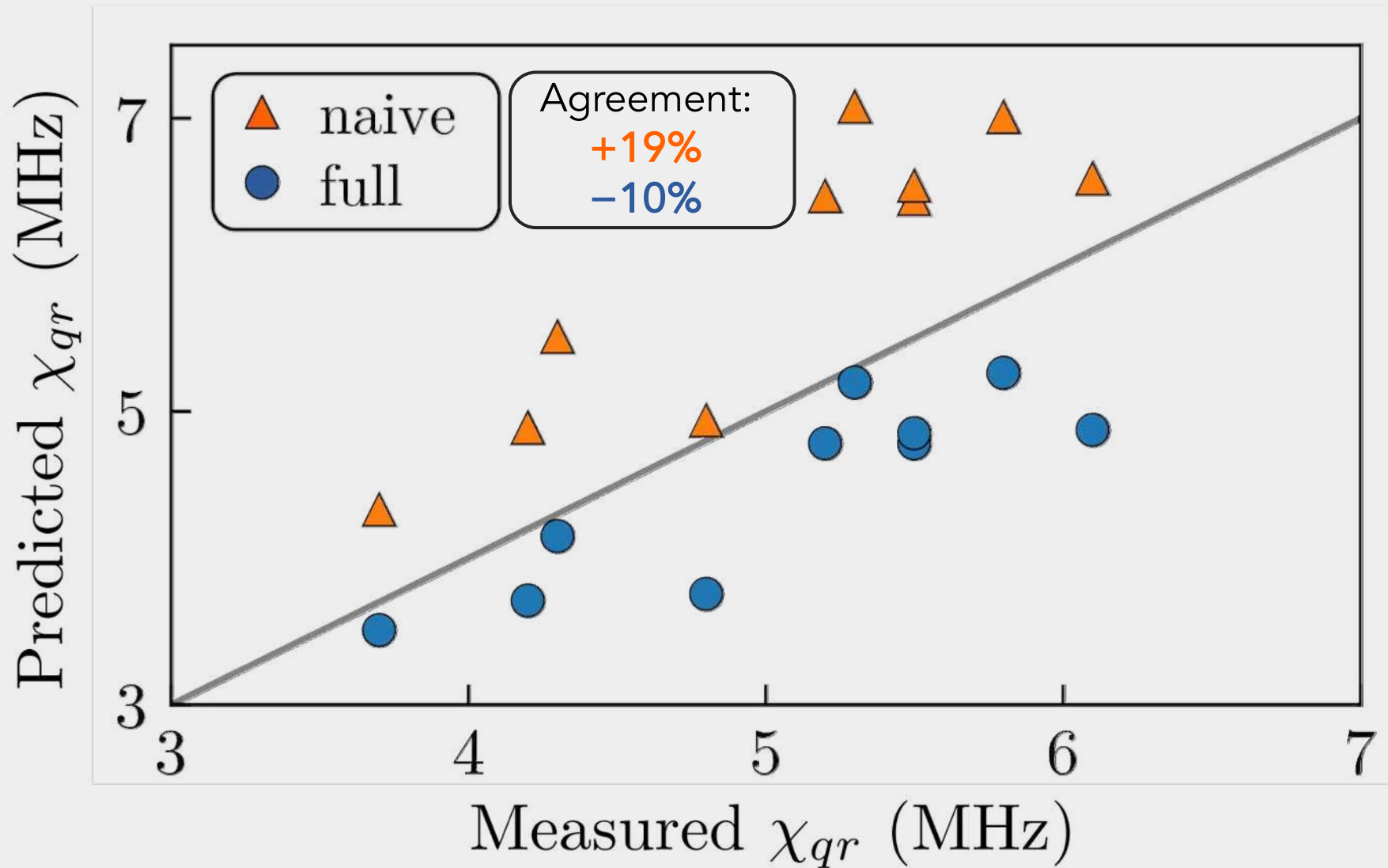
[Submitted on 2 Feb 2021]

Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits

Antonio D. Corcoles, Maika Takita, Ken Inoue, Scott Lekuch, Zlatko K. Mineev, Jerry M. Chow, Jay M. Gambetta



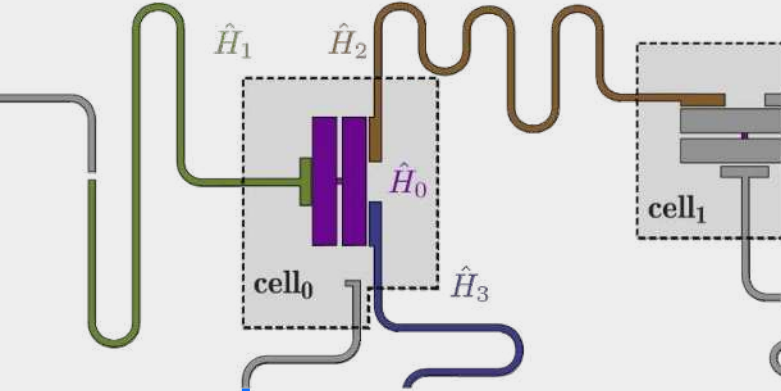
Improved precision



Data from Minev, *et al.* (to appear).

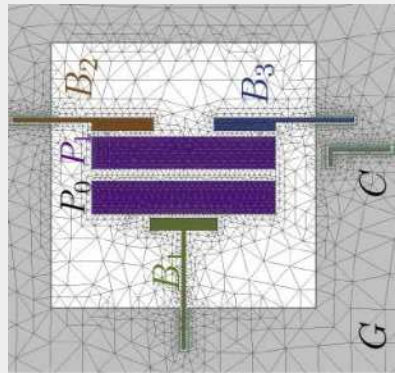
Tested on 2 larger-scale processors, see Corcoles *et al.* arXiv:2102.01682.

Summary Mineev et al. arXiv:2103.10344

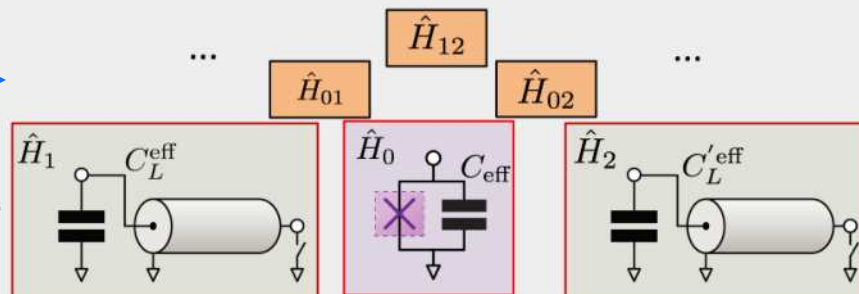
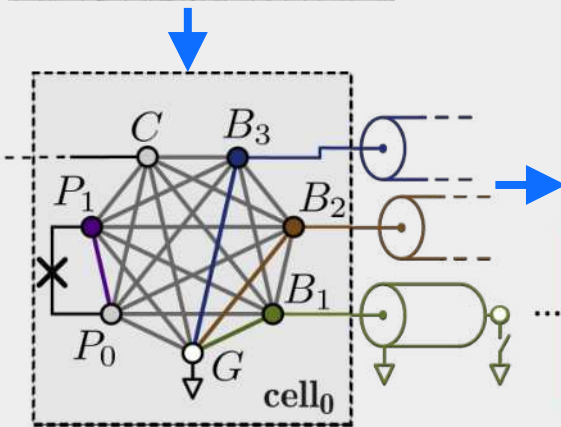


Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Mineev,* Thomas G. McConkey, Maika Takita, Antonio Corcoles, and Jay M. Gambetta
IBM Quantum, IBM T.J. Watson Research Center, Yorktown Heights, US



$$\hat{H} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$



accounts for device layout

modular

maps onto known building blocks

computationally efficient

incorporates dressings

experimentally accurate

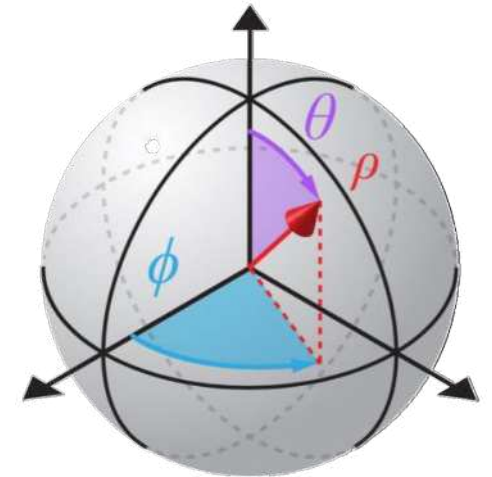
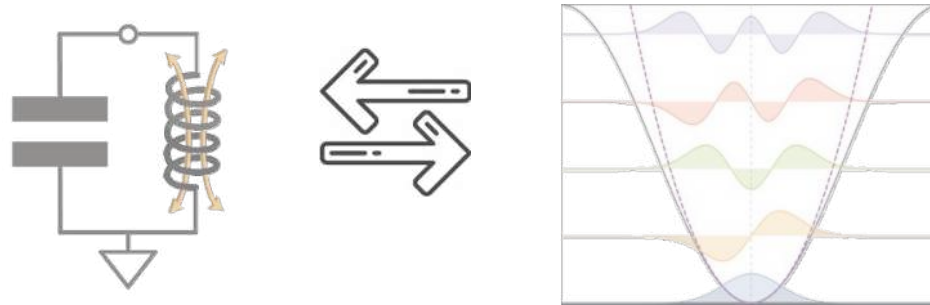
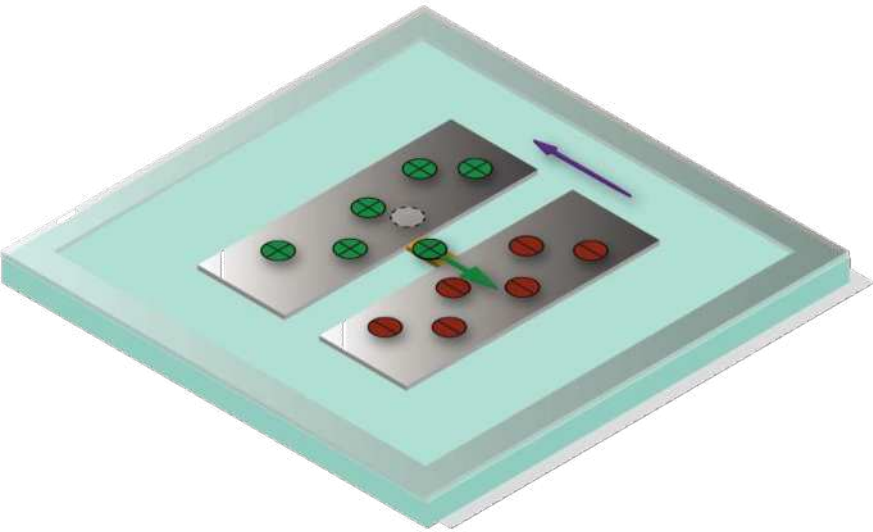


widely-applicable to planar/quasi-planar

(not suitable for 3D)

Superconducting quantum devices: design & analysis

Energy-participation-ratio (EPR)
Quasi-lumped oscillator models (LOM)
& Qiskit Metal



Zlatko K. Minev

IBM Quantum

IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev



zlatko-minev.com

Thank you

Charge dispersion

$$\omega_0 = 6.50 \text{ GHz}$$

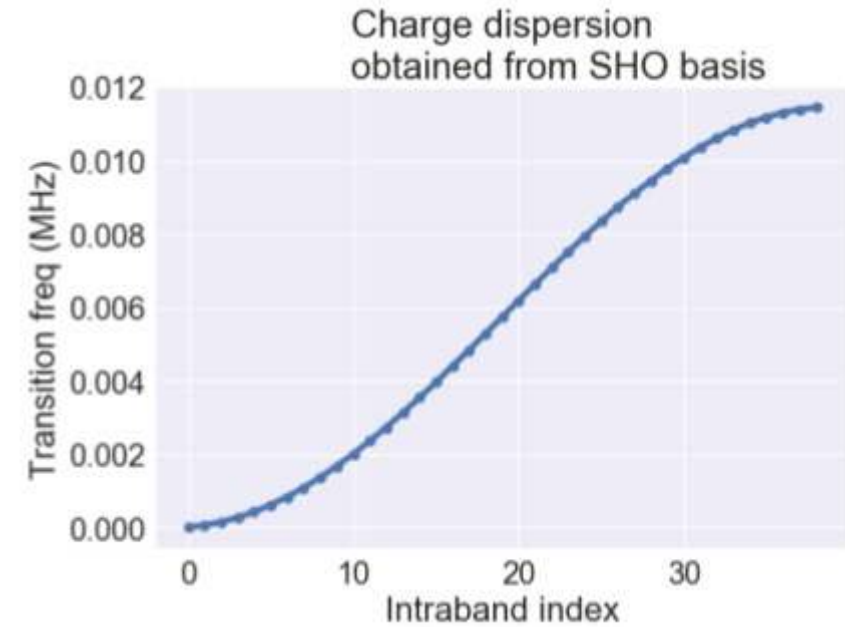
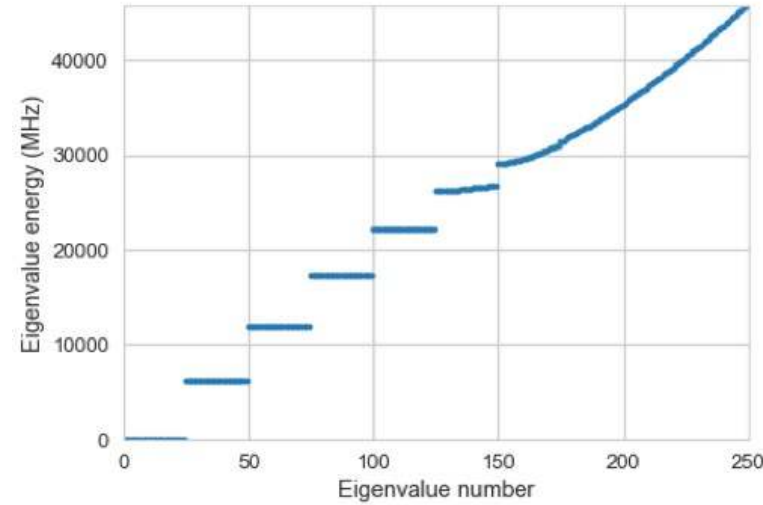
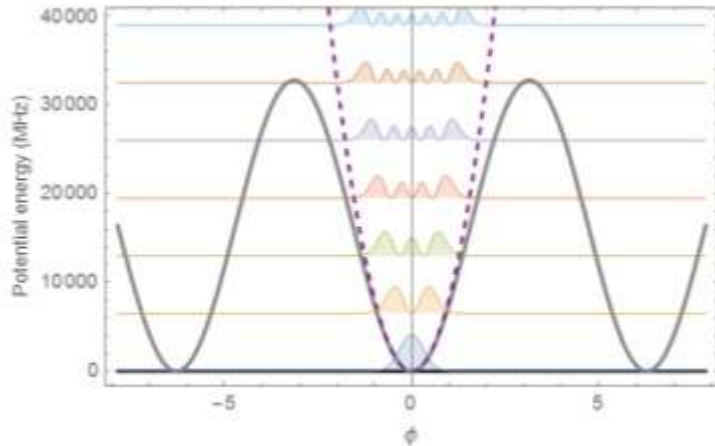
$$\alpha^{(4)} = 322.00 \text{ MHz}$$

$$\phi_{\text{ZPF}} = 0.45$$

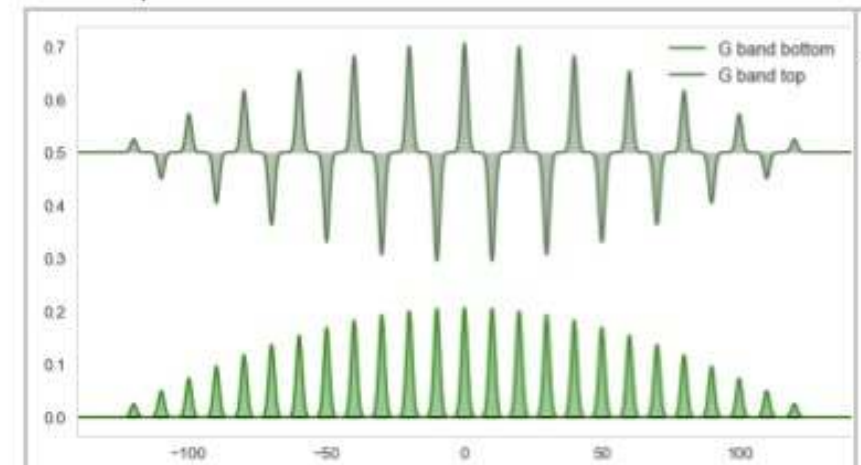
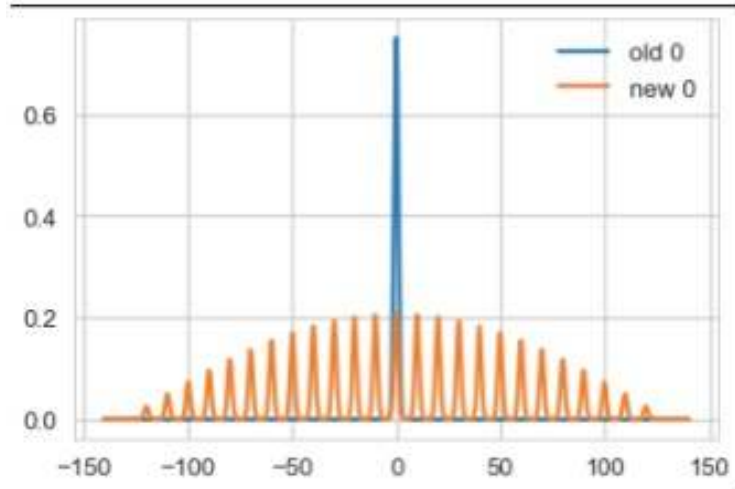
$$E_J = 16.40 \text{ GHz}$$

$$E_C = 0.32 \text{ GHz} \quad (E_J/E_C = 50.90)$$

$$\text{Dispersion} = -12.90 \text{ KHz}$$



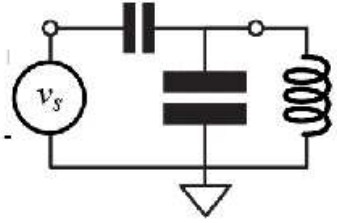
$|\psi|^2$



compare to charge basis numeric

$\backslash \phi$

a) external voltage bias



The Hamiltonian of a transmon qubits subject to a voltage bias, coupled by a coupling capacitance C_c and shunted by a capacitance C_s , is

$$H(\Phi, Q) = \frac{(Q + C_c \dot{\Phi}_s(t))^2}{2(C_s + C_c)} - \frac{C_c \dot{\Phi}_s(t)^2}{2} - E_J \cos(\Phi/\phi_0), \quad (10.1)$$

where the canonically conjugate flux and charge variables are Φ and Q , resp, such that their Poisson bracket is $\{\Phi, Q\}_p = 1$, E_J is the Josephson junction tunneling energy. and $\dot{\Phi}_s(t)$ is the voltage source flux bias, determined by the voltage source. The voltage of the source is $v_g = \dot{\Phi}_s(t)$, and the 'offset charge' is hence $Q_g := C_c v_g$, or in terms of cooper pairs, in units of $2e$, $n_g := Q_g/(2e)$; for consistence with other literature, we will choose the orientation such that there is a minus sign to the offset charge. Letting the total effective transmon capacitance be $C := C_s + C_c$,

$$\hat{H} = \frac{(\hat{Q} - Q_g)^2}{2C} - \frac{Q_g^2}{2C_c} - E_J \cos(\Phi/\phi_0). \quad (10.2)$$

Note that there are now two partially counterbalancing constant terms in the Hamiltonian; however, the terms can vary in time.

Suppose that we fix the offset charge somehow and re-write the Hamiltonian in terms of a and $a^\dagger$. Define:

$$\begin{aligned} \hat{\varphi} &:= \frac{\hat{\Phi}}{\phi_0}, & \hat{n} &:= \frac{\hat{Q}}{2e}, & n_g &:= \frac{Q_g}{2e}, \\ E_J &:= \frac{\phi_0^2}{L}, & E_C &:= \frac{e^2}{2C}, \\ Z_0 &:= \sqrt{L/C}, & \omega_0 &= 1/\sqrt{LC}, \\ \Phi_{\text{ZPF}} &:= \sqrt{\frac{\hbar}{2}} Z_0, & Q_{\text{ZPF}} &:= \sqrt{\frac{\hbar}{2}} Z_0^{-1}, \\ \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}), & \hat{\varphi}_{\text{ZPF}} &:= \Phi_{\text{ZPF}}/\phi_0, \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}), & \hat{n}_{\text{ZPF}} &:= Q_{\text{ZPF}}/(2e). \end{aligned}$$

In terms of these

$$\begin{aligned} \hat{H} &= \frac{(\hat{Q} - Q_g)^2}{2C} - \frac{Q_g^2}{2C_c} - E_J \cos(\Phi/\phi_0) \\ &= \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}} + \hat{H}_{\text{lin,ofst}}, \end{aligned}$$

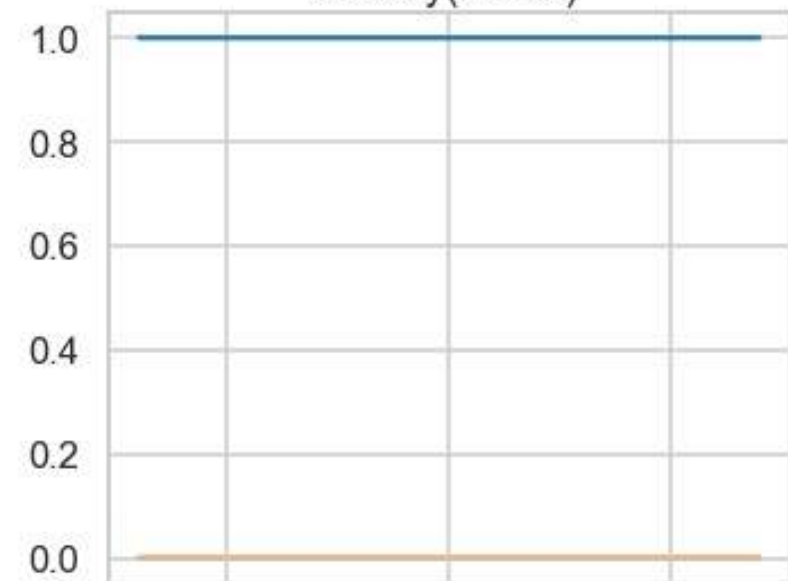
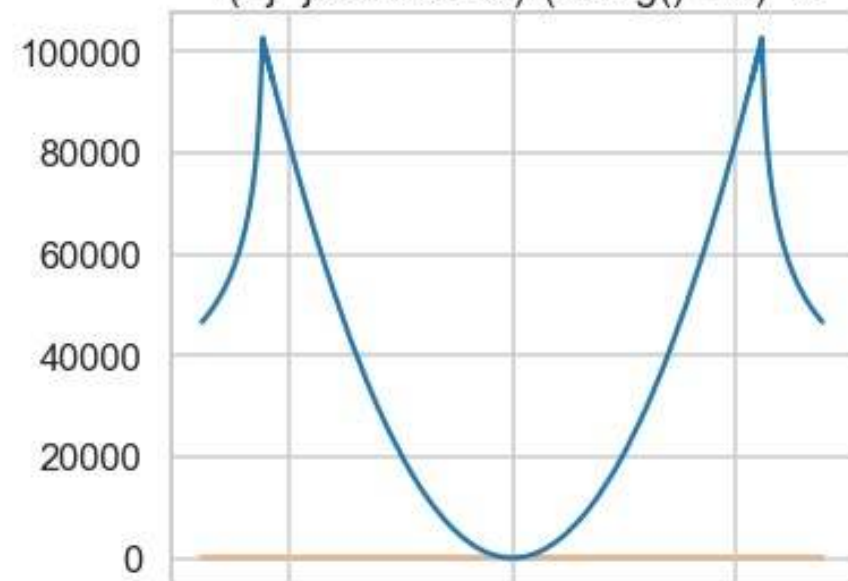
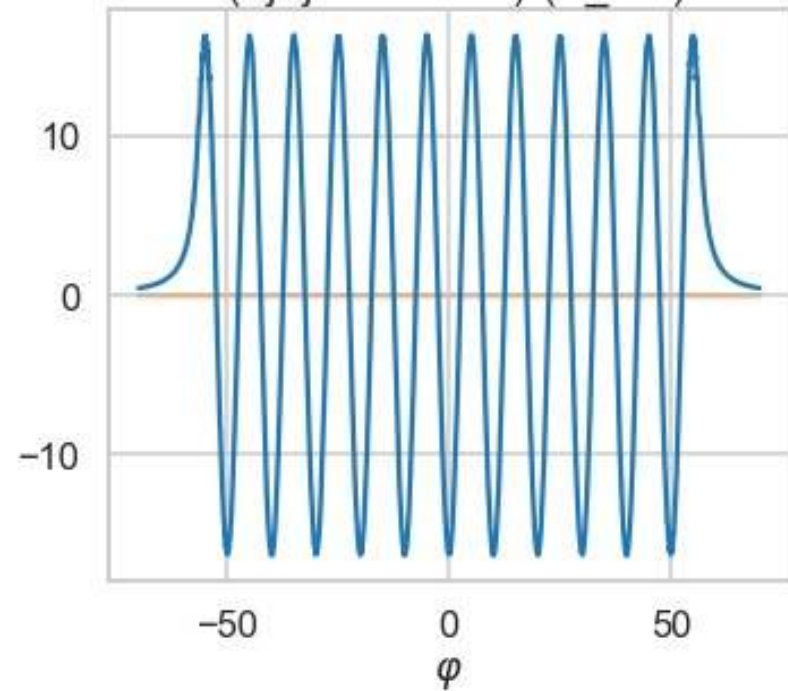
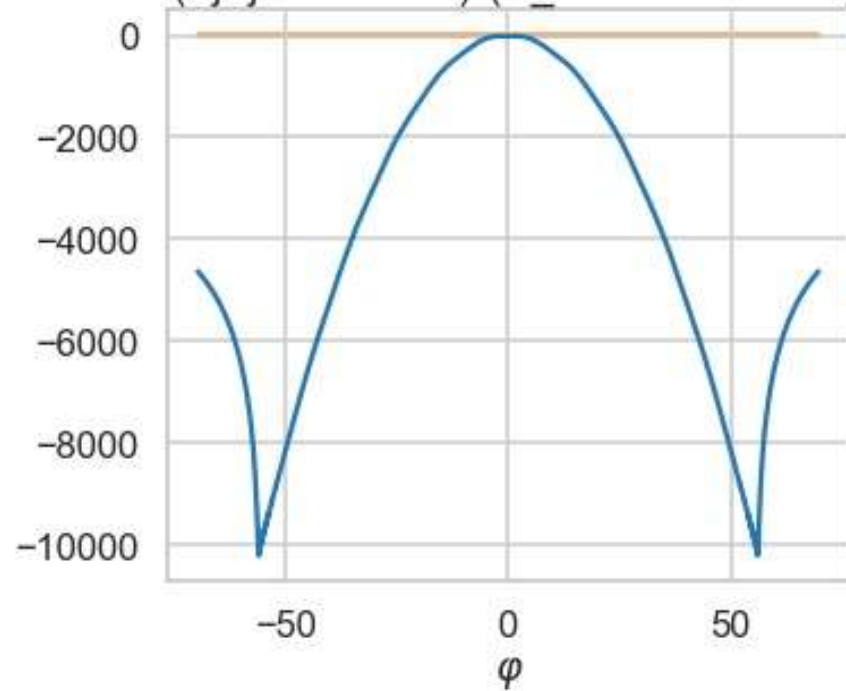
where

$$\begin{aligned} \hat{H}_{\text{lin}} &:= \frac{1}{2C} \hat{Q}^2 + \frac{1}{2L} \hat{\Phi}^2, \\ &= 4E_C \hat{n}^2 + E_J \hat{\varphi}^2, \\ &= \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right), \\ \hat{H}_{\text{lin,ofst}} &:= -\frac{Q_g}{C} \hat{Q} + \frac{Q_g^2}{2} \left(\frac{1}{C} - \frac{1}{C_c} \right), \\ &= \boxed{-8E_C n_g \hat{n}} + 4E_C n_g^2 \left(1 - \frac{C}{C_c} \right) \\ \hat{H}_{\text{nl}} &:= -E_J \left[\cos(\hat{\varphi}) + \frac{1}{2} \hat{\varphi}^2 \right]. \end{aligned}$$

Note that in this form $\hat{H}_{\text{nl}}$ actually has a constant $\cos(x) = 1 - x^2/2 + x^4/4! + \mathcal{O}(x^6)$.

$$= \boxed{-8E_C n_g \hat{n}} +$$

identity(Nmax)

 $\text{phiZPF} * (\text{a.dag}() + \text{a})$  $+(E_j * j2\text{MHz/GHz}) * (\text{a.dag}() + \text{a})^2$  $-(E_j * j2\text{MHz/GHz}) * (\text{H_cos})$  $-(E_j * j2\text{MHz/GHz}) * (\text{H_cos} + \text{Phi} * \text{Phi} / 2.)$ 

1.0

0.8

0.6

0.4

0.2

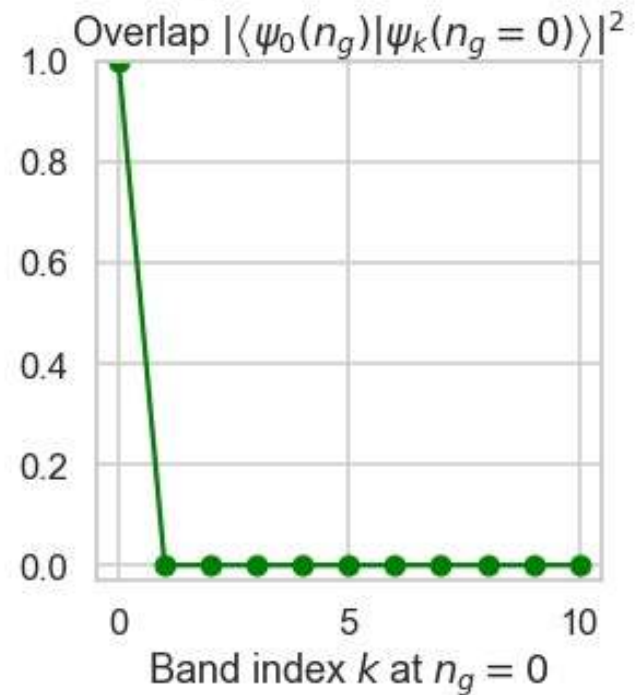
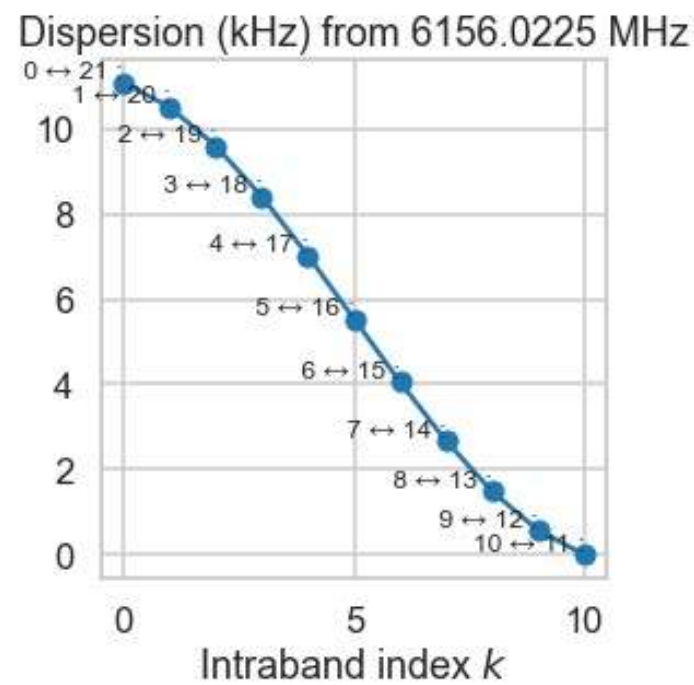
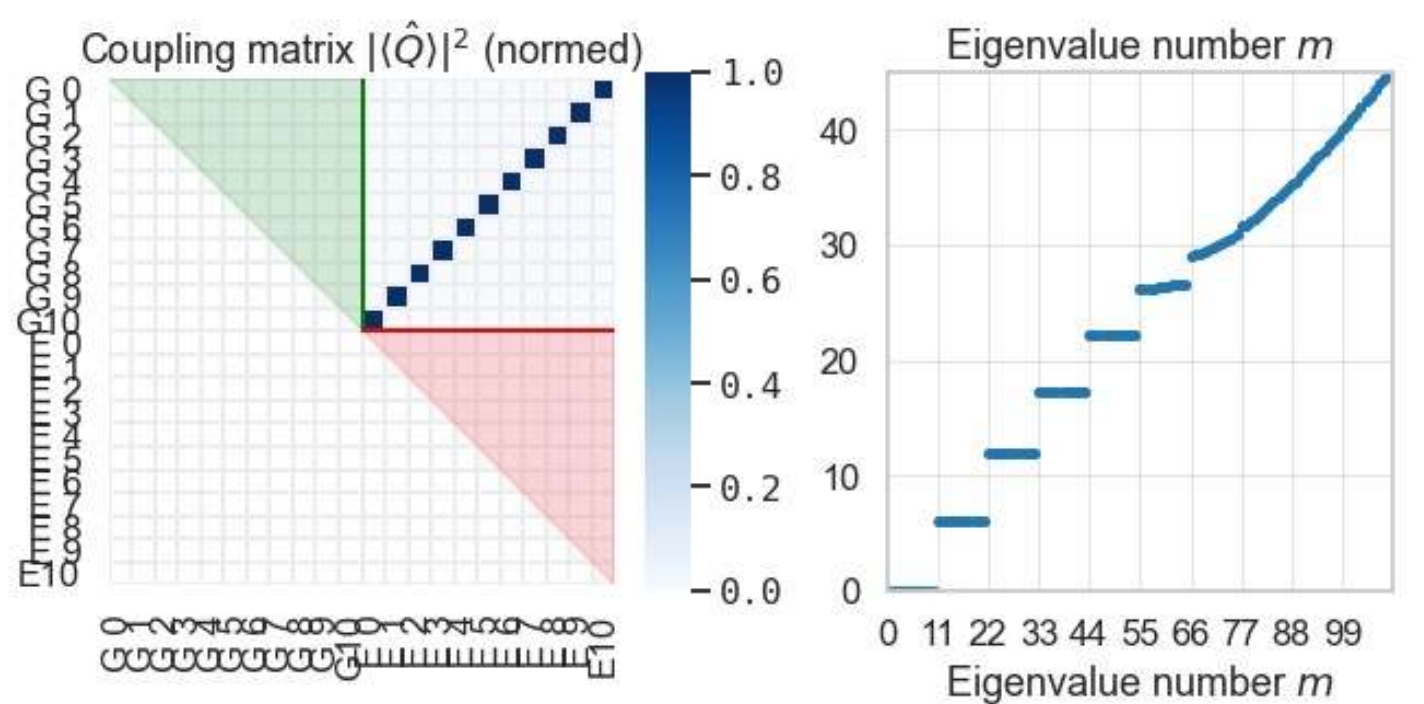
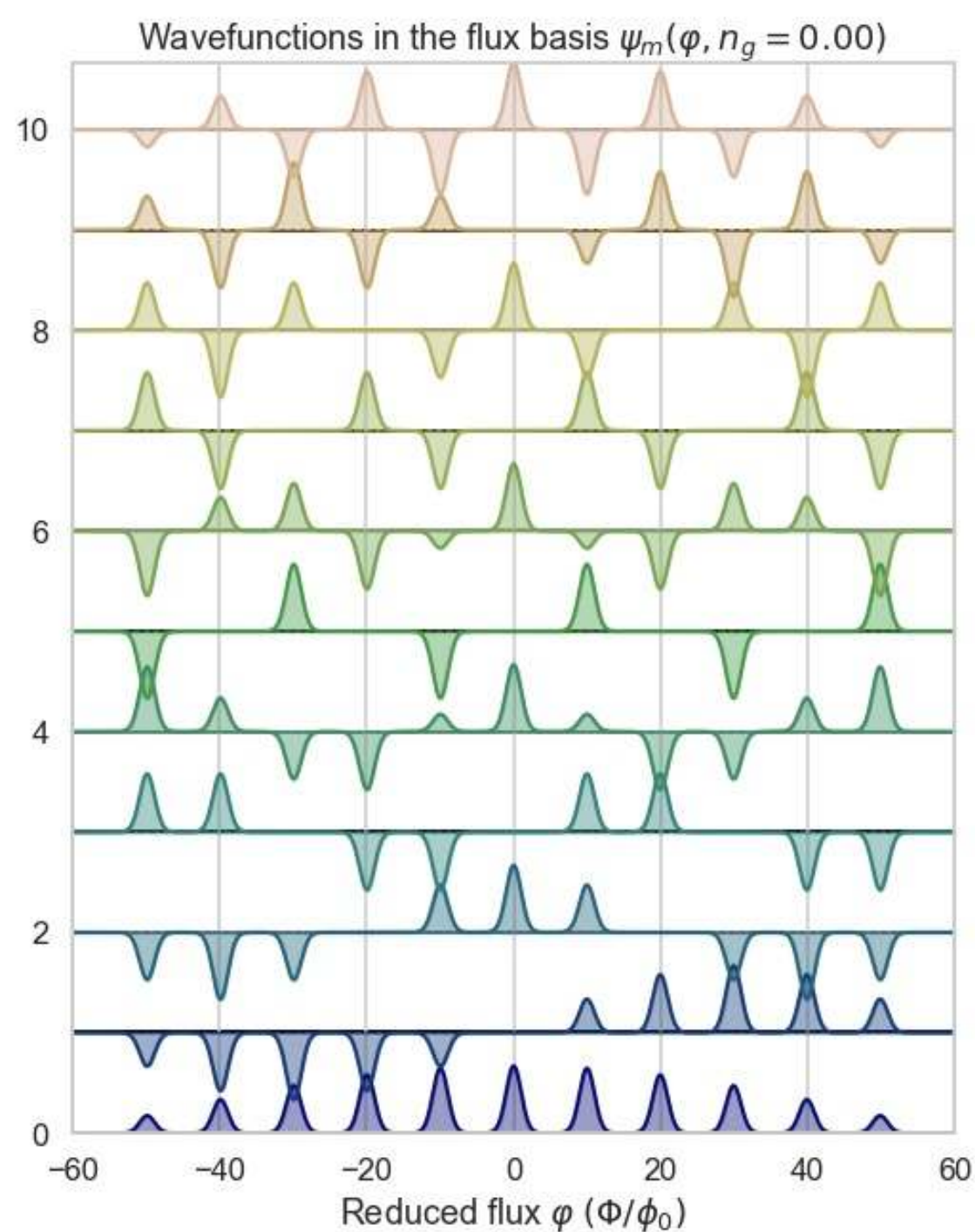
0.0

-50

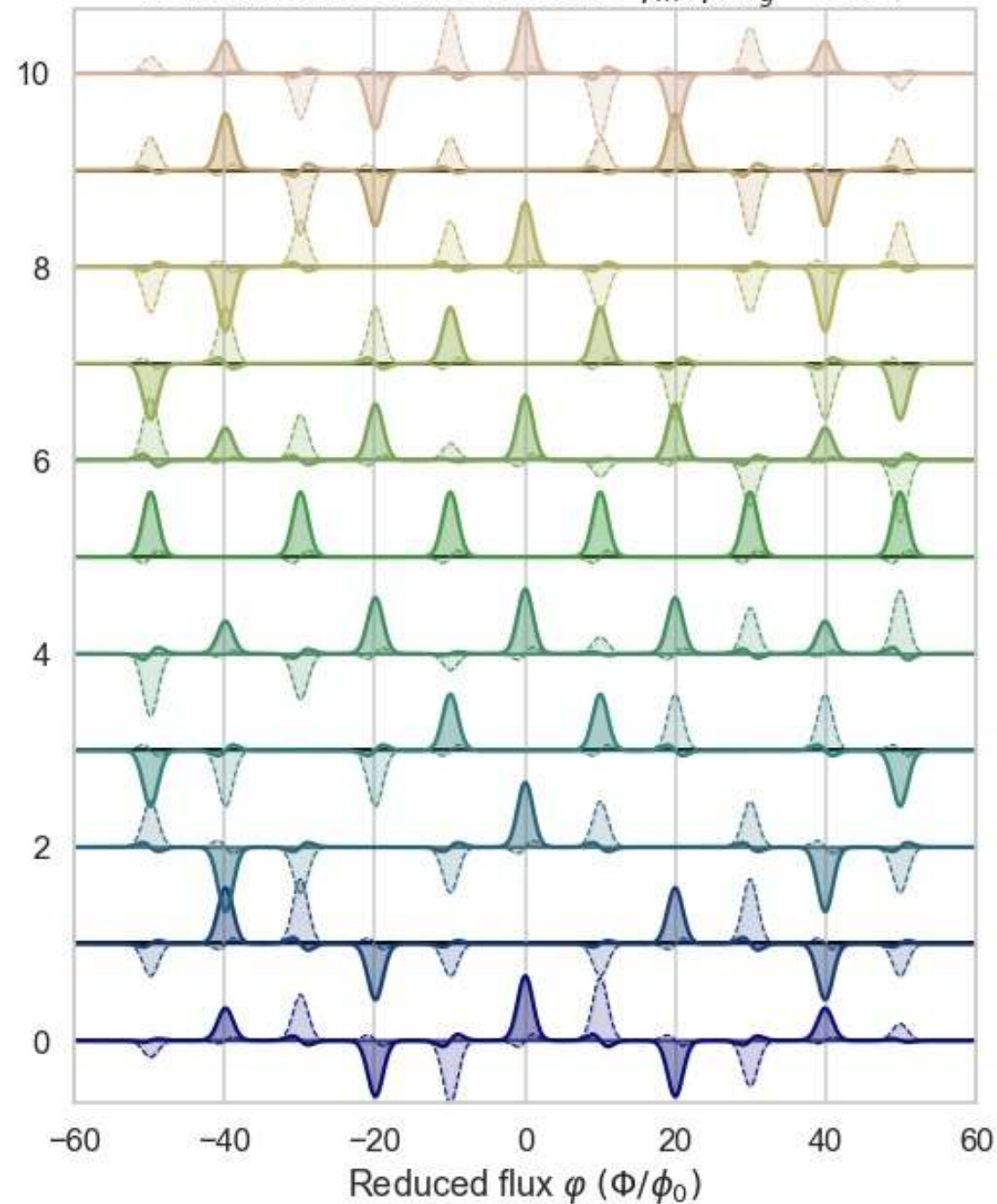
0

50

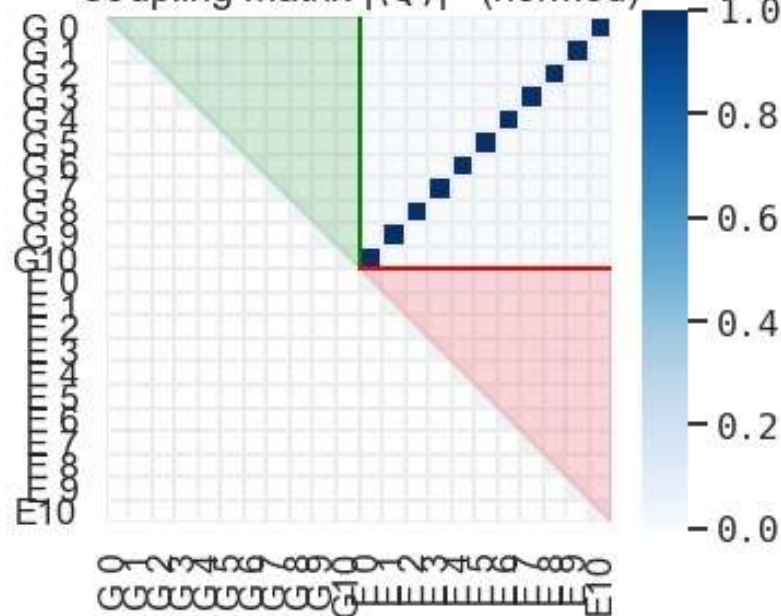
 ϕ



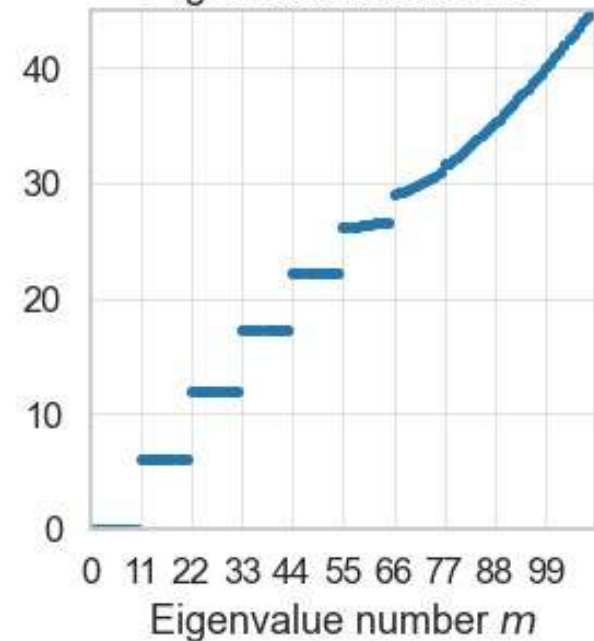
Wavefunctions in the flux basis $\psi_m(\varphi, n_g = 0.25)$



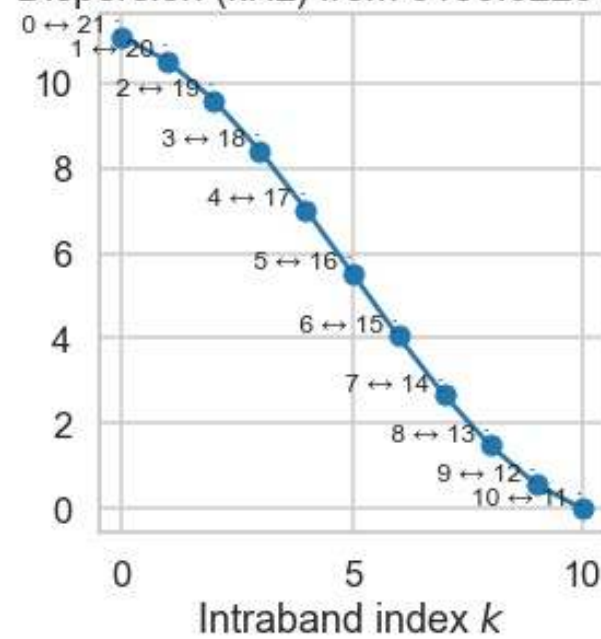
Coupling matrix $|\langle \hat{Q} \rangle|^2$ (normed)



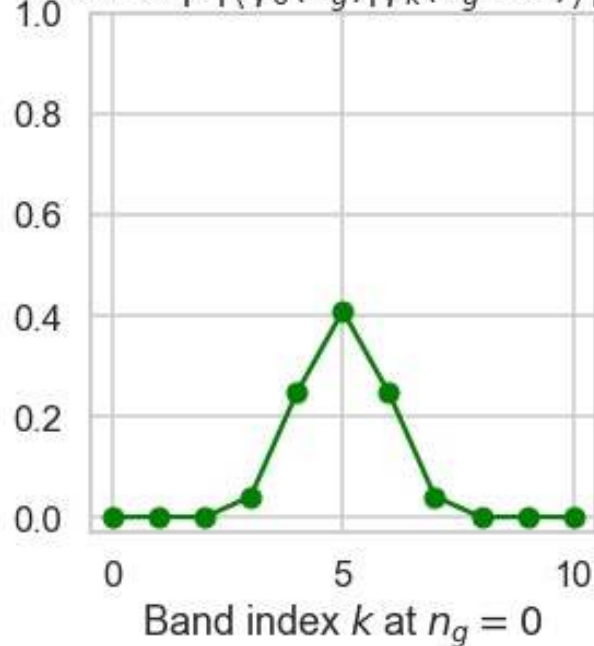
Eigenvalue number m



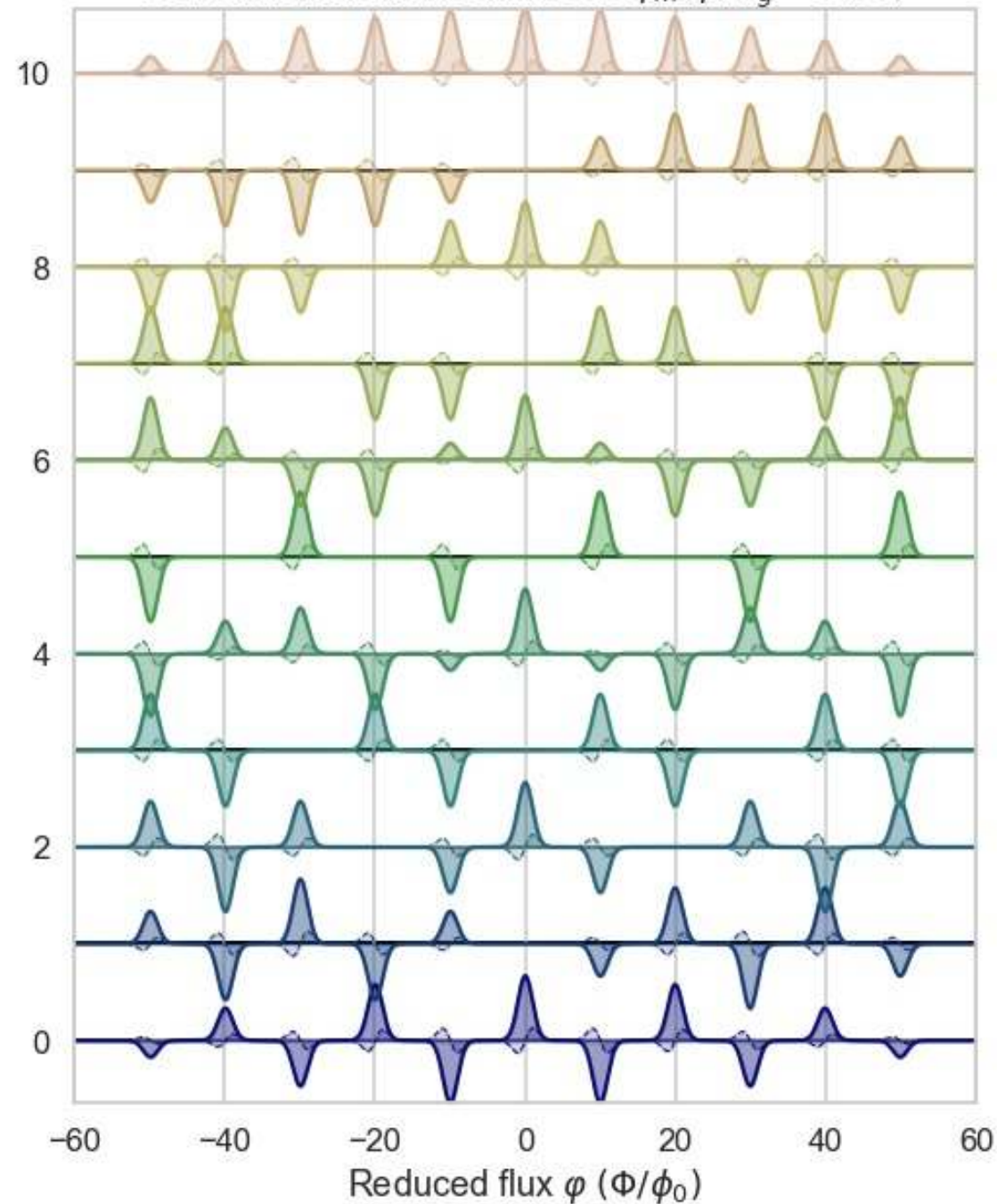
Dispersion (kHz) from 6156.0225 MHz



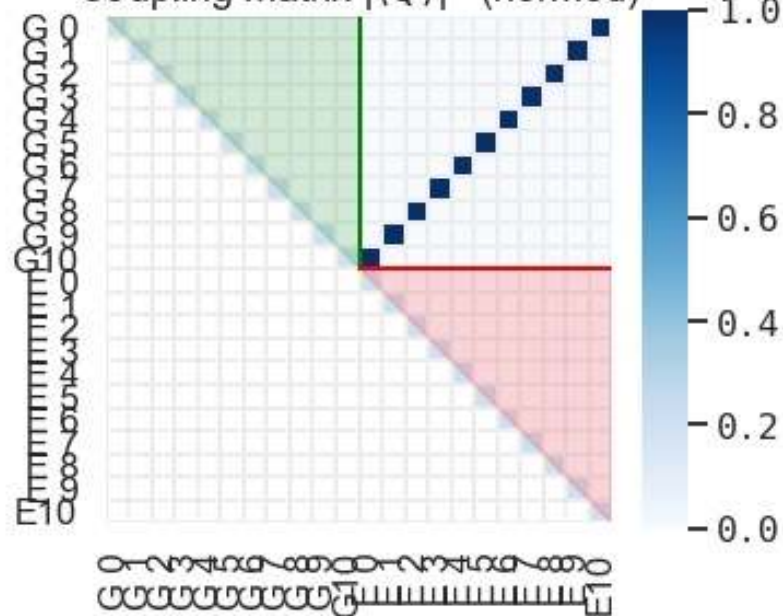
Overlap $|\langle \psi_0(n_g) | \psi_k(n_g = 0) \rangle|^2$



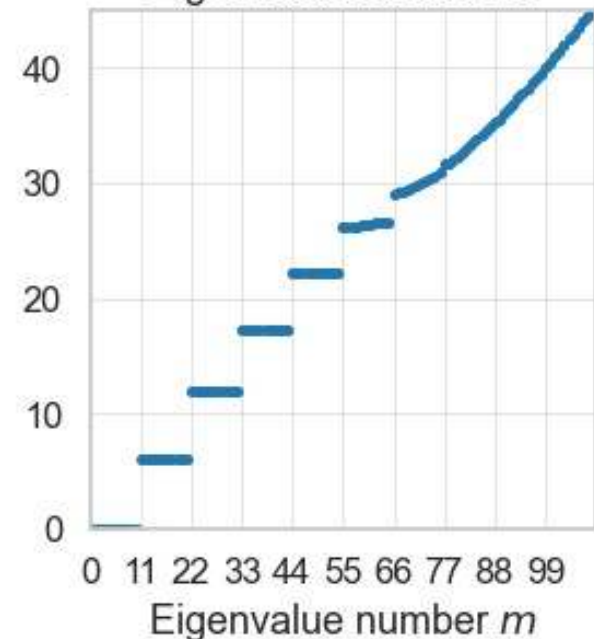
Wavefunctions in the flux basis $\psi_m(\varphi, n_g = 0.50)$



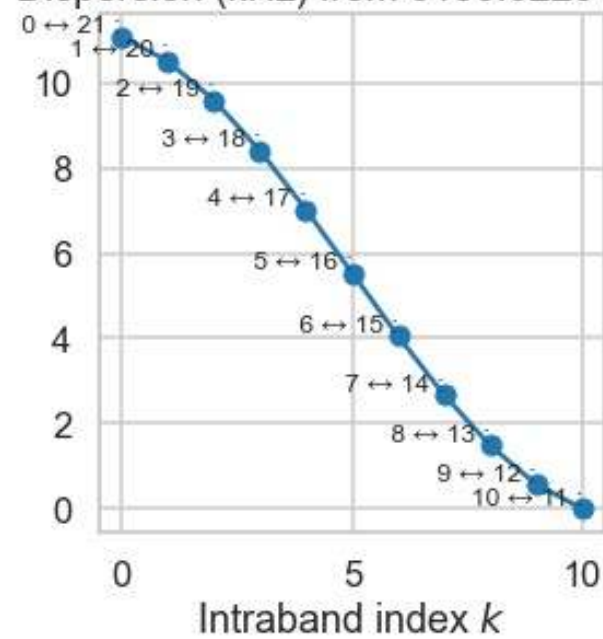
Coupling matrix $|\langle \hat{Q} \rangle|^2$ (normed)



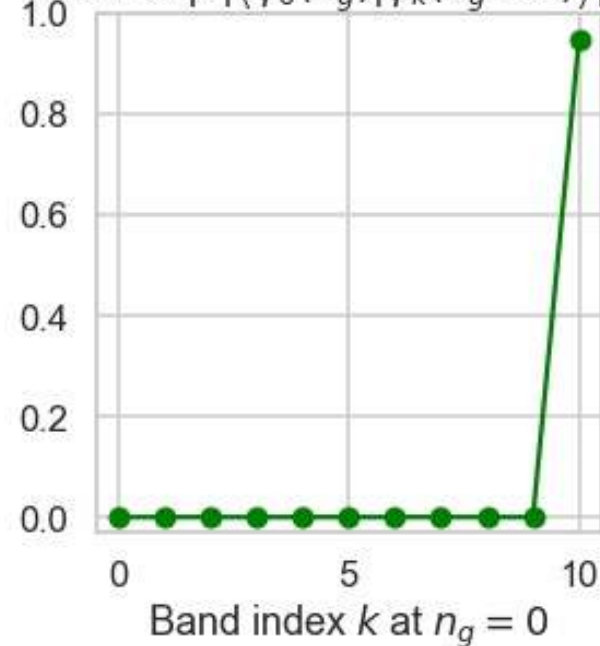
Eigenvalue number m



Dispersion (kHz) from 6156.0225 MHz



Overlap $|\langle \psi_0(n_g) | \psi_k(n_g = 0) \rangle|^2$



Summary

Base $H(\Phi, Q) = H_{lin}(\Phi, Q) + H_{nl}(\Phi)$

$$= \left[\frac{1}{2} Q^T C^{-1} Q + \frac{1}{2} \Phi^T L^{-1} \Phi \right] + \left[- \sum_{j=1}^J \varepsilon_j^{nl}(\Phi_j) \right]$$

$\uparrow \downarrow S$

CLT
Canonical linear transform

Eigenmode $\tilde{H}(\Phi_m, Q_m) = \tilde{H}_{lin}(\Phi_m, Q_m) + \tilde{H}_{nl}(\Phi_m)$

$$= \left[\frac{1}{2} Q_m^T \Omega^2 Q_m + \frac{1}{2} \Phi_m^T I \Phi_m \right] + \left[- \sum_{j=1}^J \varepsilon_j^{nl}(\Phi_j) \right]$$

$\sum_{m=1}^M \hbar \omega_m a_m^\dagger a_m$

$\Phi_j = \sum_{m=1}^M S_{jm} \Phi_m$

Note

All we have done is a canonical linear transform
No loss of information or approximations
No assumption of strength or weakness of NL

The EPR helps find S_{jm}
The eigenmode simulation yields Ω

If we now want we can add back in the linear part and recombine to find ε_j^{nl} in full form

$$\begin{aligned} \tilde{H} &= \tilde{H}_{lin} + \sum_j \varepsilon_j^{nl} + \varepsilon_j^{lm} - \varepsilon_j^{lm} \\ &= \left(\tilde{H}_{lin} - \sum_j \varepsilon_j^{lm} \right) + \sum_j \varepsilon_j^{lm} + \varepsilon_j^{nl} \\ &= \tilde{H}_{lin}' + \sum_{j=1}^J \varepsilon_j \left(\sum_{m=1}^M S_{jm} \Phi_m \right) \end{aligned}$$

Full NL func
recombined
eg., $\cos(\hat{\phi})$

in this normal mode basis

$$H_{lin}(\Phi, Q) \rightarrow \tilde{H}_{lin}(\Phi_m, Q_m) = \frac{1}{2} Q_m^T \underbrace{\Omega^2}_{\substack{\text{Diagonal} \\ \text{matrix}}} Q_m + \frac{1}{2} \Phi_m^T \underbrace{I}_{\substack{\text{identity} \\ \downarrow}} \Phi_m \quad (\text{Diagonal})$$

$$H_{nl}(\Phi) \rightarrow \tilde{H}_{nl}(\Phi_m) = - \sum_{j=1}^J \varepsilon_j^{nl} \left(\sum_{m=1}^N S_{jm} \Phi_m \right)$$

$\uparrow$ Nyl vector
 $\nwarrow$ key
 $\searrow$ SCALAR here, not vector

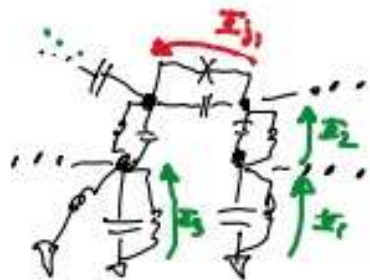
$S_{jm} = j\text{th row } m\text{th column element of } S$

This expression gives NL dipole flux Φ_j in terms of normal-mode fluxes Φ_m

this is essentially the equation from the talk

$$\hat{\phi}_j = \sum_{m=1}^M \phi_{mj}^{ZPF} (\hat{a}^+ + \hat{a})$$

Total-sys Lagrangian



in basis that includes all NL dipole fluxes

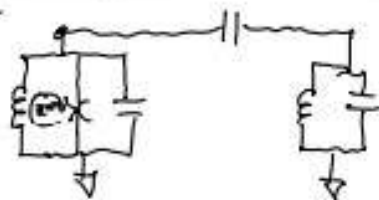
$$\Phi \approx \begin{pmatrix} \Phi_1 \\ \vdots \\ \Phi_j \\ \vdots \\ \Phi_N \end{pmatrix} \begin{array}{l} \text{5x1 vec of } \Phi_j \text{ fluxes} \\ (N-5) \times 1 \text{ vec of non } \Phi_j \text{-fluxes} \end{array}$$

minimal spanning tree vector

$$\dim \Phi = N \times 1$$

Example: Fluxonics

RAU

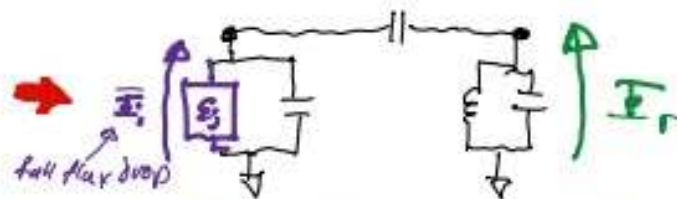


$$\Phi_j = \Phi_j - \Phi_0$$

total flux

deviation away from equilibrium

equilibrium point

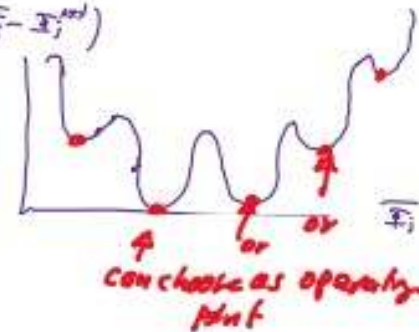


$$\mathcal{E}_j(\bar{\Phi}_j; \bar{\Phi}_j^{ext}) = \frac{1}{2} L \bar{\Phi}_j^2 - \mathcal{E}_j \cos(\bar{\Phi}_j - \bar{\Phi}_j^{ext})$$

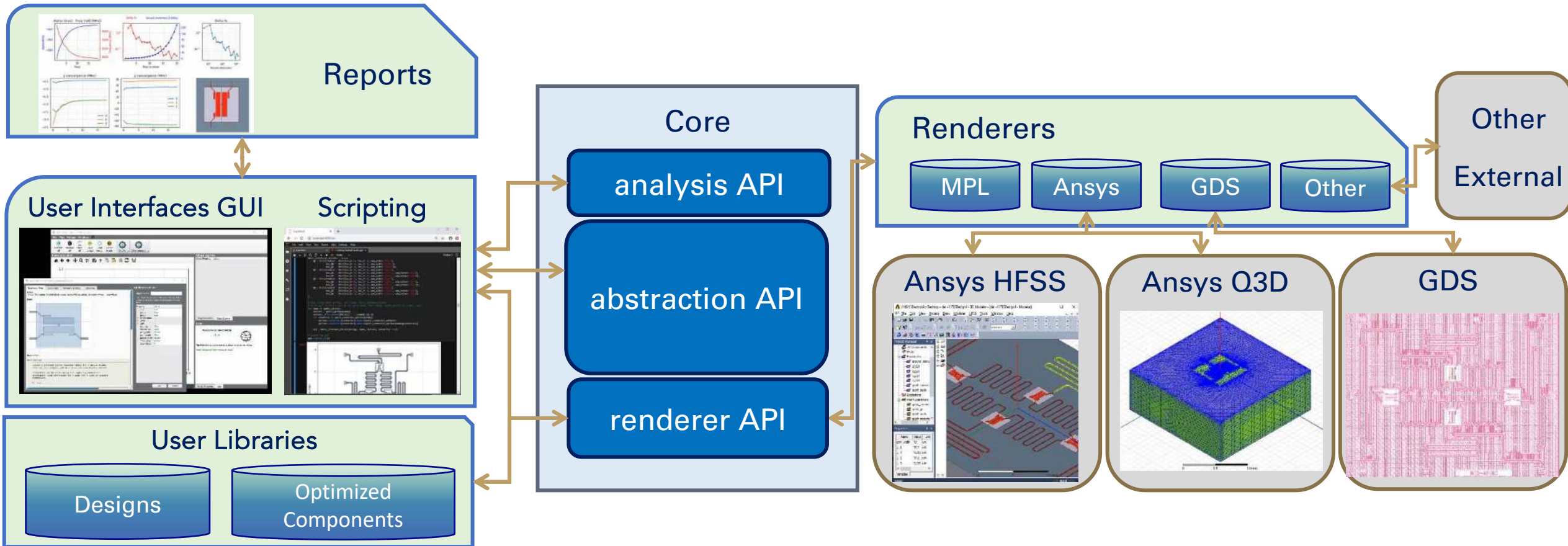


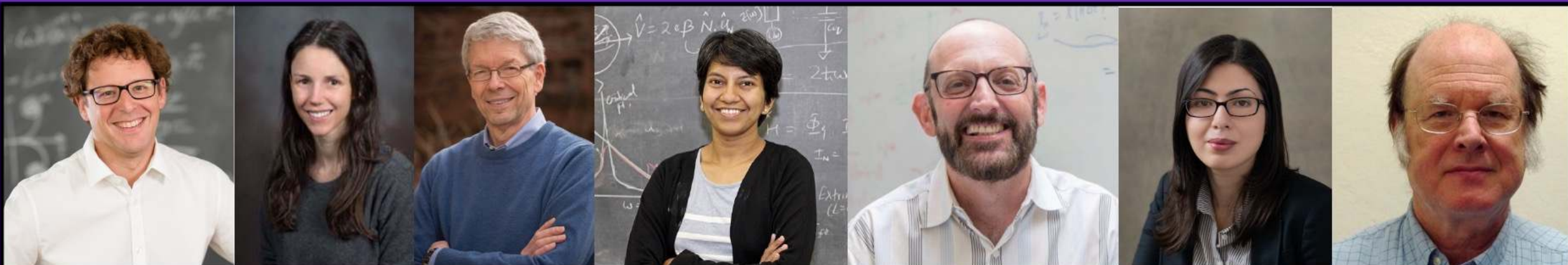
can choose as operating point

we will use later



Qiskit Metal Container Diagram





#Quantum Seminar



Qiskit



Qiskit | quantum device design



Medium

Starting Today, Anyone Can Design Quantum Hardware with ...

Qiskit Metal is the first electronic design automation (EDA) tool specifically for quantum computers, aimed to help the community innovate and ...

1 week ago



Nextgov

Program Lets Users Design Their Own Quantum Computers

It's all possible because of the new Qiskit Metal program from IBM, which is part of their effort to bring open-source tools to the world of quantum ...

6 days ago

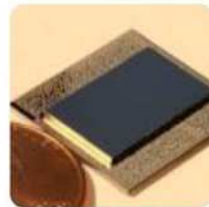


ZDNet

Quantum computing: IBM's new tool lets users design quantum chips in minutes

Qiskit Metal is an open-source platform that automates parts of the design process for quantum chips. Image: IBM Quantum. Building the ...

1 week ago



LeBigData

IBM Qiskit Metal permet à n'importe qui de concevoir une puce ...

IBM annonce la disponibilité générale de la plateforme open source Qiskit Metal, permettant de concevoir facilement des puces pour ...

1 week ago



AG Connect

IBM: Ontwerp nu je eigen kwantumchip

IBM heeft een opensourceplatform - genaamd Qiskit Metal - geïntroduceerd waarmee iedereen die dat wil, kwantumhardware kan ontwerpen.

1 week ago

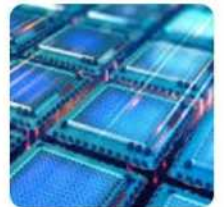


Computer World by IDG

Nova ferramenta da IBM permite que os usuários projetem chips quânticos em...

A IBM anunciou a disponibilidade geral do Qiskit Metal, plataforma de código aberto que automatiza partes do processo de design de chips ...

6 days ago



Open source EDA tool simplifies quantum device design

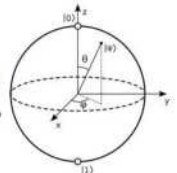
March 29, 2021 // By Rich Pell

0 Comments

QUANTUM
ZEITGEIST

BROOKHAVEN
NATIONAL LABORATORY

Quantum Computing Report
Where Qubits Entangle with Commerce



Forbes

You Don't Have To Be A Rocket (Or Quantum) Scientist To Design A Quantum Computer Chip Using IBM's New Tool Called Qiskit Metal



KeysightQuantum @KeysightQ

Replying to @zlatko_minev @Fo

🎉 Congrats! 🌟 Is there an eta



Analyst Notes:

- 1.) Considering all its advantages, Qiskit Metal should be a clear long-term winner for IBM and the quantum community.
- 2.) By reducing the complexity of chip design, IBM has eliminated a significant barrier that may make quantum attractive to more people.
- 3.) Qiskit Metal makes it possible for young K-12 students to have an understandable hands-on learning experience with quantum computing. Metal can turn an impossible task into a fun learning experience. A positive early learning experience with Metal could result in thousands of future quantum researchers.